14.310x Flipped Classroom materials

Contents Outline

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Student Version

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Materials for Weeks 1-10

14.310x Flipped Classroom
Week 1 Instructions

R for data analysis and working in the cloud

Checklist

- ☐ Complete the INTRO TO R interactive course from the `Swirl` package[1] (Requirement)
- ☐ Watch the Getting started with Google Colab notebooks video tutorial (Requirement)
- ☐ Create or set up a personal Google account
  (you must be able to use Drive and Colab). (Requirement)
- ☐ Create your first Colab notebook (Session 1)
- ☐ Complete Coding Lab 1. (Session 1)
- ☐ Complete Coding Lab 2 (Session 2)

L.1.0

Your first Colab notebook

Instructions: Read and follow the steps below **before proceeding with the activity**. After reading the instructions, access the notebook link and complete the exercises in Colab. This is an individual task, but you will collaborate on the final question.

Notebook: Your first Colab notebook

[1] **Module 1 > Introduction to R** in the online component of the course.

We will cover the essentials of working with Jupyter Notebooks on Google Colab — this resource will be an important tool throughout. Before you can begin working on the coding labs in Colab, make sure to:

- **Create or use a personal Google account**. While it is possible for you to download the Jupyter Notebooks we will work on, and manage them locally in your own machine, we strongly advise you to work on Colab since it will make it easier for your classmates and the instructor to interact with your work when needed. Colab has many features Google Drive already offers: comments, real-time colaboration in the same document or folder, etc.
- **Save the shared notebooks and documents to your own drive.** The notebooks you will have access to are read-only; in order to work on them you will have to copy them to your drive. It is recommended that you maintain a per-week structure in your folders as this will make it easier to follow instructions, especially when reading files or collaborating with others. To save the notebook to your Drive, go to the File menu: `File > Save a copy in Drive`.
- **(When creating a new notebook) Change Colab Notebook's runtime type to R.** By default, when you create a new notebook in Colab, the virtual machine Colab sets up for you is a `Python` installation. This means all the cells in the notebook will only recognize and run `Python` syntax or commands. To switch to an `R` language setup:
  1. Open the notebook menu and go to `Runtime > Change runtime type`.
     - Or click the toggle in the upper-right corner (as shown below) and select `Change runtime type`
  2. In the dropdown labeled `Runtime type`, select `R`.
  3. Click `Save`.

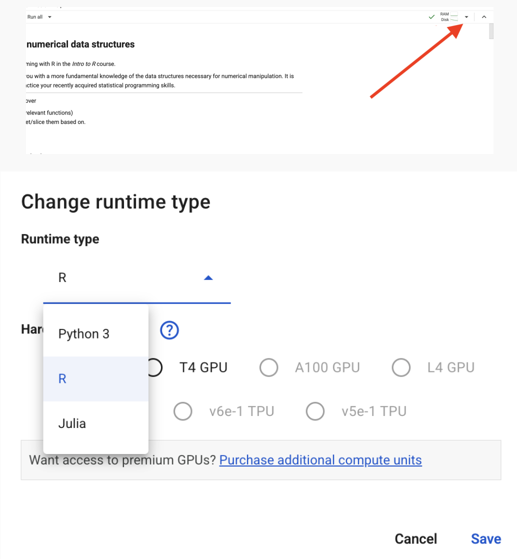

Fig. Changing the runtime type

You may now create your first Colab notebook

L.1.1

Coding Lab 1— Numeric data structures in R

Instructions: Work individually. Solve all exercises in the corresponding Colab notebook. Record your answers and/or code in your copy of the notebook.

Notebook: Coding Lab 1

Submit your work in the format required by the instructor.

L.1.2

Coding Lab 2 — Data manipulation with `dplyr`

Instructions: Work individually. Solve all exercises (sections 0 to 3) in the corresponding Colab notebook; record your answers and/or code in your own copy.

Notebook: Coding Lab 2

Optional **Section 4** allows work in pairs or teams of 3.

Upon completion, submit your work in the format required by the instructor.

14.310x Flipped Classroom
Week 2 Instructions

Discrete random variables, joint and conditional distributions

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Checklist

- ☐ Complete the ADVANCED R interactive course from the `Swirl` package and watch the `ggplot` tutorial[2] (Requirement)
- ☐ Watch the **Import Data** tutorial[3] (Requirement)
- ☐ Read the note on importing `aiwars` and other datasets (Requirement)
- ☐ Complete Guided Case 1 (Session 1)
- ☐ Complete Coding Lab 3 (Session 2)

Importing `aiwars` and other datasets

Depending on flipped classroom logistics of your group, your access to the sessions' assets (datasets, figures, scripts, etc.) will come in one of two forms:

- **Through a direct URL** for instance, the original `aiwars.csv` dataset lives directly in this URL: `https://docs.google.com/spreadsheets/d/1NeZZWI2fT71M9QD8zjnz817T1B3XBM6lolqArSe9CwQ/export?format=csv`
- **Your instructor will provide them to you privately** and they will either:
  - Share them via a URL similar to the one above.
  - Share the files through other means.

[2] **Module 2 > R Course and R Tutorial: ggplot** in the online component of the course.
[3] **Module 3 > R Tutorials: Basic Functions** in the online component of the course.

As they may want to slightly modify the original files for grading purposes, or have any other goal in mind.

If a dataset's URL is provided, you can read the data directly into R with the URL and the appropriate reading function; simply provide the URL as the path. For instance, `aiwars` is in `.csv` format:

```
URL_aiwars <- "https://docs.google.com/spreadsheets/d...."

aiwars <- read.csv(URL_aiwars)
```

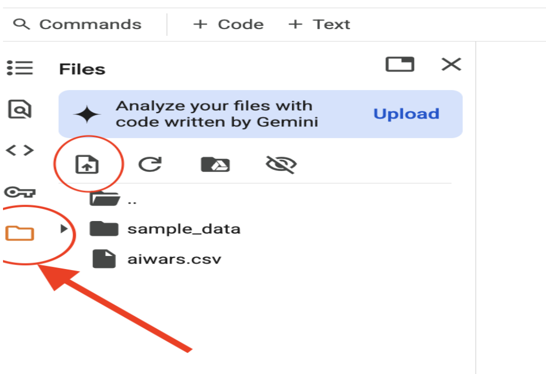
And similarly for other formats. Suppose we had a single Excel sheet:

```
install.packages("readxl")
library(readxl)

URL_aiwars_xl <- "https://...some-URL"

aiwars <- read_excel(URL_aiwars_xl)
```

If the file is shared in any other way, we will first need to upload it to the virtual machine's disk in Colab and read it from there, providing the `path` to it — as you would do if you were reading data to R in your own computer. To upload a file in Colab, click the folder icon in the leftmost menu bar, as shown in the screenshot. Then click the upload icon (circled in red); as you can see, we already uploaded the file.



Once uploaded the file can be read as a if in your machine:

```
aiwars <- read.csv("aiwars.csv")
```

Important: Colab runtime limitations

While your Colab notebook — including all code cells, text, and outputs — will remain saved, the underlying *runtime* (i.e., the virtual machine that executes your code) is temporary. After a period of inactivity or when a time limit is reached, the runtime will automatically disconnect.

When this happens:

- All variables and objects stored in memory (data frames, vectors, functions) will be lost.
- Any files you uploaded manually will be erased.

When reconnecting, a fresh virtual machine will be started, and you'll have to re-run your code, and re-upload any needed files. Disconnects may happen occasionally as you work on off-notebook tasks; re-reading or re-uploading files should not take more than 30 seconds.

Coding exercises in case studies

Throughout the course, the case studies you will work on contain a mix of conceptual and coding questions. To simplify your workflow, all the questions where you're required to write code are specially labeled. They appear in this format:

Question 5. [2.3] calculate the probability of event A...

The small icon signals that the question requires coding. The number inside the brackets (2.3 in this example) is a numbering **within the Session's dedicated Colab notebook**, which will often be different from the overall question number for the activity.

Further, if you click on the orange icon, it will take you directly to the question's cell within the Notebook by opening a new browser tab – remember you must work on your own copy of the notebook. These links are simply provided as a convenience to help you locate and visualize the relevant task quickly.

Apart from being the place where you are expected to code your answers, the corresponding cell usually includes a placeholder for your code, often accompanied by additional hints, extended context, or partially completed code to help you get started. When working on a case study, keep both this PDF and Colab notebook open and use these references to move smoothly between the written materials and your code.

Case study G.2.1

Reddit posts: PMFs and PDFs

Instructions: Work in pairs or groups of three. Answer the questions

Notebook: Case Study 1

Dataset: aiwars.csv

One of you must share editor access to their notebook with the rest.

Scenario: We will cover this week's lecture contents using a real-world dataset from Reddit. You will put your data manipulation (with dplyr and base R) skills to practice

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Context: The `AIwars` dataset consists of a collection of posts scraped from the `r/AIwars` subreddit, a forum where users debate the societal implications of AI. These range from predictions about AI-driven job loss and technological conflict to satire, trolling, and speculation. The dataset captures a rich period of discussion and polarization. Each observation corresponds to a single post, with information about its author, contents and engagement.

Key Variables:

- `post_index` — Unique identifier for each post (starts at 1 and consecutively numerates all posts).
- `author` — Reddit username of the post author (may be `[deleted]`).
- `post_date` — an R `Date`
- `fulltext` — Full text of the Reddit post in format
  `TITLE: [some post title] TEXT: [some post text]`
- `post_length` — the net number of characters in `fulltext` (excluding the TITLE and TEXT headers).
- `post_upvotes` — Number of upvotes (user endorsements or *likes*) the post received.
- `comments_number` — Total comments the post received (includes replies to other comments).

Part 1. Basic dataset facts

1.1 [1.1] Install and load the `tidyverse` packages. Create `aiwars_URL`, a `character` with the download URL for `aiwars.csv`. Then read in the dataset as `aiwars`, a data frame. Use `glimpse()` to answer:

(a) How many posts are we working with?
(b) How many variables does the dataset have?
(c) How many are *true* `character` types?
(d) How many are numeric?
(e) Which characters are better described as factors? Why?

1.2 [1.1.1] For the entire case study, will use only the **Key Variables** described above. While `select()`ing columns from the current data frame is possible, a memory-efficient alternative is to read in only those columns we need.

Use the `col_select` argument in `read_csv()` to create `aiwars_short`, a data frame containing only the columns described as **Key Variables**.

1.3 [1.1.2] Use `select()` to create `aiwars_short2`, a dataframe with only the

key variables.

1.4 Moving forward we will work with `aiwars_short` only. You may delete `rm(aiwars, aiwars_short2)`.

1.5 Use `summary()` or other summarizing methods to answer the following questions:

(a) What is the time span of the posts?
(b) On average, how long (in words) is a post on this subreddit?
(c) How many posts don't have any replies?
(d) What is the largest number of upvotes in a post?
(e) Which author has posted the most? (Consider `as.factor()`).
(f) How many distinct users have posted in this subreddit? (Consider `unique()`).

Part 2. Counting posts

2.1 [2.1] Create the following variables in `aiwars_short`:

(a) `popular`, takes value 1 when the post has 29 or more upvotes. Takes value 0 when the post has less than 29 likes.
(b) `text_classification`, which takes value `mostly title` if it is less than 70 characters long, `short` if it has 70 or more but less than 110 characters, `common` if it has 110 or more but less than 900 characters, and `long` if the post has over 900 characters.

2.2 How many `common` posts are there?

2.3 If we draw a post at random, it will be a popular one with what probability?

2.4 If we instead sample 10 posts at random **with replacement**, what is the probability 3 of them are popular?

2.5 [2.2] Suppose random variable X; $\Omega_X = \{0, 1, 2, ..., N\}$ represents the number of popular posts in $N = 10$ draws with replacement from `aiwars_short`. For the following probability statements describing various events, first write them down as a probability and then use R to compute them.

(a) What is the probability of getting at most 4 popular posts?
(b) What is the probabililty of getting at least one but at most 3?
(c) What is the probability of getting a number of popular posts that is not 0 or 4?

2.6 Similarly, we can treat the texts' classification as random variable Y. Is it continuous or discrete? What is Ω_Y?

2.7 [2.3] Express $f_Y(y)$ as a data frame `f_Y` with as many rows as elements in Ω_Y and two columns: `y`, the value taken by Y and `p` its mass or density.

2.8 What is the probability a post is not `mostly title` ?

2.9 What is the probability a post is either `common` or `long`?

Are popular posts different?

2.10 What is the probability a post is popular **AND** `long`?

As we learned from the lecture, if two events are **not independent**, the additional information one of them provides should *update* the probabiility of the other. We will examine this fact by calculating the probability of a post being long **GIVEN that we know it is popular**:

$$P(Y = long \mid popular = 1)$$

As a reminder:

$$P(long \mid popular = 1) = \frac{P(long \ \cap \ popular = 1)}{P(popular = 1)}$$

2.11 Use your answers to questions 2.3 and 2.10 to compute the probability a post is long GIVEN it is popular. How does this probability compare to the unconditional probability $P(Y = long)$?

2.12 We now know that the event $(Y = long)$ has two mutually exclusive and exhaustive partitions: $(popular = 1)$ and $(popular = 0)$. Following the lecture, Show that the Law of Total Probability verifies for this event.

2.13 What is $P(popular = 1 \mid long)$? Is your answer the same as in 2.11 ? Should it be? Why?

2.14 [2.4] Create `joint_pmf`, a data frame similar to the one you created for 2.7; in this case, it should display the probability for all possible combinations of $Y \cap Popular$. The sample consists of the ordered pair $\Omega_Y \times \Omega_X = (long, 1), (long, 0), ..., (\text{mostly title}, 1), (\text{mostly title}, 0)$

2.15 [2.4.1] Use `joint_pmf`, $f_X(x)$ and $f_Y(x)$ to compute multiple conditional probabilities at once in R:

(a) $P(Y \mid Popular = 1)$, a vector of length 4. Check that the vector adds up to 1.

(b) Get $P(Popular \mid Y = common)$. What is its length? Check it adds up to 1.

2.16 Generally, considering your answers to 2.15a, 2.15b or 2.11, are the popularity of a post and its length classification independent? Why?

L.2.3

Coding Lab — Row and Column operations

Instructions: Work individually. Solve all exercises in the corresponding Colab notebook. Record your answers and/or code in your copy of the notebook, or in the format required by the instructor.

Notebook: Coding Lab 3

14.310x Flipped Classroom

Week 3 Instructions

Continuous random variables; conditional and joint pdfs.
More advanced data manipulation and visualization in R

Checklist

- ☐ Complete Guided Case Study 2 (Sessions 1 & 2)
  - ☐ We will continue to work with `aiwars`
  - ☐ Get the `aiwars_embeddings` dataset here
- ☐ Complete Open Case Study 1 (Session 2)

Case study G.3.2

Analyzing text similarity:
applying row operations to Natural Language Processing

Instructions: Work on your own for Session 1, and in pairs or groups of 3 for Session 2.

Notebook: Case Study 2

Dataset: `aiwars_embeddings.csv`

Part 1 Part 2 Part 3

Dataset: `speech_anchors.csv`

The `aiwars_embeddings` dataset (we split it in 3) contains 1024 variables associated with each post's `fulltext` in `aiwars`. You will learn what these embedding variables `V1`, `V2`, ..., `V1024` represent below, but keep in mind `post_index` in both datasets uniquely links each post in `aiwars` to its embedding in `aiwars_embeddings`. Dataset `speech_anchors` is a 2-row data frame with the same structure but `labels` instead of post indices.

Part 0. Required data and packages

0.1 [0.1] Load/install the `tidyverse` packages.
0.2 [0.2] Read in `aiwars`, the same dataset as last week. Also, read in `speech_anchors.csv` as `anchors`, which should be a data frame with two rows and 1025 columns.
0.3 [0.3] Read in parts 1,2 and 3 of `aiwars_embeddings` as `part1`, `part2` and `part3` respectively. Then use `rbind()` to vertically merge the three parts into `aiwars_embeddings`. Use `arrange()` to sort the dataset in `post_index`-ascending order. **Reading takes over a minute.*
0.4 Remove the individual chunks: `rm(part1,part2, part3)`.

Scenario:

Natural Language Processing (NLP) focuses on enabling machines to understand and work with human language. One of the key advances in NLP has been the development of methods that represent text as vectors—documents, social media posts, chatbot requests, etc. These vector representations, often referred to as *embeddings*, capture aspects of a text's semantic content: its meaning, tone, and topical focus.

To illustrate the idea, suppose we have the following texts:

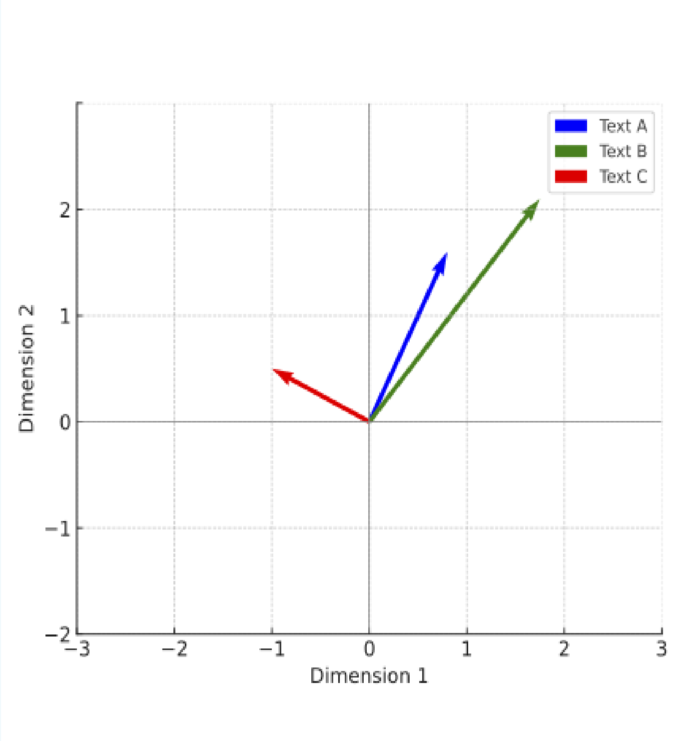
- Text A: "I love how these new AI tools can generate art!"
- Text B: "AI-generated images are cool, but I'm worried about copyright."
- Text C: "I painted this myself — no software involved."

We embed these texts in 2-dimensional[0] vectors from the origin, plotted below:

- $\vec{a} = \begin{pmatrix} x_a \\ y_a \end{pmatrix} = \begin{pmatrix} 0.8 \\ 1.6 \end{pmatrix}$
- $\vec{b} = \begin{pmatrix} x_b \\ y_b \end{pmatrix} = \begin{pmatrix} 1.7 \\ 2 \end{pmatrix}$

- $\vec{c} = \begin{pmatrix} -1 \\ 0.3 \end{pmatrix}$

Figure 1: **2D text embeddings (vector from the origin)**



Despite their differences in tone, $\vec{a}$ and $\vec{b}$ in Figure 2 are close to each other because the texts they represent use AI-related language, whereas $\vec{c}$ is far removed given the lexical differences. In part 2, we will explain what "close to each other" means for vectors, with two common semantic similarity measures: **direction** (i.e., the *angle* they form) and **distance** between them.

Part 1. Coding vector operations

You will start by programming functions necessary for this analysis, such as the *magnitude* of a vector, the *distance* between two vectors, their *dot product*, etc.

Distance between two vectors

The distance from $\vec{a}$ to $\vec{b}$ in Figure 2 is defined as:

$$\| \vec{a}, \vec{b} \| = \sqrt{(x_a - x_b)^2 + (y_a - y_b)^2}$$

This is the sum of the squared differences in x and y-coordinates (in any order). Then we take the square root of that sum. Substituting the values in the scenario description:

$$\cdots = \sqrt{(0.8 - 1.7)^2 + (1.6 - 2)^2}$$

$$\cdots = \sqrt{0.81 + 0.16} = 0.985$$

1.1 [1.1] Save $\vec{a}, \vec{b}, \vec{c}$ as numeric vectors `a`, `b` and `c` respectively. Compute the distances for each pair of vectors and save them accordingly: `dist_ab`, `dist_ac`, `dist_bc`. Hint: you can apply operations on all the coordinates at once: `(a-b)`.
   – Make sure your result for `dist_ab` is the same as above.

In general, to get the distance between any two n-dimensional vectors

$$\vec{V} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}, \vec{U} = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}$$

we generalize the process with all their coordinates and take the square root:

$$|| \vec{V}, \vec{U} || = \sqrt{(v_1 - u_1)^2 + (v_2 - u_2)^2 + \cdots + (v_{n-1} - u_{n-1})^2 + (v_n - u_n)^2}$$

$$\cdots = \sqrt{\sum_{i=1}^{n} (v_i - u_i)^2}$$

1.2 [1.2] `aiwars_embeddings` contains a 1024-dimensional vector per post. Program a distance function `dist_function(V, U)` that takes any two equal-length, numerical vectors. It returns the distance between them.
   – Verify your function is correct and reproduce the results in 1.1
   – From `aiwars_embeddings`, create vectos `p1` and `p2`, the embeddings for post indices 1 and 2, respectively (don't forget to exclude the `post_index` and coerce with `as.numeric`). Check that $||\vec{p}_1, \vec{p}_2|| = 0.96028211$

Distance from one to multiple vectors

Working with individual, separate objects such as `dist_ab`, `dist_ac`, and so on can be difficult and tedious. In R, we will often want all the distances with respect to $\vec{a}$ in a single "distances" vector or variable:

$$D_a = \begin{pmatrix} || \vec{a}, \vec{a} || \\ || \vec{a}, \vec{b} || \\ || \vec{a}, \vec{c} || \end{pmatrix}$$

We will now pass `dist_function()` through all the example vectors at a time to obtain D_a — the steps should be familiar from the previous Coding Lab.

1.3 [1.2.1] Create vector `D_a` with a *for* loop by following the steps below:

  - `rbind` vectors `a`, `b`, and `c` to create `examples_matrix`, a matrix with 3 rows and 2 columns: one row for each example vector, and one column for each x, y-coordinate.
  - Declare `D_a`, an empty numeric vector of length 3.
  - Declare a `for(i in 1:nrow(examples_matrix))...` loop to iterate over `examples_matrix`.
  - Inside the *for* loop, code the following:
    - Subset/slice the rows of `examples_matrix` to obtain vector `v` on each iteration.
    - Calculate the `distance` between `a` and `v` with `dist_function`.
    - Assign the `distance` result to `D_a[i]` on each iteration.

1.4 Verify your answers with those of 1.2.

What should be the result for $||\vec{a}, \vec{v}||$ when $\vec{v} = \vec{a}$? Why?

Finally, we can wrap the code we created for 1.3 in a function that directly returns D_a when given a vector and a matrix.

1.5 [1.2.2] Create a function `batch_distances(u, M)` which takes a vector `u` of length n and a matrix `M` with an arbitrary number of rows k, and n columns. The function should `return` D_u, a vector of length k where each element is the distance between the input `vector` and a row of the input `matrix`.

1.6 Use `batch_distances(_,_)` to reproduce 1.3. Verify you get the same vectors D_a and D_p

The norm of a vector

The norm $|| \vec{V} ||$ (also called the *magnitude* or *length*) is the distance from where it starts to where it ends. For instance, in the plot above, the norms are the lengths of each arrow. Since all embeddings are vectors from the origin (all their starting coordinates are equal to 0), their norms can be expressed as: $||\vec{V}, \vec{0}||$, with $\vec{0}$ the zero vector of the same dimension as $\vec{V}$.

$$|| \vec{V}, \vec{0} || = || \vec{V} || = \sqrt{(v_1 - 0)^2 + \cdots + (v_n - 0)^2}$$

$$\ldots = \sqrt{\sum_{i=1}^{n} v_i^2}$$

1.7 [1.3] Create `norm0_function`, a function that takes `V`, a numerical vector of length n as an input, and returns its norm from the origin. Hint: you can recycle `dist_function` inside `return()` with the appropriate `U`.

  - Verify that $||\vec{a}|| = 1.788854$ and $||\vec{p}_1|| = 1$

1.8 [1.3.1] Compute $|| \vec{a} ||, || \vec{b} ||$ and $|| \vec{c} ||$. Save them to `a_norm`, `b_norm`, `c_norm` respectively.

1.9 Similar to 1.5, create a function `batch_norms0(M)` taking a matrix of n vectors, all with the same number of dimensions/columns m, and returns a vector N_0 of length n with a norm from the origin for each row in `M`.Verify your results in 1.8 with `examples_matrix`.

Dot product

The dot or *scalar* product of two vectors $\vec{U} \cdot \vec{V}$ is defined as the sum of the product of each pair of their coordinates:

$$\begin{aligned}\vec{U} \cdot \vec{V} &= u_1 * v_1 + u_2 * v_2 + \cdots + u_n * v_n \\ &= \sum_{i=1}^{n} u_i v_i\end{aligned}$$

R does element-by-element multiplication by default: `sum(u*v)`.

1.10 [1.4] Create a function `dot_product(u,v)` that takes two numeric vectors of the same length as input and returns their dot or scalar product. Check that $\vec{p}1 \cdot \vec{p}2 = 0.5389$

1.11 [1.4.1] Again, we're interested in computing a vector $\mathbb{D}_a$ of dot products. Use a *for*-loop and wrap it in a function `batch_dot_product(v, M)` taking a vector `v` of length n and a matrix `M` of as many rows as vectors and n columns, and returns a dot product $\vec{v} \cdot \vec{m}$ for each row. Verify that you get $P_1 = \begin{pmatrix} 1 \\ 0.5389 \end{pmatrix}$

Part 2. Processing text embeddings for AIwars

We prepared the `aiwars_embeddings` dataset for you by using one of OpenAI's text embedding products[1]. For each post's `fulltext` in `aiwars`, there is a vector representation of length $n = 1024$ in `aiwars_embeddings`:

Variables

- `post_index`: a consecutive index **in the same order as** `aiwars`. A given post and its embedding share the same post index.
- `V1`, `V2`, ..., `V1024`: the coordinates $v_1, v_2, ..., v_{1024}$ of each vector embedding $\vec{V}$.

You will now apply the functions you previously programmed to measure the semantic similarity of real-world embeddings.

Semantic similarity measures: *Luddites* vs *Synthists*

To illustrate semantic similarity measures, we will take the embeddings from four select posts from the AIwars subreddit, each with distinct tones and topics:

2.1 [2.1] Filter `aiwars` for posts with the following `post_index` values: $456, 584, 2397, 2625$. Select the `fulltext` and post index columns only. Assign the resulting dataframe to `tones`. Just generally, what is each post about?

2.2 To keep track of the post contents, give each a label: `post_labels <- c("luddites", "mockery", "critique", "synthists")` and add the vector as a new variable `label`. Remember, `tones` should be in `post_index` ascending order for the labels to match.

2.3 Answer the following questions regarding `tones`:

(a) Historically, what does the term *Luddite* refer to? (you may want to do a quick internet search)
(b) With posts 456 and 584 in mind, provide a short description of a Luddite (Yuddite) in the context of the AIwars subreddit.
(c) Per post 2625, what is a *Synthist*?
(d) In your opinion, which post in `tones` is most similar to post 584 in its intent to mockingly denominate other points of view and/or users?
(e) Is any of the `tones` posts more nuanced or academic in tone? (i.e., less opinionated, more descriptive).
(f) What post would be most similar to 2625 in its descriptiveness?

2.4 [2.2] Retrieve the embeddings for `tones` in a 4×1024 matrix and save it as `tones_embeddings`:

- Filter `aiwars_embeddings` for the relevant `post_index`.
- Either drop `post_index` or tidy-`select` all variables `starts_with()` "V".
  - Remember to keep track of the embeddings. We suggest you previously arrange by ascending post index.
- Coerce the resulting data frame into a matrix with `as.matrix()`

.

Distance-based semantic similarity

As exposed in the introduction, if two texts share style, vocabulary or topics, their embeddings will tend to be closer, and on the contrary, distinct semantics will produce embeddings that are farther apart.

2.5 [2.3]

Table 1: Pairwise distances between example embeddings (not the real values)

	luddites	mockery	critique	synthists
luddites	0.0000000	72.384921	18.532947	91.004327
mockery	72.384921	0.0000000	56.110239	8.231476
critique	18.532947	56.110239	0.0000000	44.873291
synthists	91.004327	8.231476	44.873291	0.0000000

Code `dist_similarity`, 2-D distance matrix like the one displayed above with the embeddings in `tones`. Hints:

- This is a 4x4 matrix with `rownames` and `colnames`
- The matrix you get should be symmetrical. The distance between *critique* and *luddites* should be the same in (row 1 column 3) and (row 3 column 1).

2.6 Which post pair is the most dissimilar? Is this coherent –i.e., are their contents and tone quite different?

2.7 Which individual embedding seems to be further away from the rest?

2.8 Which post pair is the most similar according to the metric? Does this align with your 2.3d answer ?

Direction-based measures: Cosine Similarity

An alternative approach is to compare the angle θ formed by two vectors $\vec{a}, \vec{b}$. The cosine of said angle θ is defined as:

$$cos(\theta) = \frac{\vec{a} \cdot \vec{b}}{|| \vec{a} || \; || \vec{b} ||}$$

- $\mathbf{cos}(\theta) = \mathbf{1}$ means the angle is $\theta = 0°$, indicating identical direction. These texts will be very close in meaning and topics.
- $\mathbf{cos}(\theta) = \mathbf{0}$, the angle is $\theta = 90°$, indicating the posts are orthogonal, and for the most part, semantically unrelated: different topics and intent.
- $\mathbf{cos}(\theta) = \mathbf{-1}$, with angle $\theta = 180°$, vectors point in opposite directions.

  *While possible, embeddings will rarely have a cosine similarity close to -1 because of how the OpenAI product is constructed.

2.9 (Optional) Calculate `tones_norms`, the norms for `tones_embeddings`. How does your result simplify $cos(\theta)$?

2.10 (Optional) In question 2.5, what is the maximum possible distance between vectors?

2.11 [2.4] Write a function `batch_cossim(u,M)` taking a vector of length n and a matrix with k row-vectors and n columns, returning the cosine of the angle θ between u and each row-vector. You may apply what you learned in question 2.9.

2.12 [2.5] Similar to 2.5, create a cosine similarity 2-D matrix for the `tones` embeddings: `cos_similarity`. The diagonal elements must all be equal to 1.

2.13 Did the similarity order between posts change relative to the distance-based metric? Are your answers to 2.6, 2.7 or 2.8 the same with cosine-based similarity?

AI-awareness and Simplicity measured in the *aiwars* posts

The `speech_anchors` dataset consists of two embeddings that we will use as benchmarks for all posts in `aiwars`:

- **AI awareness**. Represents how AI-related the content of a post is.
  - Greater similarity implies the post mostly discusses AI, and makes heavy use of AI-related expressions and vocabulary.
  - Smaller similarity implies the post's content is less related to AI and its semantics.
- **Simplicity**. Represents the argumentative style of the post.
  - Greater similarity implies posts are shorter, more slogan-like, and low-context statements.
  - Smaller similarity relates to longer, more evidence-seeking discussions. Posts will often be longer and more nuanced, regardless of their AI-relatedness.

In our setup, cosine similarities in the $[\mathbf{0.5}, \mathbf{0.7}]$ range usually indicate a *non-trivial* semantic or stylistic relation. Values above $\mathbf{0.8}$ often reflect very close texts (near-paraphrases). Similarities in the $[\mathbf{-0.3},\ \mathbf{0.3}]$ range are not indicative of a strong semantic link. And $\leq \mathbf{-0.3}$ cosines can occur but are very uncommon.

3.1 [3.1] From `speech_anchors`, create vectors `ai_aware` and `simple`, then compute the cosine similarities between all posts in `aiwars`, these 2 vectors: `ai_awareness` and `simplicity`, respectively. Add them as variables of `aiwars` — remember to previously arrange both datasets by ascending `post_index`. Save the new data frame as `aiwars_updated`.

3.2 Verify your answers by replicating the following data frame.

post_index	ai_awareness	simplicity	label
<dbl>	<dbl>	<dbl>	<chr>
456	0.4775886	0.3168471	luddites
584	0.2886388	0.3751917	mockery
2397	0.5400782	0.2152752	critique
2625	0.4792042	0.3614475	synthists

Do cosine similarities `ai_awareness` and `simplicity` match the descriptions provided above when it comes to the `tones` posts?

[0]These are usually very high-dimensional vectors with over 100 dimensions as in your dataset, but we will initially work with 2 to better illustrate the concepts.

[1]The coordinates for each embedding are outputs of a more complicated embedding model involving several neural network architectures and outside the scope of this course. For this week's case studies, we limit ourselves to the resulting embeddings.

Case study O. 3. 1

Continuous methods in RVs: Joint and conditional PDF/CDFs

Instructions: Work in pairs or groups of three; solve the following exercises collaboratively, and put together the deliverable specified in Part 2.

Notebook: Open Case Study 1

Dataset: `aiwars_updated.csv`

One of you must share the case study's notebook (parts 1 and 2) and a fresh slides document (part 3) with the rest.

Part 1. Discrete and continuous methods

0.1 [0.1] Read in `aiwars_updated`; load/install the `tidyverse` packages.

Consider two continuous variables related to the semantics of text posted by redditors in r/Aiwars:

- **AI-awareness**: A, the extent to which a post's vocabulary and syntax depict artificial intelligence topics. An AI-related post will tend to score 1, while a post with no significant AI-related content will score closer to 0.
- **Simplicity**: S, a post's reliance on short, declarative phrasing and minimal argumentative or conceptual complexity. A short, opinionated post will show a score closer to 1, and a post showing a longer, more complex argument will observe a score closer to 0.

1.1 Based on the information provided, what are the sample spaces Ω_A, Ω_S?

1.2 Are A, S continuous or discrete random variables? Why?

1.3 [1.1] Use `ggplot2` to create histograms for both A and S on the same plot.
   (a) Declare a `ggplot()` on your data frame.
   (b) Add a `geom_histogram()` layer with `ai_aware` as your x-`aes()`. Outside `aes()`, set the `fill` parameter to "blue", the `color` to "black", and the transparency (`alpha`) to 0.5.
   (c) In the same plot, add another histogram layer analogously for `simplicity` but fill it with "red".

1.4 Examine the plot:
   (a) What is plotted in the y-value?
   (b) Roughly, how many posts are there in the $0.47 \leq S \leq 0.53$ range?
   (c) More generally, which histogram seems more concentrated around a given range of values?
   (d) If we summed the y-value of all the bins of either the `simplicity` or the `ai_aware` histogram, what should be the result? Is the result a probability?

1.5 [1.1.1] We want the y-value to reflect a probability. Edit the `aes()` function in your histograms plot to include the following y-aesthetic: `y = after_stat(count/n)`. Compute and save `n` separately.

(a) Similar to 1.4d, what should be the sum of all the bins' y-values of either histogram now?
(b) Very roughly (no computation needed), what is x such that $F_A(x) \geq 0.5$?
(c) Again roughly, what do you think $P(0.75 \leq S \leq 1)$ approximates to?
(d) Histograms are discrete representations of a continuous RV. Change the `binwidth` of **just the red histogram** to 0.01. What can you observe? Did the distribution's relative shape change? Why?

Recall the *Summarizing and Describing Data* lecture. Kernel Density Estimation is a **continuous** estimate of our RV's distribution. Base R has a readily available function for this: `density()` — feel free to test it in your own time.

We will use density through `ggplot2`'s interface: `geom_density()` to plot the estimated pdfs $\widehat{f_A}(\cdot)$ and $\widehat{f_S}(\cdot)$ for all the x-values in the domain of our data.

1.6 [1.2] Copy and paste your first code (question 1.3). Substitute `geom_histogram` with `geom_density` in both layers.

(a) In terms of shape and position relative to each other, how do these distribution representations compare to those in 1.5?
(b) What does the y-axis represent in this plot?
(c) Roughly, what is *max* $\widehat{f_S(s)}$? What is the minimum for both A and S?

Another way to compute the probability of an event $E = [a, b] \in \Omega_X$ for a continuous RV is through its CDF. Recall from the lecture that

$$P(a \leq X \leq b) = \int_a^b f_X(x)dx = F_X(b) - F_X(a)$$

Base R allows us to create a function for the empirical CDF of continuous variables with `ecdf()`. This function also has its `ggplot2` analog in `stat_ecdf()`, which you can use just as any other `geom`.

1.7 [1.3] Use `ecdf()` to create functions `F_A` and `F_S`, the empirical CDFs of A and S respectively. Make sure you have the correct object by quickly plotting the CDFs: `plot(F_A)`, `plot(F_S)`. Get familiar with the usage by verifying that `F_A(0)` returns 0 and `F_A(1)` returns 1. Then answer the following questions:

(a) $F_S(0.25)$?
(b) $F_A(0.25)$?
(c) What is $s \in \Omega_S$ such that $P(S \leq s) = 0.8$?
(d) What is $a \in \Omega_A$ such that $1 - P(A \leq a) = 0.1$?
(e) Compute the probability that posts are either too intricate or too simplistic $P(S \leq 0.2 \cup S \geq 0.7)$
(f) Make an event for S such that posts fall in it with 80% probability. Find

any $a, b \in (0, 1)$; $a < b$ such that $P(a \leq S \leq b) = 0.8$

.

1.8 [1.4] Use `stat_ecdf()` in `ggplot2` to jointly plot the CDFs for A and S. Make sure to `color` them blue and red, respectively.
   (a) Can you conclude there's first-order stochastic dominance of one of the random variables to the other?
   (b) In the lecture, height was the construct measured for women in both Bihar and the US. Are S and A measuring the same in the posts?

Part 2. Non-independence

2.1 Within the r/AIwars subreddit, conditional on learning posts tend to be long and argumentative, would their probability of being AI-related change?

2.2 [2.1] Let's begin by examining $f_{A,S}(a, s)$. Use `ggplot2`'s function `geom_bin_2d()` to plot a 2D-histogram with `ai_aware` in the `x`-axis and `simplicity` in the `y`-axis. Set the `fill` aesthetic to `after_stat(density)`.
   (a) What does color represent in this plot? Note that the "minimal" unit colored is a rectangle tile.
   (b) Are the x and y-binwidths the same? What are they, approximately?
   (c) Generally describe the joint composition of the subreddit's posts in terms of their AI-awareness and their simplicity. Feel free to sample a few posts in various ranges:
      – Moderate AI-awareness ($[0.4, 0.6]$) and moderate simplicity ($[0.2, 0.3]$)
      – High AI awareness and low simplicity (< 0.15). Check for high simplicity (> 0.4) too.
      – Low AI awareness (< 0.3)) and varying simplicity.

      You can sample a few random rows with `dplyr`'s function `sample_n()`. For a better dislay in Colab, select only the `fulltext`, `ai_aware` and `simplicity` columns
   (d) Judging by our joint histogram plot, are A and S related directly or inversely?
   (e) Do relatively long and argumentative tend to discuss AI? Can you find low-simplicity posts unrelated to AI in the dataset?
   (f) Visually, what is the 4x4 tiles semantic region $\boldsymbol{\Omega}_S \times \boldsymbol{\Omega}_A$ most posts fall in?

2.3 **True** or **False**: if A and S were independent, we would have a perfectly squared 1×1 grid and all tiles would be of the same color, because $f_{A,S}(a, s) = f_A(a) \cdot f_B(b)$

Numerically, we can represent the joint pdf $f_{A,S}(a, s)$ as plotted in 2.1 as a matrix with rows and columns corresponding to each x (`ai_aware`) and y-bin (`simplicity`). In the next question, code is provided to create such a 30×30 matrix, `f_AS` for you.

2.4 [2.2] Run the pre-scripted code to get `f_AS`, the joint PDF $f_{A,S}(a, s)$. Then compute the marginals $h_A(a)$ and $h_S(s)$ and save them as `h_A` and `h_S`, respectively. Keep in mind these should be vectors of length 30.

2.5 Make quick base R plots of each marginal with `plot(h_A, type = "b")` and similarly for h_S in a separate plot. Do they look familiar? Why?

2.6 Use `f_AS`, `h_A` or `h_S` — and their bin `names` — to calculate the following probabilites. You may want to print the bin names beforehand:

(a) $P(A \leq 0.38 \ \cap \ S \leq 0.24)$
(b) $P(0.156 < A \leq 0.38 \ \cap \ 0.628 < S < 0.867)$
(c) $P(A = 0.188 \cap S \leq 0.927)$
(d) $P(S \leq 0.24)$
(e) $P(A > 0.667 \cap 0.27 < S \leq 0.3)$
(f) $P(A > 0.667 \mid 0.27 < S \leq 0.3)$
(g) $P(0.27 \leq S \leq 0.3 \mid A > 0.667)$
(h) $P(A \mid 0.27 \leq S \leq 0.3)$. Save as `A_cond` and plot this
(i) $P(S \mid A \in (0.795, 0.827])$. Save as `S_cond` and plot.

2.7 Show that A, S are not independent with your answers to 2.6f, 2.6g, and the elements you used to calculate them.

2.8 Plot your answers to 2.6h and 2.6i are not independent. Side by side with their marginal:
`plot(S_cond, type = "b") lines(h_S, col = "red")`
And the same for A. What differences do you notice? What do these plots show?

2.9 [2.3] Finally, directly verify the claim in 2.3. Use `h_A` and `h_S` as well as a double (nested) loop to produce $g(a, s) = h_A(a) * h_S(s)$ for the same 30×30 grid. Then use `heatmap()` with the same arguments used in the previous code snippet to examine the joint density. Does this look like a homogeneous distribution? If not, why? If it does, at what regions of $\Omega_A \times \Omega_S$ do we have missing values, if any?

Part 3. Your own joint distributions

Your team will freely analyze the relationship between either AI-awareness or Simplicity and one other feature in the `aiwars` dataset.
Create a short deliverable following these instructions:

- Choose either `ai_aware` or `simplicity` and one other key variable.
- Your instructor will
  - Approve or change your choice based on their goals and what other students are working on.
  - Inform you of the format required for the deliverable. Usually a 5-slide eck or a text document.
- Once your choice is approved:
  1. Construct the joint distribution of the two variables (Depending on your choice of discrete or continuous, use a `geom_bin_2d`, a boxplot, or any of the plots reviewed in the lecture).
  2. Obtain and interpret the marginal distributions of each variable.
  3. Compute the conditional distributions (e.g. $f_{A|X}(a)$ or $f_{S|X}(s)$) for what you consider are relevant ranges of values. Evaluate how indpendent the

analysis variables are.

4. Include at least one visualization that helps highlight the relationship (histograms, heatmap, density plot, etc.).
5. Provide 1–2 slides (or short paragraphs) summarizing the implications of the independence (or lack thereof) between variables in plain language: What does the relationship tell us about Redditor behavior?

– Submit as required. Your instructor will let you know if your team will briefly expose your deliverable to the class.

14.310x Flipped Classroom

Week 4 Instructions

Transforms of random variables
Moments of random variables
Introductory simulation in R

Checklist

- ☐ Complete Coding Lab 4 (Requirement)
- ☐ Solve Guided Case 3 (Session 1)
- ☐ Solve Guided Case 4 (Session 2)

Coding Lab L.4.4

Simulation

Instructions: Work individually. Answer all sections of the Lab.

Notebook: Coding Lab 4: Simulating RVs in R

Record your answers in your copy of the notebook, and if required, submit your work as specified by the instructor.

Guided Case G.4.3

Select transforms of random variables: theory and coding practice

Instructions: Work in pairs or teams of 3. Answer all questions 1-3 and submit your coursework as required by the instructor.

Notebook: Guided Case Study 3

In the following questions, X will be a **known** random variable with pdf/pmf $f_X(x)$ (either provided or described for you to write down).

Each question will propose a transform $Y = g(X)$ and you must derive $f_Y(y)$, the pdf/pmf associated with said transform.

All exercises will ask you to follow a few analytical steps, after which you will plot and perform simulations in R.

Question 1

$$X \sim U[-1, 1]$$
$$Y = e^X$$

1.1 What is the support Ω_x the support of X? Write down $f_X(x)$.

> $\Omega_X = [-1, 1]$
> $f_X(x) = \frac{1}{1-(-1)} = \frac{1}{2}$ for $-1 \le x \le 1$ and 0 otherwise.

1.2 What is Ω_Y, the support of Y?

1.3 As in the lecture, work out the CDF of Y first.

$$F_Y(y) \equiv P(Y \le y)$$

(a) Substitute Y for $g(X)$.

(b) Get an expression in terms of the CDF of X: $P(X \le g^{-1}(y))$.

(c) Evaluate the expression. $\int_{-\infty}^{g^{-1}(y)} f_X(x)dx$.

1.4 Get $f_Y(y)$. If the CDF is continuous you can take the derivative $\frac{\partial F_Y}{\partial y}$.

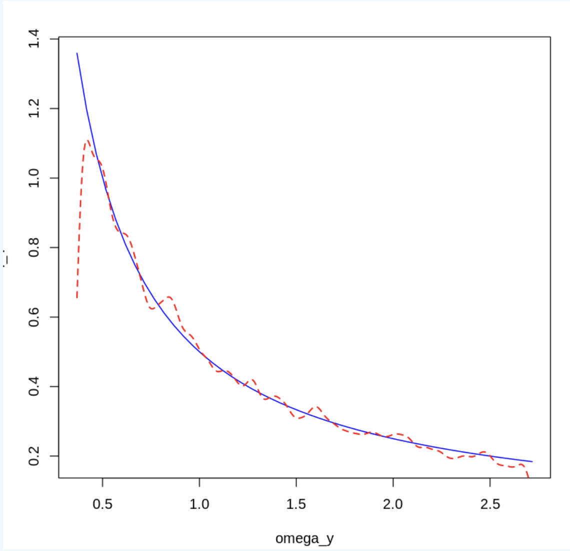
1.5 [1.1] Plot $f_Y(y)$ in R. Use `seq()` to create a vector `omega_y` from $\frac{1}{e}$ to e in increments of 0.05. Then create `f_Y`, by applying your expression for the pdf $f\_Y(y)$ to each element in `omega_y`. Plot `f_Y`.

1.6 [1.2] Now run a simulation, and plot a histogram approximation for $f\_Y(y)$:

(a) `set.seed() to 123`

(b) Use `runif()` with the relevant parameters to draw a sample with 10,000 realizations of X. Save it as vector `X_sim`.
(c) Apply the $Y = g(X)$ transformation to your sample. Save it as `Y_sim`.
(d) Make a quick histogram plot of `Y_sim`. How well does it resemble the theoretical pdf you derived?

1.7 [1.3] Run the code provided to create a side-by-side plot of both the theoretical (blue) and the estimated (dashed red) pdfs. Make sure you followed the naming conventions so far.



Note the density estimation fails when it gets close the lower limit of Ω_Y. Feel free to use the `from` and `to` and `bw` arguments, or change the `kernel` in `density()` to improve the plot.

Question 2

$$X \sim U[-1, 1]$$
$$Y = \frac{X^2}{1 + X^2}$$

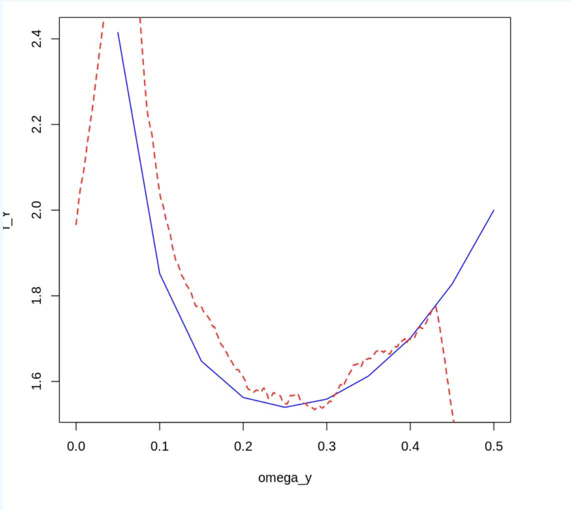
2.1 What is Ω_Y?

2.2 Get $F_Y(y)$. *Be very careful with the inequalities.*
 – Verify the following: $F_Y(0) = 0, \quad F_Y(\frac{1}{2}) = 1$

2.3 [2.1] Get $f_Y(y)$ and plot it along the support of Y as in 1.5.

2.4 [2.2] Similar to 1.6, simulate Y and plot its histogram with `breaks = 100`.

2.5 [2.3] Plot the exact and estimated densities side by side like you did in 1.7. This time, set the bandwidth argument `bw` in `density()` to 0.04, modify the robjectkernel to rectangular and specify the `from` and `to` arguments to match the sample space of Y.



Question 3: Probability integral transform

$$f_X(x) = \begin{cases} 4x & 0 \le x \le \sqrt{\frac{1}{2}} \\ 0 & \text{otherwise} \end{cases}$$

$$Y = F_X(x)$$

3.1 Derive $F_X(x)$.
3.2 What is the range of any CDF? (What is $\mathbf{\Omega}_Y$?)
3.3 According to the lecture, what should $f_Y(y)$ be? — Review the PIT slides if this is not clear.

In the lecture, Prof. Ellison mentioned that we can sample from any distribution by sampling from the distribution you got in 3.3 and transforming it by the inverse CDF of X:

$$Y = F_X(X)$$

$$\implies X = F^{-1}(Y)$$

3.4 From 3.1, solve for X as above —you should end up with a function in terms of Y.
3.5 [3.1] Confirm you get $f_X(x)$ back with a simulation:
- Draw 10,000 observations from the distribution of Y: `Y_sim`.
- Translate them to X draws by applying them function $F^{-1}(y)$ you found in the previous question: `X_sim`.
- Plot the histogram of `X_sim` with `breaks` set to 100. It should resemble $f_X(x) = 4x$.

[Harder] Question 4: Convolution

$$X_1 \sim U[0, 1]$$

$$f_{X_2}(x_2) = \begin{cases} 2x_2 & 0 \le x_2 \le 1 \\ 0 & \text{otherwise} \end{cases}$$

$$Y = X_1 + X_2$$

$$X_1, X_2 \text{ independent.}$$

$$f_Y(y) = f_Y(x_1, x_2) = \int\limits_{r_1 \in \Omega_{X_1}} \int\limits_{r_2 \in \Omega_{X_2}} f_{X_1}(x_1) \cdot f_{X_2}(x_2) dY$$

4.1 Given that $\Omega_{X_1} = \Omega_{X_2} = [0, 1]$ what is Ω_Y?

Even in simple convolutions, finding the correct integration region is challenging. The following exercises are meant to guide you in setting up the correct bounds for f_{X_1} and f_{X_2}, in order to find $F_Y(y)$.

Given Y and X_2 below, write down the possible values of X_1:

4.2 $Y = 1.5$ and $X_2 = 0.9$

4.3 $Y \ge 0.5$, and $X_2 = 0.5$

4.4 Let $0 \le Y < 1$ and $\frac{1}{4} \le X_2 \le \frac{1}{2}$.

(a) What values of Y are events in the sample space of Y that are not possible?

(b) Write down the range of values X_1 can take.

(c) Now suppose we have $y \in Y$, a specific value in $[\frac{1}{4}, 1)$. Write down the possible values for X_1 **in terms of number** y and the bounds of X_2 as before.

4.5 Let $0 \le Y < 1$ and $0 \le X_2 \le 1$. Consider numbers $y \in Y$ and $x_2 \in X_2$. Write down the bounds for X_1 in terms of these numbers (particularly, find the upper bound).

4.6 Let $1 \le Y \le 2$ and $0 \le X_2 \le 1 \mid y \in Y,\ x_2 \in X_2$. For the bounds on X_1 you now have to consider two events:

(a) When $y - x_2 \ge 1$.

(b) When $y - x_2 < 1$. *Hint: if x_2 is "too large", then x_1 can't be just any value, it will be restricted. And vice-versa.*

With the previous answers in mind, now derive the pdf of Y:

$$F_Y(y) \equiv P(Y \le y) = P(X_1 + X_2 \le y) = P(X_1 \le y - X_2)$$

Case 1: $0 \le y < 1,\ 0 \le x_2 \le y$

$$\ldots = \int_0^{\square} \int_0^{\square} f_{X_1}(x_1) \cdot f_{X_2}(x_2) dx_1 dx_2$$

4.7 Fill out the upper limits of the integrals and solve for $f_Y(y)$ when $0 \le y < 1$

Case 2A: $1 \leq y \leq 2, \quad y-1 \geq x_2$

$$\ldots = \int_{\square}^{y-1} \int_{\square}^{\square} f_{X_1}(x_1) \cdot f_{X_2}(x_2) dx_2 dx_1 \tag{1}$$

4.8 Fill out the limits and solve for $F_Y(y)$.

Case 2B: $1 \leq y \leq 2, \quad y-1 < x_2$

$$\ldots = \int_{\square}^{\square} \int_{\square}^{\square} f_{X_1}(x_1) \cdot f_{X_2}(x_2) dx_2 dx_1 \tag{2}$$

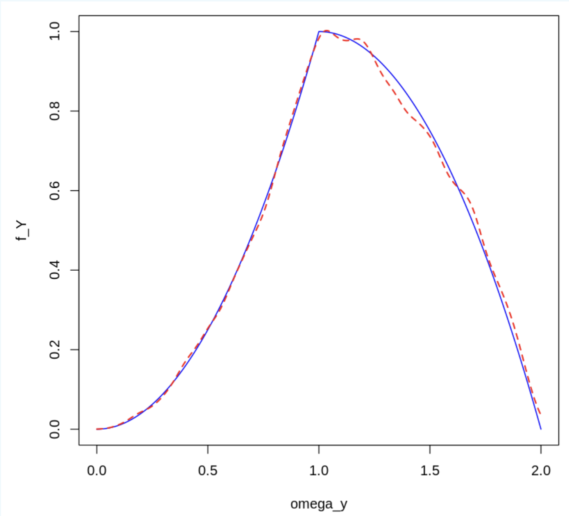
4.9 Fill out the limits and solve for $F_Y(y)$.

4.10 Write down $F_Y(y)$ and $f_Y(y)$ for all $1 \leq y \leq 2$, x_1 and x_2. *Hint: cases 2A and 2B are two (disjoint) partitions of* $1 \leq y \leq 2$.

4.11 Derive the pdf of Y for all $y \in \Omega_Y$.

4.12 [4.1] Plot $f_Y(y)$ in R. Start by defining `omega_y` as in previous questions, then apply the correct pdf for each segment. Save the pdf as vector `f_Y` . *Suggestion: use* `ifelse()`

4.13 [4.2] Now perform a simulation. Draw 10,000 random observations for both X_1 (`X1_sim`) and X_2 (`X2_sim`; as you did for 3.4 and 3.5). Create the simulated convolution: `Y_sim` and recycle the code to plot the estimated density alongside the theoretical distribution.



Guided Case G.4.4

Moments of random variables: theoretical session

Instructions: Work on your own; answer parts 1-3 and only answer part 4 if required by your instructor. Write down your answers in the format required by your instructor.

Part 1

We have (Y, X), a vector of random variables. Let:
$E(Y|X) = 3 + \frac{1}{2}X_1$, where $E(X) = 2$, $Var(X) = 3$,
And no residual variation: $Var(Y|X) = 0$.

1. What is $E(Y)$?
2. What is $Var(Y)$?
3. What is $Cov(X, E(Y|X))$?
4. Suppose $\varepsilon \sim N(0, 9)$, and $E(Y|X) = 3 + \frac{1}{2}X + \varepsilon$. What is $Cov(Y, E(Y|X)$?
5. Show that $Cov(X, Y) = Cov(X, E(Y|X))$

Now Let $E(Y|X) = 3 + \frac{1}{2}X_1 + X_2^2$ Where $E(X_1) = 2$, $Var(X_1) = 3$, $E(X_2) = 2$, $Var(X_2) = 1$. And X_1, X_2 independent.

6 What is $E(Y)$?
7 What is $Cov(Y, X_1), Cov(Y, X_2)$?
8 Let $Var(Y|X) = Var[E(Y|X)] + E[Var(Y|X)]$. Which component corresponds to the part explained by X? Which to the part that is unexplained by X? Answer in general terms.

Part 2

A policy provides fixed grants for microentrepreneurs. Previous rounds of funded businesses have received on average \$5,000 in subsequent funding opportunities. Let X be subsequent funding.

1. What is the maximum share of the microentrepreneurs that receive \$100,000 or more? ***Hint**: Markov's inequality*
2. What is the subsequent minimum funding amount over which possibly 25% of these microentrepreneurs were funded?
3. Based on previous measurements, an estimate for the standard deviation of X is 1,000. Using Chebyshev's inequality, calculate what is the maximum share of the microentrepreneurs that receive \$100,000 or more. Compare with your results from part 1. Interpret the results.

Part 3

Let $\{X_i\}_{i=1}^n \overset{\text{i.i.d.}}{\sim} U[1, 19]$.

1. What is the mean of X_i? What is its variance?
2. Is the sample mean $\bar{X}_n$ a random variable? Why?
3. What is the expectation of $\bar{X}_n$?
4. What is the probability that $\bar{X}_n$ is more than t away from the expectation of $\bar{X}_n$?
5. What happens to this probability if the sample size increases? What happens when $n \longrightarrow \infty$?

Part 4

A set of 6 grants will be awarded at random to participating firms, 4 of which are specialty grants and 2 regular grants. There are 40 prospective firms, of which 14 are microenterprises, and 26 are SMEs. Among these firms, 8 microenterprises and 8 SMEs are certified, and may receive a specialty grant.

1. What is the expected number of:
   - grants for microenterprises?
   - Grants for SMEs?
   - Grants for certified firms?
   - Grants for certified microenterprises? Grants for certified SMEs?
   - Grants for any type of firm?
   - How does the number of expected grants for microenterprises change with the number of accredited microenterprises? Is there a minimum number of accredited microenterprises which makes the expected number of grants for microenterprises higher than the expected number of grants for SMEs (holding everything else constant)?

14.310x Flipped Classroom
Week 5 Instructions

Central Limit Theorem and the Law of Large Numbers
More advanced simulation in R

Checklist

- ☐ (**Optional**) Watch Youtube video 1, an informal introduction to the Law of Large Numbers.
- ☐ We will be working with the *Weak Law of Large Numbers*. Have a simple statement of the theorem such as this one readily available.(Requirement)
- ☐ Complete Open Case Study 2 (Session 1)

Case study O.5.2

The law of large numbers (LLN)

Instructions: Work in pairs or groups of 3. Answer all questions and, when appropriate, discuss your results.

Notebook: Scrap notebook (some comments)

There are multiple ways to go about many of the questions and coding tasks, and you may solve them as you see fit. You will also need to provide open-ended answers to explain a number of the results you get. That said, some tasks will rely heavily on what we learned about *Simulation* in the Week 4 Coding Lab; your team may want to keep the notebook handy.

Scenario: We will be working with samples of five different continuous random variables as follows:

$$X_1 \sim N(5,1)$$
$$X_2 \sim N(5,9)$$
$$X_3 \sim U[2,8]$$
$$X_4 \sim Exp(\frac{1}{5})$$
$$X_5 = \frac{X_a}{X_b} \mid X_a, X_b \overset{iid}{\sim} N(0,1)$$

---

1. Get these distributions' expectations $\mu \equiv E(X)$ and variances $\sigma^2 \equiv Var(X)$.
2. For each RV, first run `set.seed(2)` **once**, then draw two samples: one of size $n = 5$ and another of size $n = 50$. Take their respective sample means:

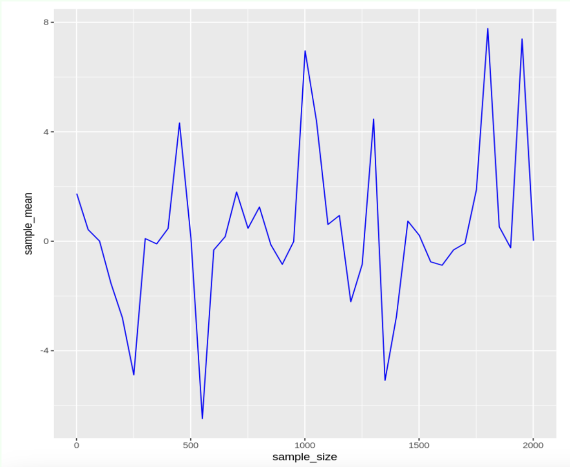
$$\bar{X}_n = \frac{x_1 + x_2 + \ldots + x_n}{n}$$

Per Chebyshev's inequality and the Law of Large Numbers (LLN) we have that

$$P(|\bar{X}_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{n\epsilon^2}$$

For arbitrary $\epsilon > 0$.

   (a) For which sample size would you expect the distance between the sample mean and its expected value $|\bar{X}_n - \mu|$ to be smaller?
   (b) Does this happen in all cases? If not, briefly discuss why.



3. For each RV, draw random samples of sizes $n = 1, 50, 100, 150, ..., 2000$ and create a plot similar to the one above — the plot must be reproducible.

   Suggestion: you can start by saving the sample means in a data frame like the one below

A data.frame: 5 × 3

RV_name	sample_size	sample_mean
<chr>	<dbl>	<dbl>
normal(0,1)	1	-0.5604756
normal(0,1)	10	0.2530814
normal(0,1)	20	-0.1478493
normal(0,1)	30	0.1767775
normal(0,1)	40	0.1003377

   (a) Is the LLN met for these RVs? If not, discuss the reason.
   (b) As n increases, how does the sequence of sample means behave? Does the trajectory *steadily* converge to μ?
   (c) Discuss how your previous answer may explain what you observed in question 2..
   (d) Plot X_1 and X_2 together. Do they converge to the same μ? How are they different? Why?
   (e) From which RV did the example plot above likely originate?
      – Suppose the sample means keep oscillating in roughly the same way around μ as $n \to \infty$. Explain precisely what part of the Weak Law or Large numbers would not be met.

4. Now consider convolution $Y = X_1 + X_2$. Compute $E(Y)$ and $Var(Y)$.
5. Simulate Y with a sample of size $n = 10,000$ in two different ways:
   - ```
     set.seed(123)
     y <- rnorm(10000, 5, 1) + rnorm(10000, 5, 3)
     ```
   - ```
     set.seed(123)
     x1 <- rnorm(10000, 5, 1)
     set.seed(123)
     ```

```
x2 <- rnorm(10000, 5, 3)
y2 <- x1+ x2
```

Compute each sample's mean $\bar{Y}_n$ and variance s_y^2. Explain the differences (if any) between $\bar{Y}_n^{(1)}, \bar{Y}_n^{(2)}, \mu_y$, and $s^2_{y,(1)}, s^2_{y,(2)}, \sigma^2_y$

Independent, but non-identically distributed samples

Sometimes we cannot directly choose the population from which we draw. A common example of this situation is when sampling is done at the level of **clusters** or **groups**, such as schools or villages, rather than individuals. Each cluster may have its own distribution, so the resulting sample is not identically distributed. Note that the draws are still independent: no observation depends on the value of any other. Drawing a cluster only adds several random observations from a distribution "at once".

The following questions are meant to help you work out how the LLN applies in these cases.

Suppose we have 2 independent random variables:

$$Y \sim N(\mu_y, \sigma_y^2)$$

$$Z \sim N(\mu_z, \sigma_z^2)$$

We have $M = \{y_1, y_2, ..., y_{n_0}, z_1, z_2, ..., z_{n_1}\}$ a random sample of size n where n_0 observations come from Y, n_1 observations come from Z, and $n_0 + n_1 = n$.

6. Write down $\bar{M}$, the sample mean:

$$\bar{M} = \frac{(\qquad) \; + \; (\qquad)}{\square}$$

7. Compute $E(\bar{M})$ in terms of $w_y \equiv \frac{n_0}{n}, w_z \equiv \frac{n_1}{n}, \mu_y$, and μ_z. What type of average is this?
8. Compute the sampling variance $Var(\bar{M})$. You should get an expression $= \frac{\square}{n}$, where $\square$ is the same type of average as in the previous question.
9. With your previous answers in mind, suppose we now have sample R with the following mixing:
   - 30% from X_1
   - 20% from X_2
   - 40% from X_3
   - 10% from X_4

   (a) What are $E(R)$ and $Var(R)$?
10. Plot $\bar{R}_n$ for $n = 50, 100, ..., 2000$.
   (a) Does $\bar{R}_n$ exhibit the same type of convergence? *Optional:* $\bar{R}_n$ alongside $\bar{X}_n$ for $X \sim N(5, 5.8)$ for the same sample sizes for reference.
   (b) Discuss what would happen with $\bar{R}_n$ if we added X_5 draws to the mix.

14.310x Flipped Classroom

Week 6 Instructions

Estimator properties
Hypothesis testing and confidence intervals
Further simulation in R

Checklist

- ☐ Complete Coding Lab 5 (Session 1)
- ☐ Complete Guided Case Study 5 (Session 2)
  - ☐ Dataset: `consultations.csv`

Coding Lab L.6.5

An intrudction to hypothesis testing and Confidence Intervals in R

Instructions: Work individually. Solve all exercises in the corresponding Colab notebook. Record your answers and/or code in your copy of the notebook.

Notebook: Coding Lab 5

Submit your work in the format required by the instructor.

Guided case study G.6.5

Examining *Clinicia*'s financial incentives for physicians

Instructions: Work in pairs or groups of three to solve all questions.

Notebook: Case Study 5

Dataset: consultations.csv

The team or pair may work on one notebook collaboratively. One team member needs to share editor access their the Colab Notebook with the rest.

Scenario:

Medical consultations take place in an environment with substantial **information asymmetry**. Patients are usually not qualified to determine whether diagnoses are accurate or if a prescribed treatment is correct. This creates incentives for healthcare providers that can alter treatment intensity and the types of drugs prescribed, directly impacting expected **health outcomes** and total **treatment costs**. Especially when a provider stands to benefit financially from a certain treatment avenue over others.

Clinicia's hospital system is reported to have unusually high rates of **antibiotic prescription**, even for conditions where guidelines recommend more conservative treatment. Overuse of antibiotics not only raises treatment costs unnecessarily, but also accelerates **microbial resistance**, reducing the effectiveness of essential drugs. The country's Ministry of Health is aware of the reports and wants to test more formally for any differences in treatment cost (or prescription rates) in the presence of financial incentives. As part of the policy team, you are tasked with performing this testing.

Medical records in Clinicia include audio logs of consultations (the audio is altered to protect patient and provider privacy) along with administrative information on diagnoses, procedures, and prescriptions. The Ministry was granted access to a fully anonymized, random sample of these records. Your team processed all the data and produced the `consultations` dataset. For each medical consultation:

- `incentive =1`: Indicates the patient is buying the prescribed drugs at the hospital or at a commercially related pharmacy.
- `request =1`: Indicates patient requested antibiotics.
- `antibiotics =1`: Indicates the physician prescribed antibiotics.
- `totalprice`: The market price (in USD) of the drugs prescribed to the patient at the consultation.
- `multiple =1`: Indicates the physician prescribed two or more distinct drug categories (e.g., analgesic + antibiotic + expectorant)
- `grade2 = 1`: Indicates the physician prescribed a drug with a high potential

for abuse and dependence (e.g., oxycodone, metanphetamines, Aderall).

---

1. [1] Load or install the `dplyr` and `pwr` packages. Read in `consultations` using the dataset URL provided above.

Part 1

2. [2]

Table 2: Number of patients in each group

	Requested	**Did not request**	**Total**
Incentive			
No incentive			
Total			■

Fill the table above. Determine the number of patients in each group I, R, and each subgroup $n_{I,R}$, $n_{I,NR}$,$n_{NI,R}$, $n_{NI,NR}$.

3. [2.1] As you know, the Ministry is concerned with both antibiotic over-prescription and treatment costs. Start by calculating the difference in average `totalprice` and `antibiotics` with and without `incentive`. Save as `pricediff_I` and `drugdiff_I`, respectively. Are the differences consistent with the presence of a financial incentive?
4. [2.2] More formally, use `t.test()` to test the hypothesis that the average treatment cost is the same for both groups. Assume the same unknown variance. Can the price equality be rejected at the 99% confidence level?
5. [2.3] We cannot perform a similar t-test for `antibiotics`. Read about the proportion test and hand-code it to test for equality in prescription rate. Report the value of the statistic, the first four non-zero decimals of the p-value, and whether or not you would reject at 1% significance.
6. Reporting back to the ministry, how confident are you that more profitable patients also face higher antibiotic prescription and higher treatment costs in Clinicia's population?

Given the strength of the evidence, the ministry wants to assess an additional driver of microbial resistance: the culture around antibiotic prescription. The ministry would like you to answer several questions (see below,) with this additional variable in the data you already have.

7. Repeat 3., 4. and 5., this time with groups defined by `request`. What can you conclude?
8. [3] Compute the average and standard deviation of `antibiotics` and `totalprice` for each of the four patient subgroups defined by all combinations of `incentive` and `request`. Save the object as `outcomes`.
9. How sensible is it to assume our outcomes draws for the 4 patient groups are identically distributed (i.e., sampled from the same distribution)?
10. [5] With your previous answer in mind, test the hypothesis conducive to solving the following question by the Ministry:

(a) Among patients **with no financial incentives**, is there a difference in their treatment costs when they request antibiotics?
(b) Are patients with no incentives prescribed antibiotics more often if they request them?
(c) For those patients buying their prescription at the hospital, is there a difference in treatment cost if they ask for antibiotics?

For each test, report $\hat{\tau}$ the estimated difference, the value of the appropriate statistic, the p-value, and whether H_0 is rejected at $\alpha = 0.05$

11. [7] From 10.b, hand-code a function `ci_props()` that takes a significance level `alpha`, and outputs a vector of length 2; the bounds of the 95% confidence interval for the difference $\hat{\tau}$ in antibiotic prescription rates. What is the smallest α for which the CI doesn't include 0? At this alpha, what is the probability that the prescription rates are in fact equal, but we reject the equality?

Part 2

12. Suppose now you implement a new experiment, where you want to be able to detect an effect with a certain probability, if present. What is this probability commonly called in statistics?
13. Your colleagues at the policy team define an economically meaningful effect to be detected of x magnitude. What is x called?
14. Consider the distributions under H_0 (zero effect) and H_β (β effect, different from 0). What area of the curve of these distributions corresponds to the power of the test?
15. [8] For a new experiment where the control group is expected to have a total medication cost of \$100, compute the required sample size per group to detect effects of $\Delta \in \{\$10, \$20, \$40, \$80\}$ with 80% power and 5% significance (two-sided). Assume a pooled standard deviation of $\sigma = 60$.
Hint: the standardized effect size is Cohen's $d = \Delta/\sigma$. Use `pwr.t.test()`.
16. [9] Compute the required sample size per group for detecting standardized effects of $d \in \{0.05, 0.10, 0.20, 0.40\}$, with $\alpha = 0.05$ and 80% power (two-sided).
17. Comment on your results. Explain the relation between sample size and effect size. Is this relation proportional?

14.310x Flipped Classroom
Week 7 Instructions

The potential outcomes framework
Experimental design and randomized evaluation in economics

Checklist

- ☐ Set aside this Wikipedia article on testing two-proportions hypotheses, in case you need it during Session 2 (Requirement)
- ☐ Skim this Wikipedia article discussing the Local Average Treatment effect (LATE) , in advance of Session 2. (Requirement)
- ☐ Complete Guided Case Study 6 invidually (Session 1)
  - ☐ Retrieve the `students` dataset here
- ☐ Complete Open Case Study 3 in pairs or teams of 3. (Session 2)
  - ☐ You will continue to use `students`

Case study G.7.6

Evaluating AI-assisted learning on student outcomes

Instructions: Work on your own; read the scenario and answer the questions. Type your answers in the format required by the instructor.

To work on the `students` dataset you may use either a Colab notebook or your own installation of R.

Dataset: students.csv

Your code will not be evaluated, but keep your R script or notebook tidy, as you may need to review some of your answers during Session 2.

Scenario:

You are the Government of *Novaria*'s new Minister of Education. The Prime Minister has tasked you with evaluating a primary education policy recommendation: the rollout of AI-assisted learning for mathematics curricula in grades 5-8.

The proposed program, *Project Mentor*, involves deploying a large language model (similar to ChatGPT) named *AlgebrAI* specifically trained and fine-tuned for elementary math tutoring. AlgebrAI's interface is tailored to deliver interactive, one-on-one tutoring sessions to students. The AI mentor adapts to each student's skill level and provides problem-solving guidance, hints, and feedback designed to help the students master their grade's math curriculum.

Each participating school receives a number of tablets with AlgebrAI pre-installed, configured for offline-first use and automatically synced with central servers when internet is available. Students selected for treatment attend 20-minute tutoring sessions per day under the supervision of a facilitator.

The Prime Minister believes Project Mentor can boost test scores nationwide, but political opponents have raised concerns over cost and long-term efficacy. You are now in charge of evaluating the impact of the program in grades. Your team provides you with:

1. A 6th-grade math test designed to perfectly measure domain of the curriculum in a scale from 0 to 100.
2. A list of 1,000 students enrolled in 6th grade across Novaria, picked at random — part of the `students` dataset. This list contains only the following variables:
   - `unit`: a consecutive number assigned to the student.
   - `W_school`: Indicates whether the student's school is managed by the government ($W_{school} = 1$) or if it is privately managed ($W_{school} = 0$)

You have authority to apply the exam to any 6th grader in Novarnia, and you can implement the program (tablet usage and monitor time) in all government-managed schools, but to include any students attending a private school to the program you must first obtain authorization from their school board.

Exercises

Consider $T_i \in \{0, 1\}$ the treatment status of student $i = 1, 2, ..., 1000$ — $T_i = 1$ if treated $T_i = 0$ if not treated. Potential outcomes $y_i(T_i)$ in `students`, measured in test results (grades 0 to 100) are defined:

- `y0`: vector $Y(0)$, assume we can't observe it unless specified.
- `y1`: vector $Y(1)$, assume we can't observe it unless specified.

1.1 What is the value of $y_3(1)$? Describe its meaning —in terms of the potential outcomes framework.
1.2 What is the value of $y_5(0)$? Describe its meaning.
1.3 Compute $\bar{Y}(1)$. Describe its meaning.
1.4 Suppose $T = 0$. What is the value of y_{20}^{obs} and y_{40}^{miss} ? Briefly explain why.
1.5 Suppose $T_i = 1$ for all $i = 1, 2, 3, ..., 1000$. What is $\bar{Y}^{obs}$?
1.6 Imagine you can observe both potential outcomes.
   (a) What is the treatment effect of Project Mentor in student 245?
   (b) What is the estimated Average Treatment Effect (ATE) of Project Mentor?
   (c) Imagine the ATE in 1.6b is a valid estimate of the treatment effect across Novarna. Would it support the Prime Minister's claims?

In practice, only Y^{obs} will be available after applying the exam. You will observe **one test score per student**, as well as the students' treatment status: either treated or untreated.

After careful consideration, your team assigned N_0 students to the control group and N_1 to the treatment group, out of the $N_0 + N_1 \equiv N = 1000$ sampled. The assignment criteria included logistics, the school-year calendar, operation costs and potential political opposition. Treatment was allocated amongst students according to the rule

$$T = W_{school}$$

2.1 Explain the assignment rule in simple words
2.2 For each of the assignment criteria, provide a brief circumstance that likely motivated Novaria's government to conclude this was the best allocation.
2.3 What is the value of N_0?
2.4 What is the value of N_1?
2.5 Create a variable for $Y^{obs}(W)$, and name it `yobs_w`. With this variable:
   (a) Compute the value of $\bar{Y}^{obs}(1)$.
   (b) Compute the value of $\bar{Y}^{obs}(0)$.
2.6 Write down both expressions $\bar{Y}^{obs}(\cdot)$ more formally, in terms of summations.
2.7 Your team knows that $ATE = E(y_i^{obs}|W = 1) - E(y_i^{obs}|W = 0)$ but they don't know how to estimate these expected values from our sample.
   (a) What is the estimated ATE, $\widehat{ATE}$, in terms of the units observables $y_i, w_i;\ \ i = 1, 2, ..., N$ in our sample?
   (b) Justify your answer in terms of a famous mathematical theorem:
      i. $\square \xrightarrow{\square} E(y_i|W = 1)$
      ii. ...
      iii. Therefore the estimate ...
   (c) Compute $\widehat{ATE}$ using R.
   (d) **Reflect on the result**. How does it compare to your answers in 1.6b and 1.6c? At this point, do we have any way to diagnose the accuracy of this result?

2.8 Again, let's imagine we can observe potential outcomes. In the lecture, it was shown that the ATE can be decomposed in *treatment on the treated* and *selection bias*. Write down that expression and estimate the values of:

(a) treatment of the treated

(b) selection bias

2.9 What do these values imply for the experiment's design? Is the value for 2.8a a potentially good $\widehat{ATE}$? What would be omitted if we were to only consider this value?

Case study O.7.3

Evaluating AI-assisted learning on student outcomes (continued)

Instructions: Work in pairs or groups of 3; answer the questions as concisely as possible. Type your answers in a single shared document or in the format required by the instructor.

One of you must set up a blank Colab notebook to work on, and share it with the rest. Save any figures or output, and incorporate as required by the instructor.

This is a direct continuation of G.7.6, thus we will work under the context you already have. Continue to use `students` when necessary to answer the questions.

Scenario: You let the Prime Minister know the treatment allocation for the Project Mentor experiment you had previously agreed on is problematic. He hires a team of consultants to help you sort this design problem out, as well as polish other details of the experiment. The following exercises contain some of the questions asked during several policy meetings.

Meeting 1

1.1 Firstly, you are asked to explain generally why you cannot get a credible average treatment effect from the current treatment allocation. How would outcomes differ from our estimates if we scaled up the program nationally?
(use your "secret" knowledge of the potential outcomes, and your answers to the previous Guided Case).

1.2 You are asked to provide an alternative assignment that would create two groups equally representative of 6th-graders nationally. Create such assignment variable `t`, and also create `yobs_t`, the outcome we would observe under this assignment. Calculate the $\widehat{ATE}$. How does this result compare to 2.8a in G.7.6? Without making any further calculations, what do you think the treatment effect among private schoolers will be?

1.3 The Prime Minister doesn't believe that your new assignment created comparable groups. One way to provide evidence of balance, is showing the groups have the same composition of public and private schoolers. Formally show there is no evidence to reject the composition is the same. Be as conservative as possible with the variance.

1.4 Imagine everyone can observe potential outcomes. From the definition of ATE, show treatment and control are comparable more decisively.

Meeting 2

While more convinced of your new assignment, the Pime Minister still insists asking for permission to private schools is impractical and will delay matters. Conveniently, the consulting team asks two questions:

- Whether attending a private/public school in Novartia actually creates systematic differences between students; particularly, differences related to the outcome. This has not been formally shown.
- Whether there may be differences in treatment effects between public and private students (e.g. the mean effect is larger for any), as these differences would justify different rollouts.

To provide evidence in favor or against these questions, bear in mind we have to start from the following assumption:

$$y_i(0)|W = 0 \sim Distr(\mu_0,\ \sigma^2_{0,0})$$
$$y_i(1)|W = 0 \sim Distr(\mu_1,\ \sigma^2_{1,0})$$
$$y_i(0)|W = 1 \sim Distr(\nu_0,\ \sigma^2_{0,1})$$
$$y_i(1)|W = 1 \sim Distr(\nu_1,\ \sigma^2_{1,1})$$

Where $\sigma^2_{0,0} \neq \sigma^2_{0,1} \neq \sigma^2_{1,0} \neq \sigma^2_{1,1}$

2.1 In your own words what do these assumptions mean? What do they entail when it comes to testing hypotheses? According to the lecture, what should we assume about the correlation among these random variables if we want to be conservative?

2.2 Answer the question about systematic differences in outcomes between public and private schools.
   (a) State the appropriate null hypotheses and their alternates.
   (b) Test them with the appropriate statistic (estimate any parameters you don't know).
   (c) Tie your conclusions directly to the question with 95% confidence.

2.3 Answer the question about differences in treatment effects.
   (a) State the appropriate null hypotheses (test equality).
   (b) Make inference on $\hat{\nu}_1, \hat{\nu}_0, \hat{\mu}_1, \hat{\mu}_0$ accordingly.
   (c) Tie your conclusions directly to the question with 95% confidence.

Meeting 3

Satisfied with your answers, the Prime Minister and consultants approach you with some final questions about the program's broader implications:

3.1 **SUTVA violations**
In your own words, briefly explain the Stable Unit Treatment Value Assumption (SUTVA). Could implementing Project Mentor violate this assumption in Novaria? Provide a specific scenario illustrating such a violation clearly. How might these externalities affect the accuracy of your estimates?

3.2 **Alternative policies with proven outcomes (e.g. TaRL)**
The consultants suggest evaluating cheaper alternatives, like Teaching at the Right Level —targeting teaching to each student's current skill level without advanced technology; human mentors, complementary material, smaller traditional groups (more teachers), among others. These alternatives currently have more robust evidence of their effects.

   (a) What, if any, are some differences between the theory of change underlying TaRL and that of Project Mentor?
   (b) If differences exist, briefly discuss how you would test them within an experimental design, clearly describing treatments, assignment and measured outcome(s).
   (c) Apart from the treatment effects, what else is necessary if we wanted to fairly compare any known TaRL intervention to Project Mentor in terms of efficiency?
   (d) In this comparison, how important do you think scale would be? Briefly describe how costs for one and the other would behave.

3.3 **Non-compliance** Not every school or student may strictly follow the treatment assignment. Describe one realistic scenario of non-compliance in this project. Explain briefly how such non-compliance might bias the estimated treatment effect. Suggest one practical strategy to reduce or mitigate non-compliance.

14.310x Flipped Classroom

Week 8 Instructions

The linear regression model
Linear models in R: `lm` class, parsing outputs with `broom`

Checklist

- ☐ Watch the linear models with R: `lm` tutorial[4] (Requirement)
- ☐ Complete Coding Lab 6 (Session 1)
  - ☐ Datasets: `wages`, `gap`, and `air`. In-lab.
- ☐ Complete Guided Case Study 7 (Session 2)
  - ☐ Dataset: `nlsy_merged.csv.gz`
- ☐ (Optional) Explore **Review Notes** and further **Readings** available.

Coding Lab L.8.6

Linear regression in R

Instructions: Work on your own. Complete coding sections 1 to 5 and submit as required by your instructor.

Notebook: Coding Lab 6

[4] **Module 8 > Introduction to the Class lm** in the online component of the course.

Guided case study G.8.7

Replicating (Deming, 2017): returns to social skills in the labor market

Instructions: Work in pairs or groups of three to solve all questions. One team member should share editor access to the Colab Notebook with the rest.

Notebook: Case Study 7

Dataset: nlsy_merged.csv.gz

Scenario:

Advances in computing and automation have fostered profound shifts in the structure of labor demand. Routine, codifiable tasks such as assembly-line manufacturing or basic data entry have witnessed decreases in labor intensity. On the contrary, roles that require either high cognitive complexity or interpersonal coordination have expanded. A large literature links this *routinization* trend to widening wage inequality and rising returns to education and cognitive skills.

Deming (2017) studies the role played by **social skills** in this transformation, and argues they have become an increasingly important determinant of labor market success over the past thirty years. One of the paper's insights is that social and cognitive skills are **complements**: a balance of cognitive and social skills will earn more than either skill alone in large amounts — optionally review the results of the 2-worker task model (pp. 12–16). This complementarity is more pronounced in diverse team environments, where members can benefit from coordinating their specialized work.

The paper builds on three groups of regression specifications:

- Wages or employment explained by workers' social and cognitive skills [Tables 1 & 4].
- Workers' occupational task intensity explained by their social and cognitive skills [Table 2].
- Wages explained by task intensity interacted with skills [Tables 3 & 5].

We will replicate selected results from Tables 1, 2 and 4 of Deming (2017) using an extract of the author's dataset `nlsy_merged`, and develop an understanding of the implications.

Dataset: `nlsy_merged`

The dataset is a merged, cleaned panel drawn from the National Longitudinal Survey of Youth 1979 (NLSY79) and 1997 (NLSY97). Each row is a (person–year) observation. Key variables:

- `pubid` — Unique person identifier.
- `sample` — Cohort indicator: = 1 for NLSY97, = 0 for NLSY79.
- `year` — Survey year.
- `age` — Age at survey date.
- `complete79` / `complete97` — = 1 if the observation qualifies. Complete if the respondent is employed, has non-missing wages, skills, and location variables.
- `ln_wage` — Log hourly wage (main outcome). `emp` — Employment indicator ($\geq$ 80% of the year employed).
- `female` — = 1 if respondent is female. `black_male`, `black_female`, `hisp_male`, `hisp_female` — Race $\times$ gender dummies (non-Hispanic white male omitted).
- `div`, `metro`, `urban` — Census division, metropolitan-area, and urban/rural codes. Included as fixed effects in all regressions.
- `afqt_std` — Armed Forces Qualifying Test (AFQT) score, standardized (mean 0, SD 1). Age-at-test adjusted following Altonji et al. (2012). Proxy for cognitive skill (COG_i).
- `soc_nlsy_std` — Social skills index from *four* items: childhood/adult sociability, HS clubs count, HS athletics. NLSY79 only; used in Tables 1–3.
- `soc_nlsy2_std` — Social skills index from the *two* items available in both cohorts: childhood and adult sociability. Standardized in the pooled sample; used in Tables 4–5 (SS_i).
- `educ` — Years of schooling (highest grade completed).
- `socskills`, `routine` — O*NET social-skill and routine task intensity (percentile rank, 0–10 scale).
- `ind6090` — Industry code (1990 Census classification). Used in Table 2.
- `weight` — Analytic weight from Altonji et al. (2012), correcting for differential age-at-test in the AFQT (Tables 1–3). `lswt` — NLSY survey sampling weight (descriptive statistics).

Part A. Setup and Exploratory Analysis

1. [1] Install/load the `dplyr`, `ggplot2` and `readr` packages. Read in `nlsy_merged.csv.gz` by running the provided code. Save the data frame as `nlsy_merged`.
   - How many **unique** workers are part of the study? On average, how many observations per worker are there in the 79 and 97 surveys?
   - Display summary statistics for `ln_wage`, `afqt_std`, `soc_nlsy_std` and `soc_nlsy2_std`. What is the dataset's median hourly wage in USD?
   - Which of these variables has the most missing observations? Per the glossary, why might this be the case?
2. [2] To understand the completeness and quality of the main skill measures, create **overlapped** density plots of each of `afqt_std`, `soc_nlsy_std` and `soc_nlsy2_std`, grouped by cohort (`sample`: 0 = NLSY79, 1 = NLSY97) — add transparency with `alpha`. Create each plot *separately* to prevent `ggplot` from

dropping all rows with any missing value.
  - `afqt_std` is standardized. Why doesn't it look normally distributed?
  - Is there any substantial difference in AFQT scores between cohorts? Describe if any.
  - From the glossary, what can you confirm about `soc_nlsy_std`?
  - What is one limitation of `soc_nlsy2_std` in the NLSY79 cohort?
  - `soc_nlsy2_std` is standardized in the pooled sample. What would a score of −2 in the NLSY97 cohort map to in the NLSY79 cohort?

3. [3] Compute the weighted mean of `ln_wage` for each cohort using `weighted.mean()` with the survey sampling weights (`lswt`). Verify you get 2.24 and 2.42 for the NLSY79 and NLSY97 cohorts respectively.

   Then arrive at the same result for the NLSY97 using `mean()` and `nrow()` only — construct the normalized weights `w_97` yourself.

Part B. Social and cognitive skills complementarity: Table 1

4. [4] Use `lm()` to estimate

$$\ln(w_{it}) = \alpha + \beta_1 \, \mathrm{COG}_i + \varepsilon_{it}$$

   on the full dataset (`nlsy_merged`). Note COG_i = `afqt_std`. Do *not* include additional controls. Report $\hat{\beta}_1$ and its standard error.
   (a) What is the t-value for $\hat{\beta}_1$? Can we reject $\beta_1 = 0$ at the 95% level?
   (b) Suppose a worker starts at $\mathrm{COG}_i = 0$. What is their estimated wage increase in \$/h from a 1 SD improvement in cognitive skill?
   (c) More generally, if a worker starts at $\mathrm{COG}_i = x_0$, what is the estimated *percentage* wage increase from a 1 SD improvement?

5. [5] Replicate column 1 of Table 1. The model:

$$\ln(w_i) = \alpha + \beta_1 \, \mathrm{SS}_i + \boldsymbol{\gamma}' \mathbf{X}_i + \varepsilon_i,$$

   where SS_i = `soc_nlsy_std` and $\mathbf{X}_i$ contains the *covariates*: `female`, `black_male`, `black_female`, `hisp_male`, `hisp_female`, `age`, `year`, `div`, `metro`, `urban`.
   (a) Create `nlsy79`: filter `nlsy_merged` to NLSY79 (`sample == 0`), `complete79 == 1`, and `age >= 23`. Coerce `age`, `year`, `div`, `metro`, `urban` to factors.
   (b) Regress `ln_wage` on `soc_nlsy_std` and all covariates on `nlsy79`. Verify $\hat{\beta}_1 \approx 0.0999$.
   (c) Add `weights = weight` to replicate the paper exactly. Save as `table1_col1`. Verify $\hat{\beta}_1 = 0.107$.
   Note: the author uses **weighted** least squares to correct for differential age-at-test in the AFQT; as you can see, the unweighted result is a close approximation.

6. [6] Replicate column 3 of Table 1. The complementarity specification:

$$\ln(w_i) = \alpha + \beta_1 \, \mathrm{COG}_i + \beta_2 \, \mathrm{SS}_i + \beta_3 \, (\mathrm{COG}_i \times \mathrm{SS}_i) + \boldsymbol{\gamma}' \mathbf{X}_i + \varepsilon_i.$$

   (a) Suppose $\beta_1 = \beta_2 = \beta > 0$. A worker starts at $(\mathrm{COG}_i, \mathrm{SS}_i) = (1, 1)$. Compute the log-wage increase from moving to each of: $(3, 1)$, $(2, 2)$, $(1, 3)$.

(b) If cognitive and social skills are complements, what sign do you expect for β_3? Which of the three bundles above yields the highest wage increase?

(c) Further suppose $\beta_3 = \frac{1}{10}\beta$. A worker starts at $(\text{COG}_i, \text{SS}_i) = (4, 0.5)$ and can invest 2 additional skill units (split freely between COG and SS at equal cost). What allocation maximises their log-wage gain? Why?

(d) Create the interaction variable `afqt_soc = afqt_std * soc_nlsy_std` in `nlsy79`. Regress with `weights = weight`. Save as `table1_col3`; verify the coefficients match column 3.

(e) Using $\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3$: find the worker with `pubid == 113`. If they can allocate one additional skill unit between COG and SS, which allocation maximises their predicted wage gain?

7. The author runs 4 additional specifications, even including a cognitive × non-cognitive interaction. Skim section IV.A of the paper for the motivation.

Part C. Occupational Sorting: selection and Table 2

Per the theoretical model implications in section II.E of the paper, *workers with higher social skills sort into non-routine occupations* and *workers earn more when they switch into non-routine occupations, and their relative wage gain is increasing in social skill* (Deming, 2017). By **occupational sorting**, we refer to this selection into less routine-oriented roles by workers with better social skills.

8. [7] Replicate columns 1 and 3 of Table 2. The models:

$$T_i = \alpha + \beta_1\,\text{COG}_i + \beta_2\,\text{SS}_i + \beta_3\,(\text{COG}_i \times \text{SS}_i) + \boldsymbol{\gamma}'\mathbf{X}_i + \varepsilon_i,$$

where $T_i \in \{$`routine`, `socskills`$\}$. Include the previous *covariates* plus `educ` and `ind6090` as factors. Weight by `weight`. Restrict to `nlsy79` observations with non-missing `ln_wage` using the `subset` argument.

(a) Interpret $\hat{\beta}_3$ in both models. What does it tell us about workers who are both cognitively and socially skilled?

(b) What is the implication of a non-significant $\hat{\beta}_3$ in column 3?

(c) What is the likely justification for the industry fixed effects?

(d) Is there evidence of social-skill sorting into social occupations regardless of cognitive skill level?

9. Review Figure 5 of the paper. Which occupations saw the greatest cumulative hourly wage growth over the 1980–2000 period?

10. Recall the Potential Outcomes Framework, and suppose "high social skills" is a treatment. We have no control over its assignment, but we would like to compute the average wage premium associated with having high social skills. Explain how selection bias arises in this setting.

11. Workers with higher social skills may sort into socially-intensive occupations that pay well for reasons unrelated to their social skills (e.g., their preferences, location, industry). Using the selection bias decomposition from 10., argue whether the coefficient $\hat{\beta}_2$ in 6. is likely above or below the causal wage effect of social skills. Identify one type of worker for whom $Y_i(0)$ — their counterfactual wage in a routine job — is already high.

Part D. Cross-cohort pay for skills: Table 4

The author argues that, despite a general trend toward non-routine occupations, social skills are increasingly rewarded over time.

12. [8] Replicate columns 1 and 4 of Table 4. The cross-cohort specification for worker i in cohort $s \in \{79, 97\}$:

$$\begin{aligned} y_{is} = \alpha &+ \beta_1\,\mathrm{COG}_{is} + \beta_2\,\mathrm{SS}_{is} + \beta_3\,(\mathrm{COG}_{is} \times \mathrm{SS}_{is}) \\ &+ \beta_4\,97_s + \beta_5\,(\mathrm{COG}_{is} \times 97_s) + \beta_6\,(\mathrm{SS}_{is} \times 97_s) \\ &+ \beta_7\,(\mathrm{COG}_{is} \times \mathrm{SS}_{is} \times 97_s) + \boldsymbol{\gamma}'\mathbf{X}_{is} + \varepsilon_{is}, \end{aligned}$$

where y_{is} is `emp` (col 1) or `ln_wage` (col 4), 97_s = `sample`, and SS_{is} = `soc_nlsy2_std`.
    (a) Create `nlsy_both`: filter `nlsy_merged` to `age` 25–33. Coerce `age`, `year`, `div`, `metro`, `urban` to factors.
    (b) Create the interactions: `afqt_sample`, `soc2_sample`, `afqt_soc2`, `afqt_soc2_sample`.
    (c) Estimate the models. For column 4, additionally restrict to `complete97 == 1`. Save as `table4_col1` and `table4_col4`.
13. Interpret $\hat{\beta}_7$ from 12.. How does a positive and significant three-way coefficient strengthen the paper's argument about the growing importance of *combined* cognitive and social skills, beyond what the cross-cohort SS premium alone ($\hat{\beta}_6$) can establish?

Part E. Hypothesis tests

14. [9] Using `table1_col3`, test the joint hypothesis $H_0 : \beta_2 = \beta_3 = 0$ — that social skills and their interaction with cognitive ability are jointly irrelevant, controlling for cognitive skill alone. Use `linearHypothesis()` from the `car` package. Report the F-statistic and p-value.
15. [10] Using `table4_col4`, test the joint hypothesis $H_0 : \beta_6 = \beta_7 = 0$ — that the *cross-cohort changes* in both the social skill premium and the cognitive–social complementarity are jointly zero. This directly tests the paper's central claim that the NLSY97 labor market rewards social skills differently from the NLSY79. Use `linearHypothesis()`. Report the F-statistic and p-value.

Additional resources

– Drive: Review Notes

14.310x Flipped Classroom

Week 9 Instructions

Regression Discontinuity Design (RDD)
RDD modeling in R

Checklist

- ☐ Watch the regression discontinuity lecture notes[5] (Requirement)
- ☐ Complete Guided Case 8 (Sessions 1 & 2)
- ☐ Dataset: `schools.csv`
- ☐ Paper: (Pop-Eleches & Urquiola, 2013)
- ☐ (Optional) Review additional materials (Optional) -

Guided Case Study G.9.8

Regression Discontinuity:
Replicating *Going to a Better School*
(Pop-Eleches & Urquiola, 2013)

Instructions: Work in pairs or groups of three to solve all questions. One team member should share editor access to the Colab notebook with the rest.

[5]**Module 9 > Regression Discontinuity Design** in the online component of the course.

Notebook: Case Study 8 — RDD replication

Dataset: `schools.csv`

Paper: (Pop-Eleches & Urquiola, 2013)

Scenario:

In 2002, Romania reformed its high school admissions system. Students take a standardised national examination at the end of middle school, and schools receive a centralised ranking based on their historical average admission scores. Each school announces a minimum admission score — its **cutoff** — and students are matched to schools in descending order of their exam scores. In towns with multiple high schools, adjacent schools share a cutoff: the top student who did not qualify for school k is the marginal student for school $k - 1$.

This system generates a compelling natural experiment. Students who scored just above and just below a given cutoff are, by construction, nearly identical in ability — yet one group attends school k while the other attends $k - 1$. Pop-Eleches and Urquiola (2013) exploit this discontinuity to study whether attending a *higher-quality* school causally affects students' academic outcomes and, equally importantly, how families and peers behaviorally respond to the assignment.

In this case study you will replicate and interpret core results of Pop-Eleches and Urquiola (2013) using `schools.csv`, a student-level survey extract from the authors' replication package. The dataset covers students in towns with two to four secondary schools — localities where the school-quality contrast is sharpest and direct surveys of students, parents, and teachers were conducted.

Dataset: `schools.csv`

Each row is one student. Key variables:

- `score_dist` — Distance of the student's admission score to the relevant school cutoff. Negative values indicate the student scored *below* the cutoff (attended the lower school); positive values indicate the student scored *above* (attended the higher school).
- `above_cutoff` — Binary treatment indicator: = 1 if the student is above the cutoff and attends the higher-ranked school.
- `score_dist_above` = `score_dist`×`above_cutoff` — Slope interaction, allowing different running-variable slopes on each side of the cutoff.
- `score_dist_r05` — `score_dist` rounded to the nearest 0.05, used for bin-scatter plots.
- `school_avg_score` — The average admission score of the school the student attends (0–10 scale). Captures the **school peer quality** — the key first-stage outcome showing that better-school students face a higher-achieving peer group.

- `school_own_score` — The school's own quality indicator (its admission average). Used to classify schools by tier.
- `n_schools_town` — Number of secondary schools in the student's town (2, 3, or 4 in this extract).
- `school_fe` — School × zone × wave fixed-effect identifier. Always include as `factor(school_fe)` in regressions.
- `cluster_id` — School-year identifier for clustering standard errors.
- **Teacher quality** (from school-level records): `teacher_certified` (share of certified teachers, 0–1), `teacher_experience` (average years of experience), `teacher_novice` (share with < 2 years of experience, 0–1).
- **Parental behavior** (from parent survey): `parent_volunteer` (= 1 if parent volunteers at school), `parent_tutoring` (= 1 if parent pays for tutoring), `parent_help_hw` (= 1 if parent helps with homework).
- **Peer quality** (from student survey): `peer_rank` (student's self-reported rank among peers, 1–10; lower is better), `peer_bad_idx` (index of bad peer behaviors, 0–1).
- **Pre-determined characteristics** (validity checks): `female`, `age`, `head_educ_tert` (household head has tertiary education), `mom_educ_tert`, `has_internet`.

Part A. Understanding the RDD Design

1. The admissions system described above generates a Regression Discontinuity Design.
   (a) Identify the **running variable**, the **cutoff**, and the **treatment** in this setting.
   (b) State the key identifying assumption of Sharp RDD in general, and then restate it in the specific language of this paper: what must be true about students near the cutoff?
   (c) Give one concrete threat to this assumption in the school admissions context. Which type of actor would need to be able to manipulate the assignment precisely?
2. [2] Install and load `tidyverse`, `fixest`, `rdrobust`, `lmtest`, `sandwich`, and `car`. Read `schools.csv` and save as `schools`. Report:
   - Total number of students and the share with `above_cutoff` = 1.
   - Mean and standard deviation of `school_avg_score` separately for students above and below the cutoff.
   - A frequency table of `n_schools_town`.

   What does the large share above the cutoff tell you about this survey sample?
3. [3] Plot a histogram of `score_dist` with bin width 0.05 (restrict to |`score_dist`| ≤ 2). Add a vertical dashed line at zero.
   - Does the density appear continuous at zero? What would a spike or dip indicate?
   - Count observations with `score_dist` < 0 and > 0. What does the ratio suggest about the survey design?

- *(Context:)* In the full administrative data (`data-AER-1`, > 3.6 million observations), the running-variable density is smooth at the cutoff, indicating no systematic score manipulation. Why does the survey extract show a large imbalance?

4. [4] Create an RDD bin-scatter for `school_avg_score`:
   (a) Compute the mean of `school_avg_score` within each `score_dist_r05` bin. Restrict to $|$`score_dist`$| \le 2$.
   (b) Plot these bin means against `score_dist_r05`, add a vertical dashed line at zero, and overlay separate linear trend lines for $r < 0$ and $r > 0$.
   (c) Describe the discontinuity: how large does the jump look relative to the overall spread of `school_avg_score`?

Part B. Linear RDD with `lm()`

The canonical parametric RDD estimator fits separate linear regressions on each side of the cutoff:

$$Y_i = \alpha + \beta_{\text{RD}} \cdot D_i + \gamma_1 \cdot r_i + \gamma_2 \cdot (r_i \times D_i) + u_i, \tag{3}$$

where Y_i is the outcome, D_i = `above_cutoff`, and r_i = `score_dist` is the running variable (so $r_i \times D_i$ = `score_dist_above`). The coefficient β_{RD} estimates the causal effect at the cutoff.

5. [5] Estimate the RDD model (10) with `lm()`, restricted to $|$`score_dist`$| \le 1$ and `score_dist` $\neq 0$. Use `coeftest()` with `vcovCL()` clustered on `cluster_id`.
   (a) Report $\hat{\alpha}$, $\hat{\beta}_{\text{RD}}$, $\hat{\gamma}_1$, $\hat{\gamma}_2$ with standard errors.
   (b) Interpret $\hat{\alpha}$ and $\hat{\beta}_{\text{RD}}$ in words.
   (c) Interpret $\hat{\gamma}_1$ and $\hat{\gamma}_2$: what do the slopes tell you about the relationship between score and school quality on each side of the cutoff?
6. [6] Extend the model with school fixed effects using `feols()` from `fixest`:

$$\texttt{school\_avg\_score}_i = \beta_{\text{RD}}\, D_i + \gamma_1\, r_i + \gamma_2\, (r_i D_i) + \mu_j + u_i,$$

   where μ_j denotes school fixed effects (`school_fe`). Save as `m_fe`.
   (a) Report $\hat{\beta}_{\text{RD}}$ and its standard error. How does it compare to Q5.?
   (b) Why do we include `school_fe` as a fixed effect? What variation is being absorbed?
   (c) Fixed effects reduce residual variance. Does this tend to increase or decrease the standard error on $\hat{\beta}_{\text{RD}}$? Why?
7. The linear specification in (10) imposes that the relationship between Y and the running variable is the same everywhere within the bandwidth. An alternative is to add a quadratic term.
   (a) What is the motivation for preferring **local linear** regression over global polynomials in RDD? (*Hint:* think about boundary bias and the role of the bandwidth.)
   (b) In terms of bias vs. variance, what is the cost of adding more polynomial terms to the RDD specification?

(c) If you added `score_dist^2` and `above_cutoff * score_dist^2` to the model in Q6., would you expect the estimate of $\hat{\beta}_{\text{RD}}$ to change substantially? Why or why not?

Part C. Optimal Bandwidth and Replication

8. [8] Use `rdrobust()` on `school_avg_score` with the default MSE-optimal bandwidth. Report:
   (a) The MSE-optimal bandwidth $\hat{h}$ and the point estimate (conventional).
   (b) The robust 95% confidence interval and corresponding p-value.
   (c) The conceptual difference between `rdrobust` and the `lm()` approach of Q5.: what does "local linear" mean and why might it be preferred?
9. [9] Replicate the bandwidth = 1.0 specification of Table 3 from Pop-Eleches and Urquiola (2013) using `feols()` with `school_fe` fixed effects and `cluster_id` clustering. Save as `m_t3`.
   (a) Report $\hat{\beta}_{\text{RD}}$ and its clustered SE. Verify you obtain approximately 0.477 (SE $\approx$ 0.047).
   (b) The paper's Table 3 reports $\hat{\beta}_{\text{RD}} \approx$ 0.11–0.20 (depending on the specification). You will explain this difference in Q11..
10. [10] Loop over bandwidths $h \in \{0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.2, 1.5, 2.0\}$, running `rdrobust()` with `h = h_val` each time. Store the conventional point estimate and robust 95% CI bounds.
   (a) Plot the point estimates with 95% CI bands against the bandwidth value.
   (b) Does the coefficient appear stable across bandwidths? What does this stability suggest about the credibility of the RDD estimate?
11. Your coefficient on `school_avg_score` from Q9. is approximately 0.48, while Pop-Eleches and Urquiola (2013) report values around 0.11–0.20 in Table 3 across their specifications.
   (a) What does `n_schools_town` tell you about the sample in `schools.csv`? How does it differ from the full population of Romanian schools?
   (b) Propose a reason, based on the design of the paper, why the jump in school quality would be *larger* in towns with fewer schools.
   (c) Is this difference a problem for the internal validity of the RDD? What does it imply for external validity?

Part D. Behavioral Responses

Pop-Eleches and Urquiola (2013) argue that assignment to a better school triggers behavioral adjustments in teachers, parents, and peers — beyond the direct effect on peer quality. Parts D and E explore these responses.

12. [12] For each of `teacher_novice`, `teacher_experience`, and `teacher_certified`, estimate the RDD model with `feols()`, `school_fe` FEs, `cluster_id` clustering, bandwidth 1.0.

(a) Report $\hat{\beta}_{\text{RD}}$ and p-value for each. Which are significant at 5%?
(b) The coefficient on `teacher_novice` is negative ($\hat{\beta} \approx -0.036$, $p \approx 0.02$). Propose a mechanism for why assignment to a better school reduces the share of novice teachers.
(c) `teacher_certified` is insignificant ($p \approx 0.89$). Does this mean teacher certification is irrelevant here? Discuss.

13. [13] For each of `parent_help_hw`, `parent_tutoring`, and `parent_volunteer`, estimate the RDD model with `feols()`, `school_fe` FEs, `cluster_id` clustering, bandwidth 1.0.
(a) Report $\hat{\beta}_{\text{RD}}$ and p-value for each. Which parental behaviors respond significantly?
(b) The coefficient on `parent_help_hw` is negative ($\hat{\beta} \approx -0.043$, $p \approx 0.027$). Propose a mechanism: why might going to a better school reduce parental homework help?
(c) The paper interprets this as a "behavioral response." What implicit assumption is needed to interpret this RD estimate causally?

14. [14] Replicate the style of Figure 4 in Pop-Eleches and Urquiola (2013) for the `teacher_novice` outcome.
(a) Compute bin means of `teacher_novice` within each `score_dist_r05` bin ($|$`score_dist`$| \leq 2$, exclude zeros). Plot with separate linear fits on each side.
(b) Compare the visual strength of the discontinuity in `teacher_novice` vs. `school_avg_score` from Q4.. Which is more pronounced visually and why?

15. [15] For each of `peer_rank` and `peer_bad_idx`, estimate the RDD model with `feols()`, `school_fe` FEs, `cluster_id` clustering, bandwidth 1.0.
(a) Report $\hat{\beta}_{\text{RD}}$ and p-value. Interpret the sign of the `peer_rank` coefficient (lower values = better-quality peers).
(b) The coefficient on `peer_bad_idx` is *positive* ($\hat{\beta} \approx 0.045$, $p \approx 0.009$): students in the better school report *more* bad peer behaviors. Propose two competing explanations.
(c) Reconcile the peer quality findings with the school quality jump from Q9..

Part E. Design Validity and Inference

16. [16] Create bin-scatter plots for `female` and `age` (bin width 0.05, $|$`score_dist`$| \leq 2$, separate linear fits on each side).
(a) Plot both variables. Is there any visible discontinuity at zero?
(b) What would a visible jump in `female` or `age` imply for the validity of the RDD?

17. [17] Run the RDD regression for `female`, `age`, and `head_educ_tert` using `feols()` with `school_fe` FEs, `cluster_id` clustering, bandwidth 1.0.
(a) Report $\hat{\beta}_{\text{RD}}$ and p-value for each. Are any significant at 5%?
(b) What does a set of insignificant placebo tests establish? What does it fail to establish?

   (c) Could the RDD be valid even if one placebo test is marginally significant at the 5% level? Use the concept of multiple testing to answer.

18. [18] Re-estimate the model from Q6. using `lm()` with explicit `factor(school_fe)` dummies (for compatibility with `linearHypothesis()`). Test $H_0 : \beta_{\text{RD}} = 0$ using `linearHypothesis()` with `vcovCL()` clustered SEs.
   (a) Report the F-statistic and p-value.
   (b) Numerically, $F = t^2$ for a single linear restriction. Verify this relationship using the t-statistic from `m_fe`.
19. For the model estimated in Q6. (with school fixed effects), use `linearHypothesis()` to test the joint hypothesis $H_0 : \gamma_1 = \gamma_2 = 0$, i.e. that the running variable has no effect on `school_avg_score` on either side of the cutoff.
   (a) Report the F-statistic and p-value. Can you reject H_0?
   (b) Why is it important to control for the running variable in the RDD specification, even if we expect γ_1 and γ_2 to be small near the cutoff?
   (c) Suppose you omitted the running variable terms. Would $\hat{\beta}_{\text{RD}}$ be upward or downward biased? Explain.
20. The RDD identifies a **Local Average Treatment Effect** (LATE) rather than an Average Treatment Effect (ATE).
   (a) In this setting, who is the *marginal student* — the student whose school assignment is determined by the cutoff? Why is the causal effect identified for this specific group?
   (b) The authors find a jump in `school_avg_score` near the cutoff but limited effects on `baccalaureate grade` (Table 4, not in this extract). Propose an explanation based on the characteristics of the marginal student.
   (c) The dataset covers only towns with 2–4 schools. Discuss one dimension along which the LATE for this subpopulation may differ from a national-level ATE, and what additional data or design would be needed to extrapolate.

Coding Lab C.9.7

Further regression tools

Instructions: Work individually. Complete all exercises in the coding lab notebook and submit as required by your instructor.

Notebook: Coding Lab 7 — Further regression tools

Additional materials and references

References:

- (Pop-Eleches & Urquiola, 2013)
- 🗁 **Drive: Review notes and supplementary readings**

Data and replication:

- Pop-Eleches and Urquiola (2013) replication package (AER data archive). The dataset `schools.csv` is derived from `data-AER-7.dta` included in the authors' replication package. This guided case is released under **CC BY 4.0**; the underlying data are subject to the AER's data use policy.

R Packages:

- (Calonico et al., 2015) — `rdrobust` package for MSE-optimal bandwidth selection and local polynomial RDD estimation.
- (Bergé, 2018) — `fixest` package for high-dimensional fixed-effects regression with clustered standard errors.

14.310x Flipped Classroom
Week 10 Instructions

Random assignment and Instrumental Variable (IV) models
More on IV: multiple instruments, multiple stages.

Checklist

- ☐ Read the *details* on `ivreg()` in the the `AER` package. Make sure you learn how to enter a 2-stage specification. (Requirement)
- ☐ Complete Open Case 4 (Sessions 1 & 2)
  - ☐ `worms.csv`

Case Study O.10.4

RCTs as an IV problem: (Miguel & Kremer, 2004)

Instructions: Work in pairs or groups of 3. Answer the following questions collaboratively.

Notebook: Case O.4

Dataset: worms.csv

Make sure you understand how to specify IV models with function `ivreg()` in the `AER` package.

Scenario: In rural western Kenya, intestinal worms (hookworm, roundworm, schistosomiasis) are highly prevalent and infect a large share of primary school children. Worm infections can lead to iron-deficiency anemia, malnutrition or abdominal pain; all important factors in school absenteeism and inattention. Non-severe cases may present as asymptomatic, with more subtle impairments in children's educational attainment and health outcomes. Deworming treatment with Albendazole and Praziquantel pills is cheap and effective, but rarely provided at scale.

(Miguel & Kremer, 2004) designed an intervention to distribute deworming pills to schools in a region surrounding Victoria Lake. They partnered with an NGO (ICS Africa) to implement an RCT where schools were phased to treatment as follows:

- **Group 1** began treatment on 1998 while groups 2 and 3 remained controls.
- **Group 2** phased in treatment on 1999.
- **Group 3** began on 2001 — after the study window for these published results.

For simplicity, we will focus only on the short-term outcomes of the first intervention. That means only schools in Group 1 schools will be considered treated in our dataset.

Glossary for `worms`

- `pupid` — pupil ID (unique child identifier).
- `baseline_school` — ID of the child's school.
- `baseline_grade` — The student's academic grade. Encoded 1 to 11.
- `assigned_98` — Indicator:=1 the student's school is part of Group 1.
- `treated_98` — Indicator:=1 the student received **any** deworming treatment in 1998 (Albendazole or Praziquantel).
- `attendance_98` — The student's observed attendance rate (proportion of times present during surveyor's visits 1 through 8)
- `testscores_98` — Reported average test scores mid-1998 (standardized by grade).
- `sex` — Pupil's gender. Encoded 1 for male and 0 for female.
- `house_floor` — Material of household floor. Encoded 0 for mud, and 1 for cement.
- `weight` – Pupil's weight in kg.

Out of practicality, categorical variables contain a level `"N/A"`, as opposed to R's `NA`. And a small number of missing values in continuous covariates are mean-imputed.

Part 1. Random assignment as an instrument

Suppose the researchers don't provide the deworming treatment. Instead, they decide to widely disseminate deworming information across all schools, including where to purchase the drugs at low prices:

1.1 If a survey was conducted at the end of 1998, how would you expect covariates such as `house_floor`, or `weight` compare between those students with (`treated_98 == 1`) and those in the (`treated_98 == 0`) group?

1.2 As previously noted, children with heavy worm infections are prone to missing school. Search the paper's introduction for evidence on the effectiveness of Albendazole and Praziquantel in abating worm infections.
   (a) Suppose students' absenteeism is 100% due to infection symptoms. What should be the mean of `attendance_98` among treated children?
   (b) Empirically, not all school absenteeism is worm-related. Describe the covariates of the **treated** students you expect to miss school the most.

1.3 Back to (Miguel & Kremer, 2004)'s randomized assignment. How would your answers to 1.1 and 1.2 change if each value in `treated_98` were drawn iid from a Bernoulli with $p = 0.5$?

1.4 [1.1] But did every student assigned to treatment receive any? What is the average compliance for assigned and non-assigned students?

Suppose we now have the model

$$X_i = \pi_0 + \pi_1 Z_i + v_i, \quad i = 1, 2, .., N \tag{4}$$

Where $X_i = \mathbf{1}$(i treated) and $Z_i = \mathbf{1}$(i's school assigned to treatment). π_0, π_1 are coefficients, and $v_i \sim N(0, \sigma_v^2)$

1.5 [1.2] Use `lm()` to get the OLS estimates $\hat{\pi}_0$, $\hat{\pi}_1$ and answer the following:

(a) Interpret the model: which variable does this model explain in terms of which other variable?
(b) What does $\hat{\pi}_0$ represent?
(c) $\hat{\pi}_1$ can be expressed as a difference of means. Write it down.
(d) Test the hypothesis that $H_0 : \pi_0 = 0$ and $H_1 \neq 0$ with 95% confidence. What does this result imply? Do you think the data-generating process warrants it?

1.6 What percentage of students were treated? What is the attendance rate among those treated and untreated?
1.7 What is the percentage points difference in attendance between compliers and non-compliers [hint: first create `complied` ...]

We have the following model

$$Y_i = \alpha + \beta X_i + \epsilon_i, \tag{5}$$

Along with the IV assumptions:

$$\mathbb{E}[Z_i u_i] = 0, \quad \text{and} \quad \text{Cov}(Z_i, X_i) \neq 0, \tag{6}$$

$$\hat{\beta}_{\text{Wald}} = \frac{\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]}{\mathbb{E}[X \mid Z = 1] - \mathbb{E}[X \mid Z = 0]}. \tag{7}$$

Again, X_i is a treatment dummy. Y_i is the attendance rate for student i, and the error $u_i \sim N(0, \sigma_\epsilon^2)$ Constants α, β are the model's coefficients.

8. [1.4] Run the regression
9. Interpret $\hat{\beta}$.
10. Do you believe $\mathbb{E}[u_i \mid X_i] = 0$ here? Use the story about who chooses (or is able) to take the drugs.
11. In which direction would you *expect* the bias to go: would OLS overestimate or underestimate the causal effect? (Justify with a short causal story.)

1. [1.5] Wald estimator. Compute $\hat{\beta}_{\text{Wald}} = \frac{\mathbb{E}[Y|Z=1]-\mathbb{E}[Y|Z=0]}{\mathbb{E}[X|Z=1]-\mathbb{E}[X|Z=0]}$
   (a) Compute the numerator and denominator separately using group means.
   (b) Compare $\hat{\beta}_{\text{Wald}}$ to your na"ive OLS estimate from 8.. Are they similar? Which one is larger?
   (c) Explain (in words) what variation identifies $\hat{\beta}_{\text{Wald}}$.
2. [1.6] 2SLS using `ivreg()`.Run `ivreg(attendance_98 treated_98 | assigned_98, data=worms)`.

1. Verify it matches your Wald estimate (up to rounding).
2. Extract and report the first-stage coefficient $\hat{\pi}_1$ and interpret it.
3. What does the 2SLS estimate measure conceptually? (Hint: "Local" average treatment effect.)

1. [1.7] Heterogeneity teaser (no extra theory required). Split the sample by `baseline_grade` into `lower` (1–5) vs `upper` (6–11), and re-run the reduced form in each subsample.
   (a) Where do you see larger ITT effects? Provide a short hypothesis (e.g., older children, opportunity cost, etc.).

Part 2. Covariates

In a randomized experiment, adding covariates is not required for identification. However, it can (i) improve precision, and (ii) help us detect problems with data quality (e.g., missingness patterns).

2.1 [2.1] Balance checks (quick).For each covariate below, compare means between `assigned_98==1` and `assigned_98==0`. Use a difference-in-means test: `sex`, `house_floor`, `weight`.

2.2 Which covariates look well-balanced? Which look suspicious?

2.3 Explain why balance is expected under random assignment (in expectation) but not guaranteed in finite samples.

1. [2.2] Missingness patterns. For each covariate above, compute the fraction equal to `"N/A"` (for categoricals) or missing/imputed (for continuous).
   (a) Is missingness correlated with `assigned_98`? Run a regression of a missing dummy on `assigned_98`.
   (b) Why is missingness correlated with treatment assignment a problem even in an RCT?

1. [2.3] First stage with covariates.Estimate

$$X_i = \pi_0 + \pi_1 Z_i + \delta' W_i + v_i$$

   where W_i includes the covariates above.
   (a) Does $\hat{\pi}_1$ change a lot? Why might it (or might it not)?
   (b) Report the R^2. Does adding covariates make X more predictable?

1. [2.4] 2SLS with covariates. Run

$$Y_i = \alpha + \beta X_i + \theta' W_i + u_i$$

   instrumenting X_i with Z_i (and keeping W_i in both stages).
   (a) Compare your IV estimate $\hat{\beta}$ with and without covariates. Should it change much? Why?
   (b) Compare standard errors. Did precision improve?

1. [2.5] A small specification ladder. Create a table with the IV estimate $\hat{\beta}$ under:

(a) no covariates;
(b) plus demographics (`sex`, `baseline_grade`);
(c) plus SES proxies (`shoes`, `uniform`, `house_floor`);
(d) plus health (`weight`).

What pattern do you see? Provide one sentence interpreting the stability (or lack of stability).

Part II

Instructor Version

Solved Handouts & Annotations

Instructor preface + answers for Weeks 1-10

14.310x Flipped Classroom
Instructor Preface

Answers format and sessions overview

This preface describes how these materials are organized, what each visual element means, and how to use the instructor build effectively. It is included at the front of the solved handouts section. Read it once before your first session; refer back to it as needed.

The instructor build is a compiled PDF that contains **the full student handout plus inline annotations**. The student-facing content is reproduced exactly; the differences are the additional instructor elements that appear after questions, at the start of each activity, and occasionally within the question text itself. These elements are described below.

Weekly session activities

Each week consists of one or more activities. Three formats are used, each corresponding to a distinct box style.

Coding Labs (gray box)

Coding Lab example

Instructions: Work individually. Solve all exercises in the corresponding Colab notebook.

Notebook: Coding Lab 1

Submit your work in the format required by the instructor.

Coding Labs are **individual** exercises solved entirely within a Colab notebook. The PDF handout provides the activity description and a link to the notebook; all questions, code cells, and submission are handled in Colab. Coding Labs appear before or alongside case studies as skill-building exercises.

Guided Case Studies (cyan box)

Guided Case Study example

Instructions: Work in pairs or groups of three to solve all questions.

Questions are written out in this handout. Coding questions are labeled with an orange icon [2.3] and link to the corresponding cell in the session notebook.

Guided Case Studies are **structured group exercises**. Questions appear in the PDF handout in a fixed sequence, mixing written/conceptual questions with coding questions. The class works through them collaboratively, with the instructor guiding the pace. Some weeks' guided cases are contained entirely within the Colab notebook, in which case the PDF box provides only a brief description and the notebook link.

Open Case Studies (green box)

Open Case Study example

Instructions: Work in pairs or groups of 3.

Answer the following questions collaboratively; one member of the team shares their notebook with the group.

Open Case Studies are **less structured** than guided cases. They typically involve a research paper, a real dataset, or a policy question, and ask students to apply methods from the module in a setting where the right approach is not pre-specified. They are suitable for in-session discussion and allow more instructor discretion over which parts to cover.

The instructor build

The instructor build is generated from the same source as the student handout but with two additions enabled:

1. **Inline answers** appear after each question that has a worked solution, inside a light blue-gray box called the *answer box*.
2. **Instructor annotations** appear at the start of each activity and occasionally inline, inside a warm orange-tinted box called the *instructor line box*.

In the compiled PDF, these elements appear exactly where the question or activity is — no separate answer key section, no page flipping.

Instructor-only elements

1. Instructor notebook box

At the start of each activity that has an associated Colab notebook, an orange-accented box appears linking to the **instructor's answer notebook**. This box does not appear in the student handout.

> **Instructor Notebook: Answers to G.2.1**
>
> Find answers to all code-related questions in this Guided Case. Each coding-relevant question links to its corresponding answer section in this notebook.

The link opens the instructor's copy of the Colab notebook. For activities where all exercises are in Colab, the instructor notebook is the primary answer reference; the answer boxes in the PDF will direct you there. For activities with written questions, the notebook handles only the coding sub-questions.

2. Answer boxes

Each written question with a worked solution is followed by an answer box:

> $\bar{X}_n \xrightarrow{p} \mu$ as $n \to \infty$ by the Weak Law of Large Numbers, provided $\text{Var}(X_i) < \infty$.
> **Instructor note:** Students often conflate convergence in probability with almost-sure convergence. If this comes up, the distinction is worth a brief mention but is not required at this level.

Answer boxes contain the **worked solution** to the question — numerical result, derivation, or explanation — followed optionally by an *Instructor note* (in bold). Instructor notes call out common errors, pedagogical tips, or alternative approaches. They are not part of the answer itself and need not be communicated to students.

3. Colab answer links

For questions that also have a coding component, a blue link appears inside or adjacent to the answer box:

Code in Colab ✔ >_ [2.3] **Answer**

> `mean(popular)`: 0.1032 (`sum(popular)`: 1,004 out of 9,724 posts)

The blue icon and [N/A] **Answer** link open the relevant cell in the instructor notebook directly. The number in brackets (e.g., **2.3**) matches the question's Colab numbering, which may differ from its overall question number in the handout. This link is the fastest way to navigate from a written answer to the corresponding code.

4. Inline instructor notes

Some questions carry only an instructor note and no full solution — typically for questions where the answer is evident from context or where the intent is pedagogical framing rather than a specific result. These appear as:

> **Instructor note:** The goal here is for students to recognize that the sample mean is itself a random variable. The numerical answer is less important than the reasoning.

Notes of this kind appear frequently in the glossary sections of case studies (where variable definitions carry contextual commentary) and in questions that ask students to *discuss* or *interpret* rather than compute.

Session checklists and labels

Each week opens with a checklist. Items are tagged with a small gray label indicating when they are expected to be completed:

- **Requirement** (Requirement) — pre-class preparation (readings, tutorials, package installs). Students are expected to arrive having completed these.
- **Session 1** (Session 1) — scheduled for the first in-class session of the week.
- **Session 2** (Session 2) — scheduled for the second in-class session of the week, or a later portion of the same session.
- **Optional** (Optional) — supplementary material, review notes, or further readings. Not required.

These labels are a guide, not a constraint. Reorganizing activities across sessions, or splitting a single case study across both sessions, is entirely appropriate depending on group pace.

Pacing, calibration, and instructor flexibility

Materials often exceed session time

For most weeks, the volume of material — questions, coding exercises, and discussion points — **exceeds what a typical session can cover in full**. This is by design: the materials are written to give the instructor choices, not to impose a fixed agenda. In practice, you should expect to leave some questions unfinished or to assign them as follow-up.

A practical approach is to review the week's materials before the session and decide:

- Which questions are **required** for all students to attempt during the session.
- Which questions are **candidates for in-session discussion** if time allows.
- Which questions can be **worked outside class** or skipped entirely.

Instructors teaching in different contexts (lecture length, class size, student background) will calibrate differently. The materials are designed to support a range of choices without requiring modification.

All answers are worked out for instructor flexibility

Every answered question has a complete worked solution, even for questions that are intentionally open-ended or where multiple valid approaches exist. These solutions are **reference points**, not scripts. You are not expected to walk through them verbatim; rather, they are provided so you can:

- Quickly verify student answers during the session.
- Anticipate where students are likely to get stuck.
- Choose how much of the solution to reveal, and when.
- Skip a question confidently, knowing the solution is available if needed later.

For open-ended questions — interpretive, conceptual, or design-oriented — the answer box typically provides one defensible response plus an instructor note highlighting alternative angles or common misconceptions. These answers deliberately leave room for class discussion.

Incomplete weeks

A small number of weeks have partially completed materials — typically weeks where additional case study parts are planned but not yet written, or where answer notebooks are still being developed. In these cases, the PDF may include placeholder content in the instructor boxes (visible only in the instructor build) indicating that a section is pending. Student-facing content is complete.

Colab notebooks: student and instructor versions

Most activities reference one or more Colab notebooks. There are two versions:

- **Student notebook** — linked in the student handout via the icon. Contains question prompts, starter code, and blank cells for student answers. **Students work in their own copy of this notebook.**
- **Instructor answer notebook** — linked in the orange instructor box at the start of each activity (instructor build only). Contains completed answer cells and, for some questions, extended explanatory notes. This is the reference for all code-based solutions.

The two notebooks are structurally identical: every cell in the student notebook has a corresponding answer cell in the instructor notebook, identified by the same Colab question number (e.g., [**2.3**]). The blue

Code in Colab **[2.3] Answer**

link in the instructor PDF navigates directly to the answer cell.

For activities without a separate answer notebook — typically weeks where the Colab exercises are straightforward or where the notebook is still being finalized — the instructor box links to the student notebook as a placeholder.

Quick reference: instructor elements at a glance

Element	Where it appears	Purpose
Instructor Notebook link	Start of each activity (instructor only)	Links to the instructor answer notebook
Worked answer	After each answered question (instructor only)	Solution, derivation, or explanation
[2.3] **Answer**	Inside answer box, for coding questions	Links to the answer cell in the instructor notebook
Instructor note: pedagogical comment	Inside answer box	Tips, common errors, discussion guidance

14.310x Flipped Classroom

Week 1 Answers

R for data analysis and working in the cloud

Checklist

- ☐ Complete the INTRO TO R interactive course from the `Swirl` package[6] (Requirement)
- ☐ Watch the Getting started with Google Colab notebooks video tutorial (Requirement)
- ☐ Create or set up a personal Google account
  (you must be able to use Drive and Colab). (Requirement)
- ☐ Create your first Colab notebook (Session 1)
- ☐ Complete Coding Lab 1. (Session 1)
- ☐ Complete Coding Lab 2 (Session 2)

L.1.0

Your first Colab notebook

Instructions: Read and follow the steps below **before proceeding with the activity**. After reading the instructions, access the notebook link and complete the exercises in Colab. This is an individual task, but you will collaborate on the final question.

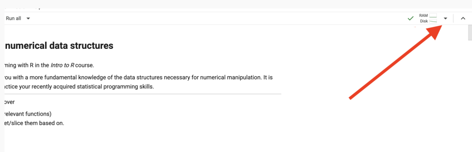
[6]**Module 1 > Introduction to R** in the online component of the course.

Note: You may want to ask students complete these requirements in advance of the session.

Notebook: Your first Colab notebook

We will cover the essentials of working with Jupyter Notebooks on Google Colab — this resource will be an important tool throughout. Before you can begin working on the coding labs in Colab, make sure to:

- **Create or use a personal Google account**. While it is possible for you to download the Jupyter Notebooks we will work on, and manage them locally in your own machine, we strongly advise you to work on Colab since it will make it easier for your classmates and the instructor to interact with your work when needed. Colab has many features Google Drive already offers: comments, real-time colaboration in the same document or folder, etc.
- **Save the shared notebooks and documents to your own drive.** The notebooks you will have access to are read-only; in order to work on them you will have to copy them to your drive. It is recommended that you maintain a per-week structure in your folders as this will make it easier to follow instructions, especially when reading files or collaborating with others. To save the notebook to your Drive, go to the File menu: `File > Save a copy in Drive`.
- **(When creating a new notebook) Change Colab Notebook's runtime type to R.** By default, when you create a new notebook in Colab, the virtual machine Colab sets up for you is a `Python` installation. This means all the cells in the notebook will only recognize and run `Python` syntax or commands. To switch to an `R` language setup:
  1. Open the notebook menu and go to `Runtime > Change runtime type`.
     - Or click the toggle in the upper-right corner (as shown below) and select `Change runtime type`
  2. In the dropdown labeled `Runtime type`, select `R`.
  3. Click `Save`.



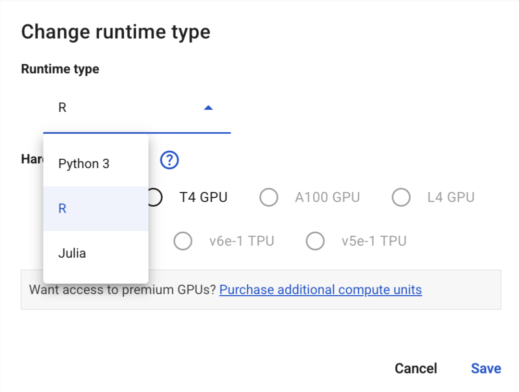


Fig.Changing the runtime type

You may now create your first Colab notebook

L.1.1

Coding Lab 1— Numeric data structures in R

Instructions: Work individually. Solve all exercises in the corresponding Colab notebook. Record your answers and/or code in your copy of the notebook.

Notebook: Coding Lab 1

Submit your work in the format required by the instructor.

Instructor Notebook: Answers to L.1.1

Find the answers to all exercises in this notebook.

L.1.2

Coding Lab 2 — Data manipulation with `dplyr`

Instructions: Work individually. Solve all exercises (sections 0 to 3) in the corresponding Colab notebook; record your answers and/or code in your own copy.

Notebook: Coding Lab 2

Optional **Section 4** allows work in pairs or teams of 3.

Upon completion, submit your work in the format required by the instructor.

Instructor Notebook: Answers to L.1.2

Find the answers to all exercises in this notebook.

14.310x Flipped Classroom
Week 2 Answers

Discrete random variables, joint and conditional distributions

---

Checklist

- ☐ Complete the ADVANCED R interactive course from the `Swirl` package and watch the `ggplot` tutorial[7] (Requirement)
- ☐ Watch the **Import Data** tutorial[8] (Requirement)
- ☐ Read the note on importing `aiwars` and other datasets (Requirement)
- ☐ Complete Guided Case 1 (Session 1)
- ☐ Complete Coding Lab 3 (Session 2)

Importing `aiwars` and other datasets

Depending on flipped classroom logistics of your group, your access to the sessions' assets (datasets, figures, scripts, etc.) will come in one of two forms:

- **Through a direct URL** for instance, the original `aiwars.csv` dataset lives directly in this URL:
`https://docs.google.com/spreadsheets/d/1NeZZWI2fT71M9QD8zjnz817T1B3XBM6lolqArSe9CwQ/export?format=csv`
- **Your instructor will provide them to you privately** and they will either:
  - Share them via a URL similar to the one above.
  - Share the files through other means.

[7]**Module 2 > R Course and R Tutorial: ggplot** in the online component of the course.
[8]**Module 3 > R Tutorials: Basic Functions** in the online component of the course.

As they may want to slightly modify the original files for grading purposes, or have any other goal in mind.

If a dataset's URL is provided, you can read the data directly into R with the URL and the appropriate reading function; simply provide the URL as the path. For instance, `aiwars` is in `.csv` format:

```
URL_aiwars <- "https://docs.google.com/spreadsheets/d...."

aiwars <- read.csv(URL_aiwars)
```

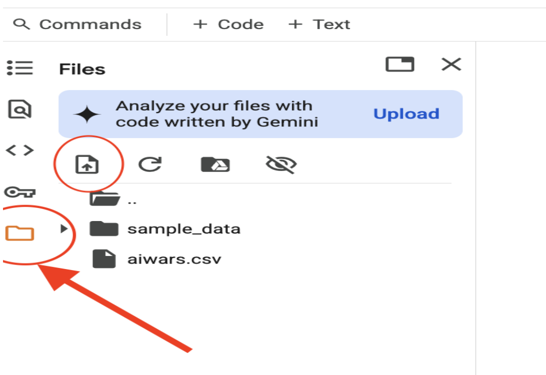
And similarly for other formats. Suppose we had a single Excel sheet:

```
install.packages("readxl")
library(readxl)

URL_aiwars_xl <- "https://...some-URL"

aiwars <- read_excel(URL_aiwars_xl)
```

If the file is shared in any other way, we will first need to upload it to the virtual machine's disk in Colab and read it from there, providing the `path` to it — as you would do if you were reading data to R in your own computer. To upload a file in Colab, click the folder icon in the leftmost menu bar, as shown in the screenshot. Then click the upload icon (circled in red); as you can see, we already uploaded the file.



Once uploaded the file can be read as a if in your machine:

```
aiwars <- read.csv("aiwars.csv")
```

Important: Colab runtime limitations

While your Colab notebook — including all code cells, text, and outputs — will remain saved, the underlying *runtime* (i.e., the virtual machine that executes your code) is temporary. After a period of inactivity or when a time limit is reached, the runtime will automatically disconnect.

When this happens:

- All variables and objects stored in memory (data frames, vectors, functions) will be lost.
- Any files you uploaded manually will be erased.

When reconnecting, a fresh virtual machine will be started, and you'll have to re-run your code, and re-upload any needed files. Disconnects may happen occasionally as you work on off-notebook tasks; re-reading or re-uploading files should not take more than 30 seconds.

Coding exercises in case studies

Throughout the course, the case studies you will work on contain a mix of conceptual and coding questions. To simplify your workflow, all the questions where you're required to write code are specially labeled. They appear in this format:

Question 5. [2.3] calculate the probability of event A...

The small icon signals that the question requires coding. The number inside the brackets (2.3 in this example) is a numbering **within the Session's dedicated Colab notebook**, which will often be different from the overall question number for the activity.

Further, if you click on the orange icon, it will take you directly to the question's cell within the Notebook by opening a new browser tab – remember you must work on your own copy of the notebook. These links are simply provided as a convenience to help you locate and visualize the relevant task quickly.

Apart from being the place where you are expected to code your answers, the corresponding cell usually includes a placeholder for your code, often accompanied by additional hints, extended context, or partially completed code to help you get started. When working on a case study, keep both this PDF and Colab notebook open and use these references to move smoothly between the written materials and your code.

Case study G.2.1

Reddit posts: PMFs and PDFs

Instructions: Work in pairs or groups of three. Answer the questions

Notebook: Case Study 1

Dataset: aiwars.csv

One of you must share editor access to their notebook with the rest.

Instructor Notebook: Answers to G.2.1

Find answers to all code-related questions in this Guided Case. Consider that each coding-relevant question has a link to its respective answer section within this notebook.

Scenario: We will cover this week's lecture contents using a real-world dataset from Reddit. You will put your data manipulation (with dplyr and base R) skills to practice

---

Context: The `AIwars` dataset consists of a collection of posts scraped from the `r/AIwars` subreddit, a forum where users debate the societal implications of AI. These range from predictions about AI-driven job loss and technological conflict to satire, trolling, and speculation. The dataset captures a rich period of discussion and polarization. Each observation corresponds to a single post, with information about its author, contents and engagement.

Key Variables:

- `post_index` — Unique identifier for each post (starts at 1 and consecutively numerates all posts).
- `author` — Reddit username of the post author (may be `[deleted]`).
- `post_date` — an R `Date`
- `fulltext` — Full text of the Reddit post in format

  `TITLE: [some post title] TEXT: [some post text]`
- `post_length` — the net number of characters in `fulltext` (excluding the TITLE and TEXT headers).
- `post_upvotes` — Number of upvotes (user endorsements or *likes*) the post received.
- `comments_number` — Total comments the post received (includes replies to other comments).

Part 1. Basic dataset facts

1.1 [1.1] Install and load the `tidyverse` packages. Create `aiwars_URL`, a `character` with the download URL for `aiwars.csv`. Then read in the dataset as `aiwars`, a data frame. Use `glimpse()` to answer:

(a) How many posts are we working with?
(b) How many variables does the dataset have?
(c) How many are *true* `character` types?
(d) How many are numeric?
(e) Which characters are better described as factors? Why?

Code in Colab [1.1] **Answer**

(a) How many posts are we working with? 9724

(b) How many variables does the dataset have? 12

(c) How many are *true* `character` types? One. The `fulltext` variable.

(d) How many are numeric? Three: `post_length`,`post_upvotes` and `comments_number`

(e) Which characters are better described as factors? Why?

`author` and `post_date` are variables accounting for a repeating qualitative feature, which can be used beyond a `character` type – that said, `post_date` may fit its own `Date` type.

`post_index` may be considered a factor, but its values do not repeat.

1.2 [1.1.1] For the entire case study, will use only the **Key Variables** described above. While `select()`ing columns from the current data frame is possible, a memory-efficient alternative is to read in only those columns we need.

Use the `col_select` argument in `read_csv()` to create `aiwars_short`, a data frame containing only the columns described as **Key Variables**.

Code in Colab [1.1.1] **Answer**

1.3 [1.1.2] Use `select()` to create `aiwars_short2`, a dataframe with only the **key variables**.

Code in Colab [1.1.2] **Answer**

1.4 Moving forward we will work with `aiwars_short` only. You may delete `rm(aiwars, aiwars_short2)`.

Instructor note: this is done mostly to avoid name confusion when creating future objects.

1.5 Use `summary()` or other summarizing methods to answer the following questions:

(a) What is the time span of the posts?

From December 30th 2022 to December 29th 2024

(b) On average, how long (in words) is a post on this subreddit?

Approximately 128 — a little more than a paragraph. The mean of `post_length` is 639 characters. Students can quickly search for the average word length in the English language; between 4 and 6 characters. Such that: $\frac{639}{5} \approx 128$.

(c) How many posts don't have any replies?

At most 832. From the `summary` output, `comments_number` has 832 NAs and no zeros. *Note: current versions of the post may now have comments; this value depends upon the moment the forum contents were extracted.*

(d) What is the largest number of upvotes in a post?

400

(e) Which author has posted the most? (Consider `as.factor()`).

User *Wiskkey*. Summary has different outputs for `factor` and `character`. For factors, R returns a count of the levels in descending order.
It is also possible to obtain this count by summarizing in dplyr style:

Code in Colab ✔ >_ [N/A] **Answer**

(f) How many distinct users have posted in this subreddit? (Consider `unique()`).

2982.

`length(unique(aiwars_short$author))` returns 2983, then discount the [deleted] tag.

Part 2. Counting posts

2.1 >_ [2.1] Create the following variables in `aiwars_short`:

Code in Colab ✔ >_ [2.1] **Answer**

Verify variables:

- `mean(popular)`: 0.1032
- Count for *short* `text_classification`: 1416

Instructor note: Remind students `text_classification` is a character; coerce to factor to get nicely printed `summary` output.

(a) `popular`, takes value 1 when the post has 29 or more upvotes. Takes value 0 when the post has less than 29 likes.

(b) `text_classification`, which takes value `mostly title` if it is less than 70 characters long, `short` if it has 70 or more but less than 110 characters, `common` if it has 110 or more but less than 900 characters, and `long` if the post has over 900 characters.

2.2 How many `common` posts are there?

3,497

2.3 If we draw a post at random, it will be a popular one with what probability?

0.1032

`sum(popular)`: 1,004

$P(popular = 1) = \frac{1004}{9724} = 0.1032$

Instructor note: If suitable, review how the mean of a dummy variable is also the proportion of 1's in it.

2.4 If we instead sample 10 posts at random **with replacement**, what is the probability 3 of them are popular?

0.0615

There is replacement, which implies $P(popular = 1)$ doesn't change across draws. All 10 draws are independent (Bernoulli trials with success probability 0.1032). The number of popular posts pop is a binomial random variable:

$$P(pop = 3|N = 10) = \binom{10}{3}(0.1032)^3(1 - 0.1032)^7 = 0.0615$$

Students can use R's `dbinom`:

```
dbinom(3,10, prob = 0.1032)
```

2.5 [2.2] Suppose random variable X; $\Omega_X = \{0, 1, 2, ..., N\}$ represents the number of popular posts in $N = 10$ draws with replacement from `aiwars_short`. For the following probability statements describing various events, first write them down as a probability and then use R to compute them.

Code in Colab [2.2] **Answer**

(a) What is the probability of getting at most 4 popular posts?

0.9981

$$P(X \leq 4) = \sum_{x=0}^{4}\binom{10}{x}(0.1032)^x(1 - 0.1032)^{10-x} = 0.9981$$

(b) What is the probabililty of getting at least one but at most 3?

0.6494

$$P(1 \leq X \leq 3) = \sum_{x=1}^{4}\binom{10}{x}(0.1032)^x(1 - 0.1032)^{10-x} = 0.6494$$

Instructor note: Avoid computing each point mass `dbinom(1, N, p) + dbinom(2, N, p) + dbinom(3, N, p)`, make emphasis on the fact that

$$P(1 \leq X \leq 3) = P(X \leq 3) - P(X = 0)$$

To get attempts similar to the solution code in Colab

(c) What is the probability of getting a number of popular posts that is not 0 or 4?

> 0.6513
>
> **Instructor note:** Here the intent is for students to work with properties of probability to reuse their answers to 2.5a and 2.5b.
>
> $$P(1 \leq X \leq 3 \cup X > 4) = P(1 \leq X \leq 3) + (1 - P(X \leq 4))$$
>
> $$\approx 0.6494 + (1 - 0.9981) = 0.6513$$
>
> Students may compute the answer directly too, but it is definitely more impractical, especially in R:
>
> ```
> (pbinom(3, 10, p) - pbinom(0, 10 p)) + (pbinom(10,10, p )-
> pbinom(4,10,p))
> ```

2.6 Similarly, we can treat the texts' classification as random variable Y. Is it continuous or discrete? What is Ω_Y?

> A **discrete** random variable with sample space equal to its possible values:
>
> $$\Omega_Y = \{\text{mostly title}, \text{short}, \text{common}, \text{long}\}$$
>
> With pmf equal to the probability of randomly drawing a post of each classification:
>
> $$f_Y(y) = \begin{cases} \frac{2700}{9724} & y = \text{mostly title} \\ \frac{1416}{9724} & y = \text{short} \\ \frac{3497}{9724} & y = \text{common} \\ \frac{2111}{9724} & y = \text{long} \end{cases}$$

2.7 [2.3] Express $f_Y(y)$ as a data frame `f_Y` with as many rows as elements in Ω_Y and two columns: `y`, the value taken by Y and `p` its mass or density.

Code in Colab [2.3] **Answer**

A tibble: 4 × 2

y	p
<fct>	<dbl>
common	0.3596257
long	0.2170917
mostly title	0.2776635
short	0.1456191

2.8 What is the probability a post is not `mostly title` ?

0.7223

This is the probability of Y being rest of classifications:

$$P(Y \neq \text{mostly title}) = P(\ Y \in \{\text{short}, \text{common}, \text{long}\})$$

$$\ldots = f_Y(\text{short}) + f_Y(\text{common}) + f_Y(\text{long}) = 0.1456 + 0.3596 + 0.2170$$

Code in Colab [N/A] **Answer**

2.9 What is the probability a post is either `common` or `long`?

0.5767

$P(Y \in \{\text{common}, \text{long}\}) = 0.3596 + 0.2170$

Or $P(Y \neq \text{mostly title}) - P(Y = \text{short})$, etc.

Are popular posts different?

2.10 What is the probability a post is popular **AND** `long`?

$\frac{224}{9724} \approx 0.0230$

Code in Colab [N/A] **Answer**

Directly count the instances with `filter()` and `nrow()`. Also possible to use group operations– see question 2.14

As we learned from the lecture, if two events are **not independent**, the additional information one of them provides should *update* the probabiility of the other. We will examine this fact by calculating the probability of a post being long **GIVEN that we know it is popular**:

$$P(Y = long \mid popular = 1)$$

As a reminder:

$$P(long \mid popular = 1) = \frac{P(long\ \cap\ popular = 1)}{P(popular = 1)}$$

2.11 Use your answers to questions 2.3 and 2.10 to compute the probability a post is long GIVEN it is popular. How does this probability compare to the unconditional probability $P(Y = long)$?

Code in Colab [N/A] **Answer**

$$\frac{0.0230}{0.1032} \approx 0.2232$$

$P(Y = \text{long}) = 0.2171$. Conditional on a post being popular, it does not change the post's probability of being long by much —a change of less than 2%.

2.12 We now know that the event ($Y = long$) has two mutually exclusive and exhaustive partitions: ($popular = 1$) and ($popular = 0$). Following the lecture, Show that the Law of Total Probability verifies for this event.

The law of total probability:

$$P(\text{long}) = P(\text{long} \mid \text{popular} = 1) \cdot P(\text{popular} = 1) + P(\text{long} \mid \text{popular} = 0) \cdot P(\text{popular} = 0)$$

From question 2.10, we have 224 popular and long posts, and from 2.1, there are 2111 long posts. The remaining 1887 long posts are not popular:

$$P(\text{long} \cap \text{popular} = 0) = \frac{1887}{9724} = 0.1941$$

And the conditional probability:

$$P(\text{long}|\text{popular} = 0) = \frac{0.1941}{(1 - 0.1032)} = 0.2164$$

Computing the law of total probability:

$$(0.2232) * (0.1032) + (0.2164) * (1 - 0.1032) = 0.2171$$

Verifying that

$$f_Y(\text{long}) = \frac{2111}{9724} = 0.2171$$

2.13 What is $P(popular = 1 \mid long)$? Is your answer the same as in 2.11 ? Should it be? Why?

$$P(\text{popular} = 1 \mid long) = \frac{P(popular = 1 \cap long)}{P(\text{long})}$$

$$\ldots = \frac{0.0230}{0.2171} = 0.1059$$

Not the same: $P(popular = 1) = 0.1032$

Instructor note: Students should be able to both compute the answer in R, as in 2.11, and find it without new computations, by the definition of conditional probability.

2.14 [2.4] Create `joint_pmf`, a data frame similar to the one you created for 2.7; in this case, it should display the probability for all possible combinations of $Y \cap Popular$. The sample consists of the ordered pair $\Omega_Y \times \Omega_X = (long, 1), (long, 0), ..., (\text{mostly title}, 1), (\text{mostly title}, 0)$

Code in Colab [2.4] **Answer**

A grouped_df: 8 × 3

popular	text_classification	p
<dbl>	<fct>	<dbl>
0	common	0.32640889
0	long	0.19405594
0	mostly title	0.24640066
0	short	0.12988482
1	common	0.03321678
1	long	0.02303579
1	mostly title	0.03126285
1	short	0.01573427

Instructor note: *Video lectures and materials during Week 3 will show joint PMF/PDFs differently, as a matrix with X represented in the rows and Y represented in the columns. Such a representation will often be more convenient to compute marginal and conditional PDF/PMFs as sums of rows or columns — covered in Coding Lab 3.*

2.15 [2.4.1] Use `joint_pmf`, $f_X(x)$ and $f_Y(x)$ to compute multiple conditional probabilities at once in R:

Code in Colab [2.4.1] **Answer**

Instructor Note: The intent is to generalize into conditional PMFs (and by analogy, to PDFs) without mentioning them explicitly, by asking students to compute several conditional probabilities at once as vectors.

(a) $P(Y \mid Popular = 1)$, a vector of length 4. Check that the vector adds up to 1.

popular	text_classification	p
<dbl>	<fct>	<dbl>
1	common	0.3217131
1	long	0.2231076
1	mostly title	0.3027888
1	short	0.1523904

These are the values of the last 4 rows of `joint_pmf` divided by $P(Popular = 1) = 0.1032$

(b) Get $P(Popular \mid Y = common)$. What is its length? Check it adds up to 1.

popular	text_classification	p
<dbl>	<fct>	<dbl>
0	common	0.90763512
1	common	0.09236488

Subset the *common* rows of `joint_pmf` and divide by $f_Y(common)$:

2.16 Generally, considering your answers to 2.15a, 2.15b or 2.11, are the popularity of a post and its length classification independent? Why?

L.2.3

Coding Lab — Row and Column operations

Instructions: Work individually. Solve all exercises in the corresponding Colab notebook. Record your answers and/or code in your copy of the notebook, or in the format required by the instructor.

Notebook: Coding Lab 3

Instructor Notebook: Answers to L.2.3

Find the answers to all exercises in this notebook.

14.310x Flipped Classroom

Week 3 Answers

Continuous random variables; conditional and joint pdfs.
More advanced data manipulation and visualization in R

Checklist

- ☐ Complete Guided Case Study 2 (Sessions 1 & 2)
  - ☐ We will continue to work with `aiwars`
  - ☐ Get the `aiwars_embeddings` dataset here
- ☐ Complete Open Case Study 1 (Session 2)

Case study G.3.2

Analyzing text similarity:
applying row operations to Natural Language Processing

Instructions: Work on your own for Session 1, and in pairs or groups of 3 for Session 2.

Notebook: Case Study 2

Dataset: `aiwars_embeddings.csv`

Part 1 Part 2 Part 3

Dataset: `speech_anchors.csv`

Instructor Notebook: Answers to G.3.2

Find answers to all code-related questions in this Guided Case. Each coding-relevant question links to its corresponding answer section in this notebook.

The `aiwars_embeddings` dataset (we split it in 3) contains 1024 variables associated with each post's `fulltext` in `aiwars`. You will learn what these embedding variables `V1`, `V2`, ..., `V1024` represent below, but keep in mind `post_index` in both datasets uniquely links each post in `aiwars` to its embedding in `aiwars_embeddings`. Dataset `speech_anchors` is a 2-row data frame with the same structure but `label`s instead of post indices.

Part 0. Required data and packages

0.1 [0.1] Load/install the `tidyverse` packages.

Code in Colab [0.1] **Answer**

0.2 [0.2] Read in `aiwars`, the same dataset as last week. Also, read in `speech_anchors.csv` as `anchors`, which should be a data frame with two rows and 1025 columns.

Code in Colab [0.2] **Answer**

0.3 [0.3] Read in parts 1,2 and 3 of `aiwars_embeddings` as `part1`, `part2` and `part3` respectively. Then use `rbind()` to vertically merge the three parts into `aiwars_embeddings`. Use `arrange()` to sort the dataset in `post_index`-ascending order. **Reading takes over a minute.*

Code in Colab [0.3] **Answer**

> **Instructor Note**: Arranging intends to avoid mistakes in case the student binds the chunks in an order different from
>
> `rbind(part1,part2, part3)`
>
> Most exercises in this session rely on the ascending order of both `aiwars` and `aiwars_embeddings`.

0.4 Remove the individual chunks: `rm(part1,part2, part3)`.

Scenario:

Natural Language Processing (NLP) focuses on enabling machines to understand and work with human language. One of the key advances in NLP has been the development of methods that represent text as vectors—documents, social media posts, chatbot requests, etc. These vector representations, often referred to as *embeddings*, capture aspects of a text's semantic content: its meaning, tone, and topical focus.

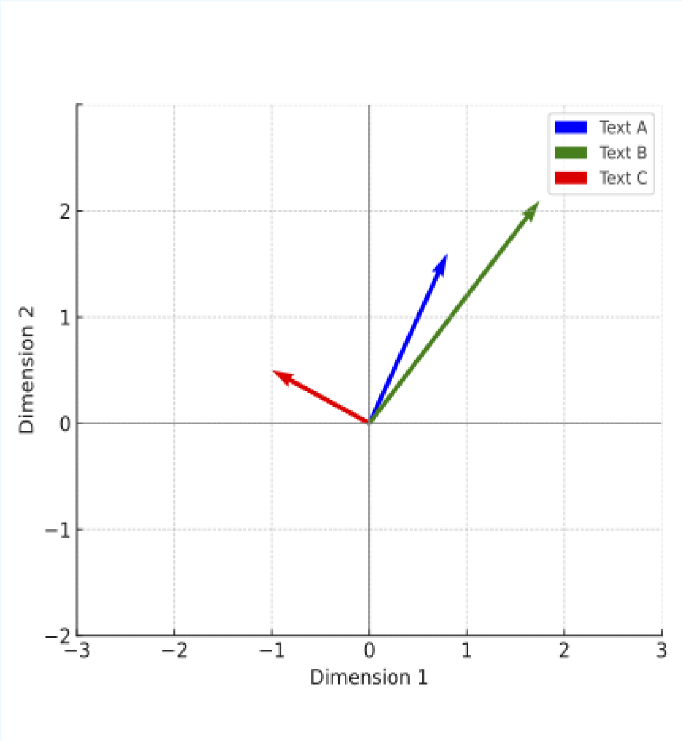
To illustrate the idea, suppose we have the following texts:

- Text A: "I love how these new AI tools can generate art!"
- Text B: "AI-generated images are cool, but I'm worried about copyright."
- Text C: "I painted this myself — no software involved."

We embed these texts in 2-dimensional[0] vectors from the origin, plotted below:

- $\vec{a} = \begin{pmatrix} x_a \\ y_a \end{pmatrix} = \begin{pmatrix} 0.8 \\ 1.6 \end{pmatrix}$
- $\vec{b} = \begin{pmatrix} x_b \\ y_b \end{pmatrix} = \begin{pmatrix} 1.7 \\ 2 \end{pmatrix}$
- $\vec{c} = \begin{pmatrix} -1 \\ 0.3 \end{pmatrix}$

Figure 2: **2D text embeddings (vector from the origin)**



Despite their differences in tone, $\vec{a}$ and $\vec{b}$ in Figure 2 are close to each other because the texts they represent use AI-related language, whereas $\vec{c}$ is far removed given the lexical differences. In part 2, we will explain what "close to each other" means for vectors, with two common semantic similarity measures: **direction** (i.e., the *angle* they form) and **distance** between them.

Part 1. Coding vector operations

You will start by programming functions necessary for this analysis, such as the *magnitude* of a vector, the *distance* between two vectors, their *dot product*, etc.

Distance between two vectors

The distance from $\vec{a}$ to $\vec{b}$ in Figure 2 is defined as:

$$|| \vec{a}, \vec{b} ||= \sqrt{(x_a - x_b)^2 + (y_a - y_b)^2}$$

This is the sum of the squared differences in x and y-coordinates (in any order). Then we take the square root of that sum. Substituting the values in the scenario description:

$$\cdots = \sqrt{(0.8 - 1.7)^2 + (1.6 - 2)^2}$$

$$\cdots = \sqrt{0.81 + 0.16} = 0.985$$

1.1 [1.1] Save $\vec{a}, \vec{b}, \vec{c}$ as numeric vectors `a`, `b` and `c` respectively. Compute the distances for each pair of vectors and save them accordingly: `dist_ab`, `dist_ac`, `dist_bc`. Hint: you can apply operations on all the coordinates at once: `(a-b)`.
   - Make sure your result for `dist_ab` is the same as above.

Code in Colab [1.1] **Answer**

$$||\vec{a}, \vec{b}|| = 0.9848858$$
$$||\vec{a}, \vec{c}|| = 2.22036$$
$$||\vec{b}, \vec{c}|| = 3.190611$$

In general, to get the distance between any two n-dimensional vectors

$$\vec{V} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}, \vec{U} = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}$$

we generalize the process with all their coordinates and take the square root:

$$|| \vec{V}, \vec{U} ||= \sqrt{(v_1 - u_1)^2 + (v_2 - u_2)^2 + \cdots + (v_{n-1} - u_{n-1})^2 + (v_n - u_n)^2}$$

$$\cdots = \sqrt{\sum_{i=1}^{n} (v_i - u_i)^2}$$

1.2 [1.2] `aiwars_embeddings` contains a 1024-dimensional vector per post. Program a distance function `dist_function(V, U)` that takes any two equal-length, numerical vectors. It returns the distance between them.

- Verify your function is correct and reproduce the results in 1.1
- From `aiwars_embeddings`, create vectos `p1` and `p2`, the embeddings for post indices 1 and 2, respectively (don't forget to exclude the `post_index` and coerce with `as.numeric`). Check that $||\vec{p}_1, \vec{p}_2|| = 0.96028211$

Code in Colab [1.2] **Answer**

https://colab.research.google.com/drive/1SlbXk1n121ks084Z-Kf_p4jTnm-2PD3t?usp=sharing#scrollTo=65010882

Distance from one to multiple vectors

Working with individual, separate objects such as `dist_ab`, `dist_ac`, and so on can be difficult and tedious. In R, we will often want all the distances with respect to $\vec{a}$ in a single "distances" vector or variable:

$$D_a = \begin{pmatrix} ||\ \vec{a}, \vec{a}\ || \\ ||\ \vec{a}, \vec{b}\ || \\ ||\ \vec{a}, \vec{c}\ || \end{pmatrix}$$

We will now pass `dist_function()` through all the example vectors at a time to obtain D_a — the steps should be familiar from the previous Coding Lab.

1.3 [1.2.1] Create vector `D_a` with a *for* loop by following the steps below:

- `rbind` vectors `a`, `b`, and `c` to create `examples_matrix`, a matrix with 3 rows and 2 columns: one row for each example vector, and one column for each x, y-coordinate.
- Declare `D_a`, an empty numeric vector of length 3.
- Declare a `for(i in 1:nrow(examples_matrix))...` loop to iterate over `examples_matrix`.
- Inside the *for* loop, code the following:
  - Subset/slice the rows of `examples_matrix` to obtain vector `v` on each iteration.
  - Calculate the `distance` between `a` and `v` with `dist_function`.
  - Assign the `distance` result to `D_a[i]` on each iteration.

Code in Colab [1.2.1] **Answer**

$$D_a = \begin{pmatrix} 0 \\ 0.984886 \\ 2.220360 \end{pmatrix}$$

1.4 Verify your answers with those of 1.2.

What should be the result for $\|\vec{a}, \vec{v}\|$ when $\vec{v} = \vec{a}$? Why?

Code in Colab [N/A] **Answer**

> The result of the first element of D_a is zero, since there is no distance between $\vec{a}$ and itself:
>
> $$||\vec{a}, \vec{a}|| = \sqrt{(a_1 - a_1)^2 + (a_2 - a_2)^2 + \ldots + (a_n - a_n)^2} = 0$$

Finally, we can wrap the code we created for 1.3 in a function that directly returns D_a when given a vector and a matrix.

1.5 [1.2.2] Create a function `batch_distances(u, M)` which takes a vector `u` of length n and a matrix `M` with an arbitrary number of rows k, and n columns. The function should `return` D_u, a vector of length k where each element is the distance between the input `vector` and a row of the input `matrix`.

Code in Colab [1.2.2] **Answer**

1.6 Use `batch_distances(_,_)` to reproduce 1.3. Verify you get the same vectors D_a and D_p

Code in Colab [N/A] **Answer**

The norm of a vector

The norm $||\ \vec{V}\ ||$ (also called the *magnitude* or *length*) is the distance from where it starts to where it ends. For instance, in the plot above, the norms are the lengths of each arrow. Since all embeddings are vectors from the origin (all their starting coordinates are equal to 0), their norms can be expressed as: $||\vec{V}, \vec{0}||$, with $\vec{0}$ the zero vector of the same dimension as $\vec{V}$.

$$||\ \vec{V}, \vec{0}\ || = ||\ \vec{V}\ || = \sqrt{(v_1 - 0)^2 + \cdots + (v_n - 0)^2}$$

$$\ldots = \sqrt{\sum_{i=1}^{n} v_i^2}$$

1.7 [1.3] Create `norm0_function`, a function that takes `V`, a numerical vector of length n as an input, and returns its norm from the origin. Hint: you can recycle `dist_function` inside `return()` with the appropriate `U`.
   - Verify that $||\vec{a}|| = 1.788854$ and $||\vec{p}_1|| = 1$

Code in Colab [1.3] **Answer**

> Previously programmed `dist_function()` can be recycled using a zeros vector of the same length as input `v`; create with `rep(0, length(v))` or similar.
>
> It is also possible to directly program from the expression above with a one-liner: `sqrt(sum(v2))`
>
> Whenever possible, highlight how a custom function can be reused inside another custom function.

1.8 [1.3.1] Compute $\| \vec{a} \|, \| \vec{b} \|$ and $\| \vec{c} \|$. Save them to `a_norm`, `b_norm`, `c_norm` respectively.

Code in Colab [1.3.1] **Answer**

$$||\vec{a}|| = 1.788854$$
$$||\vec{b}|| = 2.624881$$
$$||\vec{c}|| = 1.044031$$

1.9 Similar to 1.5, create a function `batch_norms0(M)` taking a matrix of n vectors, all with the same number of dimensions/columns m, and returns a vector N_0 of length n with a norm from the origin for each row in `M`.Verify your results in 1.8 with `examples_matrix`.

Code in Colab [N/A] **Answer**

Dot product

The dot or *scalar* product of two vectors $\vec{U} \cdot \vec{V}$ is defined as the sum of the product of each pair of their coordinates:

$$\begin{aligned}\vec{U} \cdot \vec{V} &= u_1 * v_1 + u_2 * v_2 + \cdots + u_n * v_n \\ &= \sum_{i=1}^{n} u_i v_i\end{aligned}$$

R does element-by-element multiplication by default: `sum(u*v)`.

1.10 [1.4] Create a function `dot_product(u,v)` that takes two numeric vectors of the same length as input and returns their dot or scalar product. Check that $\vec{p}_1 \cdot \vec{p}2 = 0.5389$

Code in Colab [1.4] **Answer**

1.11 [1.4.1] Again, we're interested in computing a vector $\mathbb{D}_a$ of dot products. Use a *for*-loop and wrap it in a function `batch_dot_product(v, M)` taking a vector `v` of length n and a matrix `M` of as many rows as vectors and n columns, and returns a dot product $\vec{v} \cdot \vec{m}$ for each row. Verify that you get $P_1 = \begin{pmatrix} 1 \\ 0.5389 \end{pmatrix}$

Code in Colab [1.4.1] **Answer**

Part 2. Processing text embeddings for AIwars

We prepared the `aiwars_embeddings` dataset for you by using one of OpenAI's text embedding products[1]. For each post's `fulltext` in `aiwars`, there is a vector representation of length $n = 1024$ in `aiwars_embeddings`:

Variables

- `post_index`: a consecutive index **in the same order as** `aiwars`. A given post and its embedding share the same post index.
- `V1`, `V2`, ..., `V1024`: the coordinates $v_1, v_2, ..., v_{1024}$ of each vector embedding $\vec{V}$.

You will now apply the functions you previously programmed to measure the semantic similarity of real-world embeddings.

Semantic similarity measures: *Luddites* vs *Synthists*

To illustrate semantic similarity measures, we will take the embeddings from four select posts from the AIwars subreddit, each with distinct tones and topics:

2.1 [2.1] Filter `aiwars` for posts with the following `post_index` values: $456, 584, 2397, 2625$. Select the `fulltext` and post index columns only. Assign the resulting dataframe to `tones`. Just generally, what is each post about?

Code in Colab [2.1] **Answer**

> Find post contents in Colab. Posts 456 and 584 mention and mockingly address *luddites* (see 2.3). Post 2397 is an opinionated but argumentative critique on a paper and various previous comments on said paper. Post 2526 is a humorous proposal to denominate certain views or people as *synthists*— more on this in 2.3.
>
> ---
>
> **Instructor notes:**
>
> i Question 2.2 suggests qualitative labeling of the posts that relies on the index-ascending order proposed in this question; while not essential, it is advised that students follow both the post index order and labeling for questions in Part 2. The labels are used to simplify the discussion and make it easier for students to remember the post contents.
>
> ii Because this is a suitable example, logical set operation `%in%` is suggested in the student code sketch. If students are not familiar, this may be an appropriate moment to introduce simple usage: `variable %in% c(x1, x2, x3, ..)` returns `TRUE` for the elements of `variable` matching any of the elements `x1, x2, ...` and `FALSE` if no match is found for that element.

2.2 To keep track of the post contents, give each a label: `post_labels <- c("luddites", "mockery", "critique", "synthists")` and add the vector as a new variable `label`. Remember, `tones` should be in `post_index` ascending order for the labels to match.

2.3 Answer the following questions regarding `tones`:

(a) Historically, what does the term *Luddite* refer to? (you may want to do a quick internet search)

(b) With posts 456 and 584 in mind, provide a short description of a Luddite (Yuddite) in the context of the AIwars subreddit.

(c) Per post 2625, what is a *Synthist*?

(d) In your opinion, which post in `tones` is most similar to post 584 in its intent to mockingly denominate other points of view and/or users?

(e) Is any of the `tones` posts more nuanced or academic in tone? (i.e., less opinionated, more descriptive).

(f) What post would be most similar to 2625 in its descriptiveness?

(a) The Luddites were members of a 19th-century movement of English textile workers who opposed the use of certain types of automated machinery due to concerns relating to worker pay and output quality. They often destroyed the machines in organized raids.

(b) The term is used in a humorous, mocking way to refer to skeptical, reserved, or opposing views (and their supporters) on AI-related research, work integration, the use of AI for image or sound creation, etc. It draws an analogy with the original Luddite movement, admonishing the possibility of an organized movement to "burn" AI as Luddites burned or destroyed machinery. *Yuddite* is a deformation alluding to Eliezer Yudkowsky, a proponent of the inevitable extinction of the human race due to AI development.

(c) Judging by the author's expressions ("promptist", "A Synthist who makes Synth Art", "...were not actually human"), the post purports to coin the term to refer to individuals who apply AI tools without judgment or cautiousness, or AI-optimistic opinions failing to consider the impact of AI in human activities and behavior.

(d) Posts 2526 or 445 are possible answers. Post 2526 is semantically more alike, given the usage of "name" in the sentence.

(e) 2397, a reasoned critique; while not free of opinionated and ironic sentences —and even insults— it offers an argumentation around the paper's claims it addresses and other comments on it.

(f) None or 2625 are possible answers. The tone and contents of 2397 are not clearly related to any of the other 3 posts.

2.4 [2.2] Retrieve the embeddings for `tones` in a 4×1024 matrix and save it as `tones_embeddings`:

– Filter `aiwars_embeddings` for the relevant `post_index`.

– Either drop `post_index` or tidy-`select` all variables `starts_with()` "V".

  – Remember to keep track of the embeddings. We suggest you previously arrange by ascending post index.

– Coerce the resulting data frame into a matrix with `as.matrix()`

.

Code in Colab [2.2] **Answer**

A matrix: 4 × 1024 of type dbl

V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	⋯	V1015	V1016	
0.02820631	0.03051831	-0.01078634	0.030500522	0.058511205	-0.022017285	-0.01628177	0.041758150	0.006242381	-0.037240870	⋯	0.022266270	-0.03179879	-0.0471
0.02614705	-0.01794884	-0.03656052	-0.006514959	-0.013858461	0.019134969	0.00016200	-0.005450936	0.017033087	-0.028746060	⋯	-0.021716531	0.01390207	-0.0087
-0.01002630	0.04195827	0.00234733	-0.022302737	0.021955290	0.001642097	0.02685263	0.038781613	-0.049271180	0.011606358	⋯	0.034248270	-0.04530037	-0.0444
0.02941498	0.02268429	-0.06235636	-0.001079750	-0.003755284	-0.025668170	0.01139300	0.015368690	0.048894983	-0.001348893	⋯	0.008370977	0.02015816	-0.0137

Distance-based semantic similarity

As exposed in the introduction, if two texts share style, vocabulary or topics, their embeddings will tend to be closer, and on the contrary, distinct semantics will produce embeddings that are farther apart.

2.5 [2.3]

Table 3: Pairwise distances between example embeddings (not the real values)

	luddites	mockery	critique	synthists
luddites	0.0000000	72.384921	18.532947	91.004327
mockery	72.384921	0.0000000	56.110239	8.231476
critique	18.532947	56.110239	0.0000000	44.873291
synthists	91.004327	8.231476	44.873291	0.0000000

Code `dist_similarity`, 2-D distance matrix like the one displayed above with the embeddings in `tones`. Hints:

- This is a 4x4 matrix with `rownames` and `colnames`
- The matrix you get should be symmetrical. The distance between *critique* and *luddites* should be the same in (row 1 column 3) and (row 3 column 1).

Code in Colab [2.3] **Answer**

	luddites	mockery	critique	synthists
luddites	0.0000000	1.071767	0.9498988	1.183828
mockery	1.0717666	0.000000	1.3090685	1.060917
critique	0.9498988	1.309068	0.0000000	1.216314
synthists	1.1838281	1.060917	1.2163139	0.000000

Each row or column of `dist_similarity` is the output of a single row with the whole embeddings matrix: `batch_distances(v,tones_embeddings)`.

2.6 Which post pair is the most dissimilar? Is this coherent –i.e., are their contents and tone quite different?

Mockery & critique. They are very different. Apart from different content and tone, text length probably accounts for the additional distance (larger texts typically generate relatively longer embedding vectors).

2.7 Which individual embedding seems to be further away from the rest?

We're looking for a simple metric such as total distance:
`rowSums(dist_similarity)`
Critique seems to be the furthest apart from the rest of the embeddings; however, the difference in total distance is not very substantial, particularly with respect to *synthists*.

Instructor note: All embeddings are constructed to have norm 1, which makes distance-based interpretations limited, particularly in terms of magnitude.

2.8 Which post pair is the most similar according to the metric? Does this align with your 2.3d answer ?

luddites & *critique*, likely because of the colorful language.

For the most part, the result does align with 2.3d: the next closest pair is *luddites* & *mockery*— note, *luddites* is most similar to *mockery*. Both posts menBoth posts mention Luddites and are of similar length.

Direction-based measures: Cosine Similarity

An alternative approach is to compare the angle θ formed by two vectors $\vec{a}, \vec{b}$. The cosine of said angle θ is defined as:

$$cos(\theta) = \frac{\vec{a} \cdot \vec{b}}{|| \vec{a} || \ || \vec{b} ||}$$

- $\mathbf{cos}(\theta) = \mathbf{1}$ means the angle is $\theta = 0°$, indicating identical direction. These texts will be very close in meaning and topics.
- $\mathbf{cos}(\theta) = \mathbf{0}$, the angle is $\theta = 90°$, indicating the posts are orthogonal, and for the most part, semantically unrelated: different topics and intent.
- $\mathbf{cos}(\theta) = \mathbf{-1}$, with angle $\theta = 180°$, vectors point in opposite directions.

*While possible, embeddings will rarely have a cosine similarity close to -1 because of how the OpenAI product is constructed.

2.9 (Optional) Calculate `tones_norms`, the norms for `tones_embeddings`. How does your result simplify $cos(\theta)$?

Code in Colab ✔ >_ [N/A] **Answer**

All embeddings have norm 1. This means the cosine of the angle between any two of these vectors gets simplified to their dot product:

$$\frac{\vec{a} \cdot \vec{b}}{|| \vec{a} || \ || \vec{b} ||} = \frac{\vec{a} \cdot \vec{b}}{1}$$

2.10 (Optional) In question 2.5, what is the maximum possible distance between vectors?

> 2
> Since all vectors are unitary. The maximum distance is betwee two exactly opposing vectors: $||\vec{v}, -\vec{v}||$

2.11 [2.4] Write a function `batch_cossim(u,M)` taking a vector of length n and a matrix with k row-vectors and n columns, returning the cosine of the angle θ between u and each row-vector. You may apply what you learned in question 2.9.

Code in Colab [2.4] **Answer**

> Students may use the fact that all vectors have unit norm, and in this case, the function is exactly the same as `batch_dot_product()` in 1.11.
>
> Outputs may differ slightly in each case due to the accumulation of small rounding errors (computed norms are not exactly 1, and they multiply).

2.12 [2.5] Similar to 2.5, create a cosine similarity 2-D matrix for the `tones` embeddings: `cos_similarity`. The diagonal elements must all be equal to 1.

Code in Colab [2.5] **Answer**

	luddites	mockery	critique	synthists
luddites	1.0000000	0.4256582	0.5488461	0.2992756
mockery	0.4256582	1.0000000	0.1431699	0.4372273
critique	0.5488461	0.1431699	1.0000000	0.2602903
synthists	0.2992756	0.4372273	0.2602903	1.0000000

2.13 Did the similarity order between posts change relative to the distance-based metric? Are your answers to 2.6, 2.7 or 2.8 the same with cosine-based similarity?

> The second part of 2.8 changes. Post *mockery* is now most similar to *synthists* (and not *luddites* as with distance). This similarity seems to better capture the mockingly tone in both posts.
>
> In general, the rest of the answers remains unchanged. The example's intent is to show how very distinct (dis)similarity will remain regardless of the metric. However subtler, closer similarity may be ordered in a differentiated manner.

AI-awareness and Simplicity measured in the *aiwars* posts

The `speech_anchors` dataset consists of two embeddings that we will use as benchmarks for all posts in `aiwars`:

- **AI awareness**. Represents how AI-related the content of a post is.
    - Greater similarity implies the post mostly discusses AI, and makes heavy

use of AI-related expressions and vocabulary.

- Smaller similarity implies the post's content is less related to AI and its semantics.

- **Simplicity**. Represents the argumentative style of the post.
  - Greater similarity implies posts are shorter, more slogan-like, and low-context statements.
  - Smaller similarity relates to longer, more evidence-seeking discussions. Posts will often be longer and more nuanced, regardless of their AI-relatedness.

In our setup, cosine similarities in the $[\mathbf{0.5}, \mathbf{0.7}]$ range usually indicate a *non-trivial* semantic or stylistic relation. Values above **0.8** often reflect very close texts (near-paraphrases). Similarities in the $[\mathbf{-0.3},\ \mathbf{0.3}]$ range are not indicative of a strong semantic link. And $\leq \mathbf{-0.3}$ cosines can occur but are very uncommon.

3.1 [3.1] From `speech_anchors`, create vectors `ai_aware` and `simple`, then compute the cosine similarities between all posts in `aiwars`, these 2 vectors: `ai_awareness` and `simplicity`, respectively. Add them as variables of `aiwars` — remember to previously arrange both datasets by ascending `post_index`. Save the new data frame as `aiwars_updated`.

Code in Colab [3.1] **Answer**

3.2 Verify your answers by replicating the following data frame.

post_index	ai_awareness	simplicity	label
<dbl>	<dbl>	<dbl>	<chr>
456	0.4775886	0.3168471	luddites
584	0.2886388	0.3751917	mockery
2397	0.5400782	0.2152752	critique
2625	0.4792042	0.3614475	synthists

Do cosine similarities `ai_awareness` and `simplicity` match the descriptions provided above when it comes to the `tones` posts?

> Note *critique* is substantially more AI-aware and less simple than the rest. In contrast, *mockery* is the least AI-aware and most simple post, both because of its length and its declarative style.

[0]These are usually very high-dimensional vectors with over 100 dimensions as in your dataset, but we will initially work with 2 to better illustrate the concepts.

[1]The coordinates for each embedding are outputs of a more complicated embedding model involving several neural network architectures and outside the scope of this course. For this week's case studies, we limit ourselves to the resulting embeddings.

Case study O. 3. 1

Continuous methods in RVs: Joint and conditional PDF/CDFs

Instructions: Work in pairs or groups of three; solve the following exercises collaboratively, and put together the deliverable specified in Part 2.

Notebook: Open Case Study 1

Dataset: `aiwars_updated.csv`

Instructor Notebook: Answers to G.3.1

Find answers to all code-related questions in this Guided Case. Each coding-relevant question links to its corresponding answer section in this notebook.

One of you must share the case study's notebook (parts 1 and 2) and a fresh slides document (part 3) with the rest.

Part 1. Discrete and continuous methods

0.1 [0.1] Read in `aiwars_updated`; load/install the `tidyverse` packages.

Code in Colab [0.1] **Answer**

Consider two continuous variables related to the semantics of text posted by redditors in r/Aiwars:

- **AI-awareness**: A, the extent to which a post's vocabulary and syntax depict artificial intelligence topics. An AI-related post will tend to score 1, while a post with no significant AI-related content will score closer to 0.
- **Simplicity**: S, a post's reliance on short, declarative phrasing and minimal argumentative or conceptual complexity. A short, opinionated post will show a score closer to 1, and a post showing a longer, more complex argument will observe a score closer to 0.

1.1 Based on the information provided, what are the sample spaces Ω_A, Ω_S?

$$\Omega_A = \Omega_S = [0, 1]$$

Both random variables are scores spanning from 0 to 1.
Instructor Note: If students completed the previous session on NLP, they will know that cosine similarity can generally go from -1 to 1. The original dataset included a few posts where $S < 0$ or $A < 0$ but they were intentionally left out to simplify the exercises.

1.2 Are A, S continuous or discrete random variables? Why?

Both are continuous RVs.
Instructor Note. The probability of any given score is zero. For example

$$P(A = 0.6986162) = 0$$

because cosine similarity is a continuous variable with infinitely many possible values. Instead, probabilities refer to ranges of values. For example, there is a positive probability that a post has a similarity score near this value, such as between 0.69 and 0.70

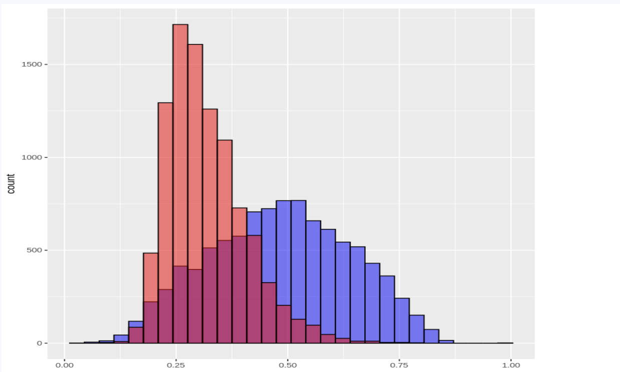
1.3 [1.1] Use `ggplot2` to create histograms for both A and S on the same plot.

(a) Declare a `ggplot()` on your data frame.

(b) Add a `geom_histogram()` layer with `ai_aware` as your x-`aes()`. Outside `aes()`, set the `fill` parameter to "blue", the `color` to "black", and the transparency (`alpha`) to 0.5.

(c) In the same plot, add another histogram layer analogously for `simplicity` but fill it with "red".

Code in Colab [1.1] **Answer**



1.4 Examine the plot:

(a) What is plotted in the y-value?

The number of posts with values A, S in the range of each bin.

(b) Roughly, how many posts are there in the $0.47 \leq S \leq 0.53$ range?

300 to 350. The bin to the left of 0.5 accounts for approximately 200, the one to the right amounts to roughly 125.

(c) More generally, which histogram seems more concentrated around a given range of values?

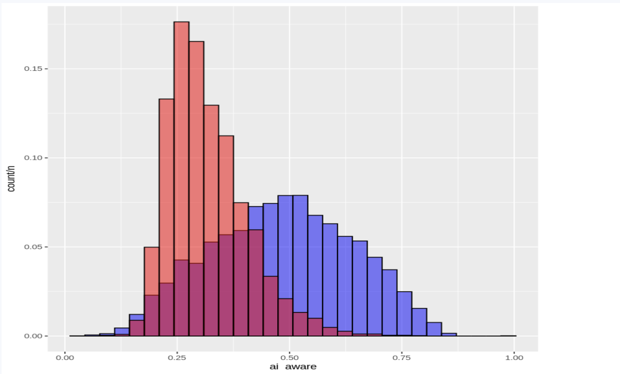
The `simplicity` histogram concentrates over 60% of the posts in the 5 bins that go from (approximately) 0.22 to 0.37
While centered around $A = 0.5$, there are no 5 bins in the `ai_aware` histogram concentrating posts so densely.

(d) If we summed the y-value of all the bins of either the `simplicity` or the `ai_aware` histogram, what should be the result? Is the result a probability?

9724 (the number of posts). This count is not a probability.

1.5 [1.1.1] We want the y-value to reflect a probability. Edit the `aes()` function in your histograms plot to include the following y-aesthetic: `y = after_stat(count/n)`. Compute and save `n` separately.

Code in Colab [1.1.1] **Answer**



(a) Similar to 1.4d, what should be the sum of all the bins' y-values of either histogram now?

The bins for each add up to 1. Note the shape and relative heights remain the same. We simply normalized the bins as a proportion of the total posts.

(b) Very roughly (no computation needed), what is x such that $F_A(x) \geq 0.5$?

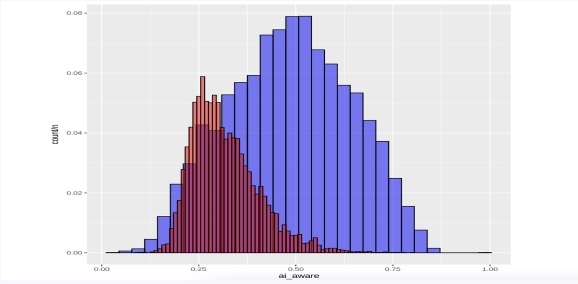
$x = 0.5$. As mentioned before, the distribution of A is roughly symmetrical and centered around 0.5; therefore, at said value, half of the distribution is to the left.

(c) Again roughly, what do you think $P(0.75 \leq S \leq 1)$ approximates to?

Zero. Less than 0.3% score above 0.75 in simplicity.

(d) Histograms are discrete representations of a continuous RV. Change the `binwidth` of **just the red histogram** to 0.01. What can you observe? Did the distribution's relative shape change? Why?

Code in Colab [N/A] **Answer**



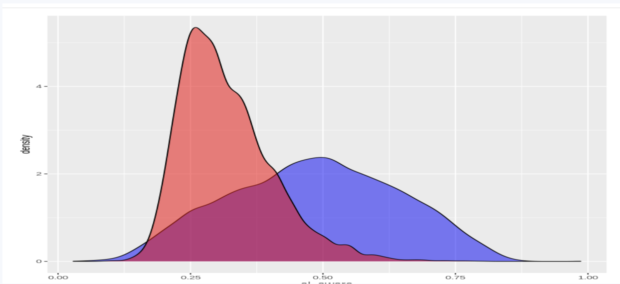
Although the bin heights changed, the relative histogram shapes did not. The new binwidth size is smaller than the default one (roughly 0.33), which makes the counts decrease per bin, but remain the same for the range; they still add up to 1 just as before.

Recall the *Summarizing and Describing Data* lecture. Kernel Density Estimation is a **continuous** estimate of our RV's distribution. Base R has a readily available function for this: `density()` — feel free to test it in your own time.

We will use density through `ggplot2`'s interface: `geom_density()` to plot the estimated pdfs $\widehat{f_A(\cdot)}$ and $\widehat{f_S(\cdot)}$ for all the x-values in the domain of our data.

1.6 [1.2] Copy and paste your first code (question 1.3). Substitute `geom_histogram` with `geom_density` in both layers.

Code in Colab [1.2] **Answer**



(a) In terms of shape and position relative to each other, how do these distribution representations compare to those in 1.5?

The relative shapes don't change, however note the heights of the plot changed, and now exceed the unit.

(b) What does the y-axis represent in this plot?

The density of the pdfs $f_A(a), f_S(s)$ at each point.
Instructor Note: This is a pertinent example to emphasize that densities are not probabilities, and they can exceed 1; probability is defined as the area under these curves. Areas with heights > 1 and "bases < 1" can still be less than 1.

(c) Roughly, what is *max* $\widehat{f_S(s)}$? What is the minimum for both A and S?

> $max \quad \widehat{f_S(x)} \approx 5.5$.
> The minimum estimated density for both is 0.

Another way to compute the probability of an event $E = [a, b] \in \Omega_X$ for a continuous RV is through its CDF. Recall from the lecture that

$$P(a \leq X \leq b) = \int_a^b f_X(x)dx = F_X(b) - F_X(a)$$

Base R allows us to create a function for the empirical CDF of continuous variables with `ecdf()`. This function also has its `ggplot2` analog in `stat_ecdf()`, which you can use just as any other `geom`.

1.7 [1.3] Use `ecdf()` to create functions `F_A` and `F_S`, the empirical CDFs of A and S respectively. Make sure you have the correct object by quickly plotting the CDFs: `plot(F_A)`, `plot(F_S)`. Get familiar with the usage by verifying that `F_A(0)` returns 0 and `F_A(1)` returns 1. Then answer the following questions:

(a) $F_S(0.25)$?
(b) $F_A(0.25)$?
(c) What is $s \in \Omega_S$ such that $P(S \leq s) = 0.8$?
(d) What is $a \in \Omega_A$ such that $1 - P(A \leq a) = 0.1$?
(e) Compute the probability that posts are either too intricate or too simplistic $P(S \leq 0.2 \cup S \geq 0.7)$
(f) Make an event for S such that posts fall in it with 80% probability. Find any $a, b \in (0, 1)$; $a < b$ such that $P(a \leq S \leq b) = 0.8$

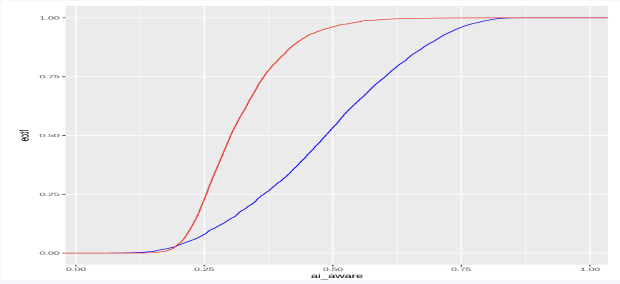
.

Code in Colab [1.3] **Answer**

> (a) 0.228918
>
> (b) 0.080111
>
> (c) $s \approx 0.385$. Experiment with a few values, knowing f_S is heavily concentrated in $[0.2, 0.4]$
>
> (d) $a \approx 0.7$. Find a such that $P(A \leq a) = 0.9$
>
> (e) 0.040621
>
> (f) Many possibilities, one of them $a = 0.9$, $b \approx 0.244$

1.8 [1.4] Use `stat_ecdf()` in `ggplot2` to jointly plot the CDFs for A and S. Make sure to `color` them blue and red, respectively.

Code in Colab [1.4] **Answer**



(a) Can you conclude there's first-order stochastic dominance of one of the random variables to the other?

> Not strictly, since there is a crossing at roughly $s = a = 0.2$.

(b) In the lecture, height was the construct measured for women in both Bihar and the US. Are S and A measuring the same in the posts?

> In this case, even if S FOSD A, the dominance is non-informative; our random variables are not necessarily measured the same; hard to describe semantic similarity with the same consistency as height. The only thing we can conclude is S takes smaller values in general.

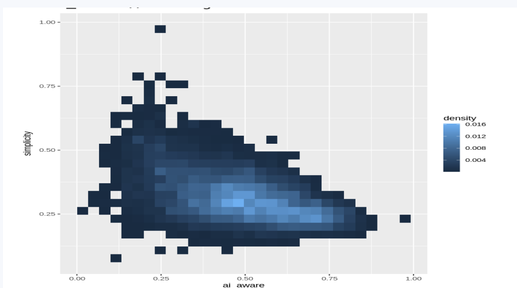
Part 2. Non-independence

2.1 Within the r/AIwars subreddit, conditional on learning posts tend to be long and argumentative, would their probability of being AI-related change?

> We can't know *a priori*. In general, we would expect longer and more substantiated posts in the AIWars subreddit to systematically contain more AI-related content, to the extent that A and S are not independent.

2.2 [2.1] Let's begin by examining $f_{A,S}(a, s)$. Use `ggplot2`'s function `geom_bin_2d()` to plot a 2D-histogram with `ai_aware` in the `x`-axis and `simplicity` in the `y`-axis. Set the `fill` aesthetic to `after_stat(density)`.

Code in Colab [2.1] **Answer**



(a) What does color represent in this plot? Note that the "minimal" unit colored is a rectangle tile.

> Each rectangle (tile) represents a bin equivalent to histogram bins in 1D. In this case, bins are 2-dimentional, covering a segment for A and a segment S. Color represents either a count or probability of posts falling in that combination of (a, s) values.

(b) Are the x and y-binwidths the same? What are they, approximately?

> The tiles are close to squares (slightly elongated), hence bin widths are similar. Bins cover roughly 0.03 units on each variable.

(c) Generally describe the joint composition of the subreddit's posts in terms of their AI-awareness and their simplicity. Feel free to sample a few posts in various ranges:

- Moderate AI-awareness ($[0.4, 0.6]$) and moderate simplicity ($[0.2, 0.3]$)
- High AI awareness and low simplicity (< 0.15). Check for high simplicity (> 0.4) too.
- Low AI awareness (< 0.3)) and varying simplicity.

You can sample a few random rows with `dplyr`'s function `sample_n()`. For a better dislay in Colab, select only the `fulltext`, `ai_aware` and `simplicity` columns

> Code in Colab ✔ >_ **[N/A] Answer**
>
> As the bullets suggest. Students may think of the pdf as quadrants of various types of posts:
>
> Most posts concentrate around substantial AI-related content, and moderate simplicity. These posts are up to a couple of paragraphs long, argumentative and not constrained to a specific topic within AI.
>
> Highly AI-aware and highly simple posts are few, and tend to be users sharing links to external AI-related content or news with little commentary. Most highly AI-related posts tend to be long and argumentative. Because of how `ai_aware` is constructed, scores over 0.7 generally discuss art topics, and the impact of AI on art.
>
> High simplicity and low AI-awareness is also dominated by link sharing, but tilts towards mocking tones and commentary that cannot be understood in the context of AI on its own. Low simplicity and low AI-awareness tend to be posts completely unrelated or only tangentially related to the subreddit (see notebok: an account of a christian saint, and a text in Chinese on a individual named Zhai Weimin.

(d) Judging by our joint histogram plot, are A and S related directly or inversely?

> For the most part, highly AI-aware posts feature low simplicity. The high A - high S quadrant is empty.

(e) Do relatively long and argumentative tend to discuss AI? Can you find low-simplicity posts unrelated to AI in the dataset?

> No.
>
> Although more concentrated above 0.5, the plot shows low simplicity across all values of AI-awareness. Long, argumentative posts unrelated to AI can be found: post indices 2107 and 3631 for instance, fit this description.

(f) Visually, what is the 4x4 tiles semantic region $\Omega_S \times \Omega_A$ most posts fall in?

> $[0.44, 0.55] \times [0.25, 0.37]$ approximately.

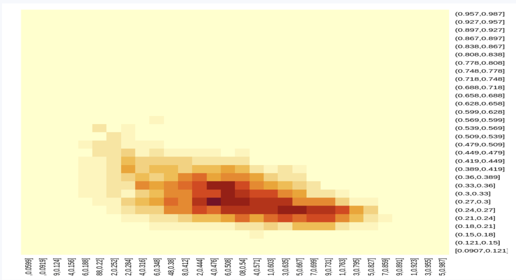
2.3 **True** or **False**: if A and S were independent, we would have a perfectly squared 1×1 grid and all tiles would be of the same color, because $f_{A,S}(a,s) = f_A(a) \cdot f_B(b)$

> False.

Numerically, we can represent the joint pdf $f_{A,S}(a,s)$ as plotted in 2.1 as a matrix with rows and columns corresponding to each x (`ai_aware`) and y-bin (`simplicity`). In the next question, code is provided to create such a 30×30 matrix, `f_AS` for you.

2.4 [2.2] Run the pre-scripted code to get `f_AS`, the joint PDF $f_{A,S}(a,s)$. Then compute the marginals $h_A(a)$ and $h_S(s)$ and save them as `h_A` and `h_S`, respectively. Keep in mind these should be vectors of length 30.

Code in Colab [2.2] **Answer**

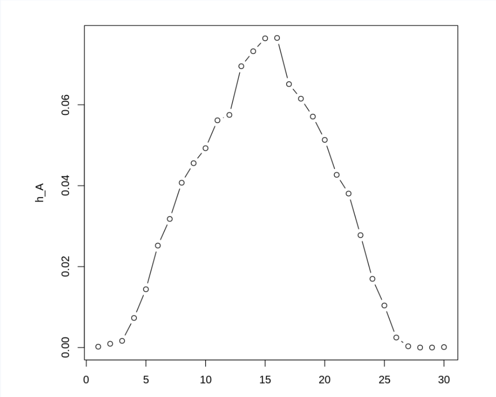


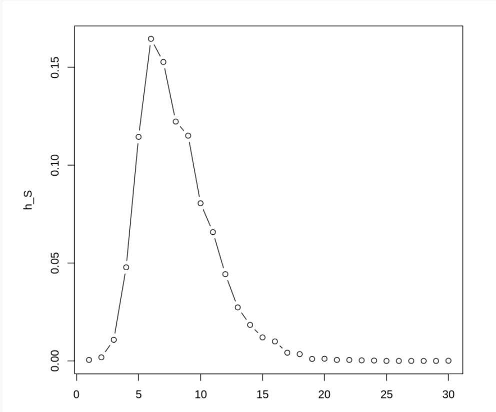
To get the marginals, use row/col sums acordingly:

- `h_A <- colSums(f_AS)`
- `h_S <- rowSums(f_AS)`

The output will be length-30 labeled vectors, with the bins as names, similar to a `hist()` object.

2.5 Make quick base R plots of each marginal with `plot(h_A, type = "b")` and similarly for h_S in a separate plot. Do they look familiar? Why?





h_A to the left and h_S to the right. Since histograms are a representation of a PDF, they are approximately equivalent to the histograms from question 1.5.

2.6 Use `f_AS`, `h_A` or `h_S` — and their bin `names` — to calculate the following probabilites. You may want to print the bin names beforehand:

Code in Colab [N/A] **Answer**

(a) $P(A \leq 0.38 \ \cap \ S \leq 0.24)$

> 0.00051419

(b) $P(0.156 < A \leq 0.38 \ \cap \ 0.628 < S < 0.867)$

> 0.0034965

(c) $P(A = 0.188 \cap S \leq 0.927)$

> 0
> The probability of any point value is zero.

(d) $P(S \leq 0.24)$

> 0.0038050

(e) $P(A > 0.667 \cap 0.27 < S \leq 0.3)$

> 0.023138

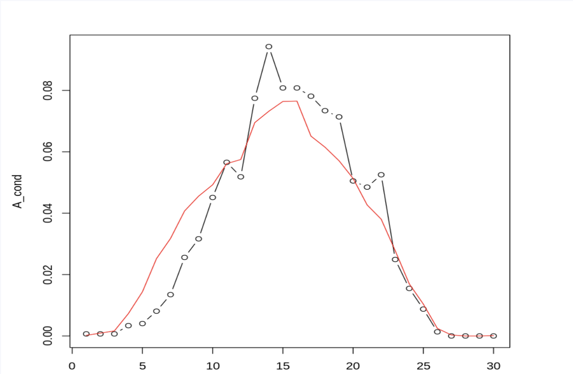
(f) $P(A > 0.667 \ | \ 0.27 < S \leq 0.3)$

> 0.15151515
> Divide the previous answer by h_S at that bin: `h_S[7]`

(g) $P(0.27 \leq S \leq 0.3 \ | \ A > 0.667)$

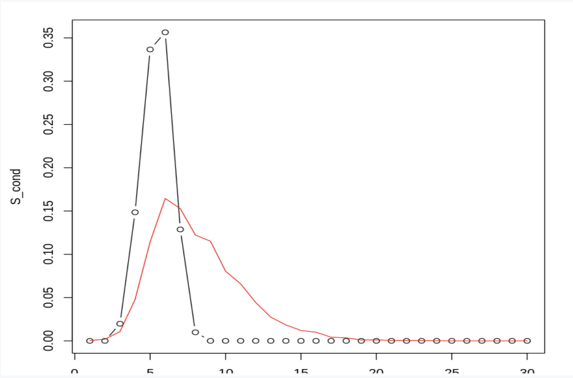
> 0.1667902

(h) $P(A \ | \ 0.27 \leq S \leq 0.3)$. Save as `A_cond` and plot this



Plotted alongside its marginal in red (see question 2.8) for reference.

(i) $P(S \mid A \in (0.795, 0.827])$. Save as `S_cond` and plot.



Plotted alongside its marginal in red (see question 2.8) for reference.

2.7 Show that A, S are not independent with your answers to 2.6f, 2.6g, and the elements you used to calculate them.

Under independence, we have that

$$f_{A|S}(a|S^*) = \frac{f_{A,S}(A, S^*)}{h_S(S^*)} = \frac{h_A(a) \cdot h_S(S^*)}{h_S(S^*)} = h_A(a)$$

Show the marginal at those bins is not equal to the answer:

$$h_A(A > 0.667) \neq \frac{f_{A|S}(A > 0.667, 0.27 \leq S \leq 0.3)}{h_S(0.27 \leq S \leq 0.3)}$$

Which in code translates to
`sum(f_AS[7, 21:30])/sum(h_S[7]) != sum(h_A[21:30])`
Or analogously for $S|A$

2.8 Plot your answers to 2.6h and 2.6i are not independent. Side by side with their marginal:

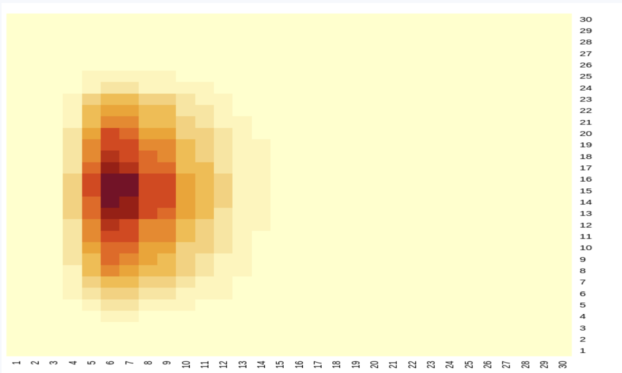
```
plot(S_cond, type = "b") lines(h_S, col = "red")
```

And the same for A. What differences do you notice? What do these plots show?

See answers 2.6h and 2.6i for the plots.
The plots show 2.7 more generally, for all values of the non-conditioned RV. Additionally, note the conditional plot for A is more alike its marginal than the conditional for S and its marginal.

2.9 [2.3] Finally, directly verify the claim in 2.3. Use `h_A` and `h_S` as well as a double (nested) loop to produce $g(a, s) = h_A(a) * h_S(s)$ for the same 30×30 grid. Then use `heatmap()` with the same arguments used in the previous code snippet to examine the joint density. Does this look like a homogeneous distribution? If not, why? If it does, at what regions of $\Omega_A \times \Omega_S$ do we have missing values, if any?

Code in Colab [2.3] **Answer**



It doesn't look homogeneous because not all values A and S have equal density as we saw in the first part. Furthermore, the plot doesn't look close to the joint plot in 2.2, corroborating the joint pdf is not the result of multiplying the marginals.

Part 3. Your own joint distributions

Your team will freely analyze the relationship between either AI-awareness or Simplicity and one other feature in the `aiwars` dataset.
Create a short deliverable following these instructions:

- Choose either `ai_aware` or `simplicity` and one other key variable.
- Your instructor will
  - Approve or change your choice based on their goals and what other students are working on.
  - Inform you of the format required for the deliverable. Usually a 5-slide eck or a text document.
- Once your choice is approved:

1. Construct the joint distribution of the two variables (Depending on your choice of discrete or continuous, use a `geom_bin_2d`, a boxplot, or any of the plots reviewed in the lecture).

2. Obtain and interpret the marginal distributions of each variable.
3. Compute the conditional distributions (e.g. $f_{A|X}(a)$ or $f_{S|X}(s)$) for what you consider are relevant ranges of values. Evaluate how indpendent the analysis variables are.
4. Include at least one visualization that helps highlight the relationship (histograms, heatmap, density plot, etc.).
5. Provide 1–2 slides (or short paragraphs) summarizing the implications of the independence (or lack thereof) between variables in plain language: What does the relationship tell us about Redditor behavior?

– Submit as required. Your instructor will let you know if your team will briefly expose your deliverable to the class.

14.310x Flipped Classroom

Week 4 Answers

Transforms of random variables
Moments of random variables
Introductory simulation in R

Checklist

- ☐ Complete Coding Lab 4 (Requirement)
- ☐ Solve Guided Case 3 (Session 1)
- ☐ Solve Guided Case 4 (Session 2)

Coding Lab L.4.4

Simulation

Instructions: Work individually. Answer all sections of the Lab.

Notebook: Coding Lab 4: Simulating RVs in R

Record your answers in your copy of the notebook, and if required, submit your work as specified by the instructor.

Instructor Notebook: Answers to L.4.4

Find the answers to all exercises in this notebook.

Guided Case G.4.3

Select transforms of random variables: theory and coding practice

Instructions: Work in pairs or teams of 3. Answer all questions 1-3 and submit your coursework as required by the instructor.

Notebook: Guided Case Study 3

Instructor Notebook: Answers to G.4.3

Find answers to all code-related questions in this Guided Case. Each coding-relevant question links to its corresponding answer section in this notebook.

In the following questions, X will be a **known** random variable with pdf/pmf $f_X(x)$ (either provided or described for you to write down).

Each question will propose a transform $Y = g(X)$ and you must derive $f_Y(y)$, the pdf/pmf associated with said transform.

All exercises will ask you to follow a few analytical steps, after which you will plot and perform simulations in R.

Question 1

$$X \sim U[-1, 1]$$
$$Y = e^X$$

1.1 What is the support Ω_x the support of X? Write down $f_X(x)$.

$\Omega_X = [-1, 1]$

$f_X(x) = \dfrac{1}{1-(-1)} = \dfrac{1}{2}$ for $-1 \le x \le 1$ and 0 otherwise.

1.2 What is Ω_Y, the support of Y?

$$\Omega_Y = [\frac{1}{e}, e]$$

e^x is a continuous function on the $[-1, 1]$ interval.

1.3 As in the lecture, work out the CDF of Y first.

$$F_Y(y) \equiv P(Y \le y)$$

(a) Substitute Y for $g(X)$.

$$F_Y(y) \equiv P(Y \leq y)$$
$$\implies P(e^X \leq y)$$

(b) Get an expression in terms of the CDF of X: $P(X \leq g^{-1}(y))$.

$$P(e^X \leq y)$$

Taking natural logarithms on both sides of the inequality:

$$\implies P(X \leq ln(y)) \equiv F_X(ln(y))$$

(c) Evaluate the expression. $\int_{-\infty}^{g^{-1}(y)} f_X(x)dx$.

$$F_X(ln(y)) = \int_{-\infty}^{ln(y)} \frac{1}{2}dx$$

Since $f_X(x) = 0$ for $x < -1$, we modify the integral's lower limit:

$$\int_{-1}^{ln(y)} \frac{1}{2}dx = \frac{1}{2}[x]_{-1}^{ln(y)} = \frac{1}{2}(ln(y) + 1)$$

Note that the CDF of X in terms of y coincides with Ω_Y:

$$F_X(\frac{1}{e}) = \frac{1}{2}(ln(\frac{1}{e}) + 1) = \frac{1}{2}(-1 + 1) = 0$$

$$F_X(e) = \frac{1}{2}(ln(e) + 1) = \frac{1}{2}(1 + 1) = 1$$

1.4 Get $f_Y(y)$. If the CDF is continuous you can take the derivative $\frac{\partial F_Y}{\partial y}$.

$$F_Y(y) = \frac{1}{2}(ln(y) + 1)$$

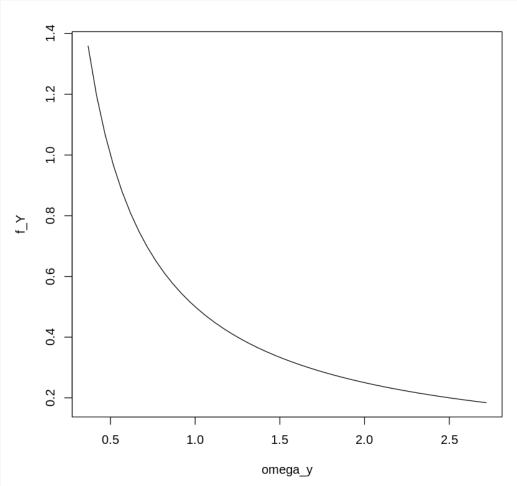
A continuous function on the $[\frac{1}{e}, e]$ interval:

$$\frac{\partial F_Y(y)}{\partial y} = \frac{1}{2}(\frac{1}{y}) = \frac{1}{2y}$$

$$\implies f_Y(y) = \frac{y+1}{2y} \text{ for } \frac{1}{e} \leq y \leq e \text{ and 0 otherwise.}$$

1.5 [1.1] Plot $f_Y(y)$ in R. Use `seq()` to create a vector `omega_y` from $\frac{1}{e}$ to e in increments of 0.05. Then create `f_Y`, by applying your expression for the pdf $f\_Y(y)$ to each element in `omega_y`. Plot `f_Y`.

Code in Colab [1.1] **Answer**



1.6 [1.2] Now run a simulation, and plot a histogram approximation for $f\_Y(y)$:

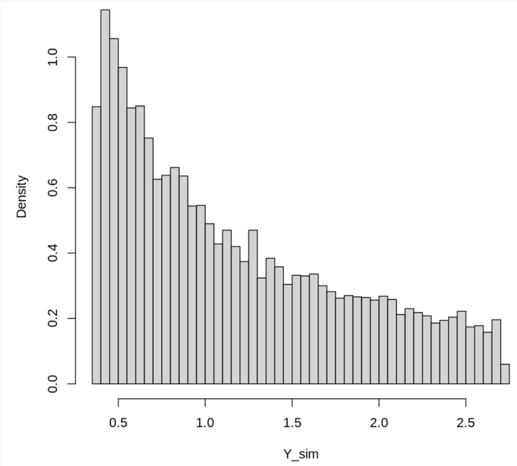
(a) `set.seed() to 123`

(b) Use `runif()` with the relevant parameters to draw a sample with 10,000 realizations of X. Save it as vector `X_sim`.

(c) Apply the $Y = g(X)$ transformation to your sample. Save it as `Y_sim`.

(d) Make a quick histogram plot of `Y_sim`. How well does it resemble the theoretical pdf you derived?

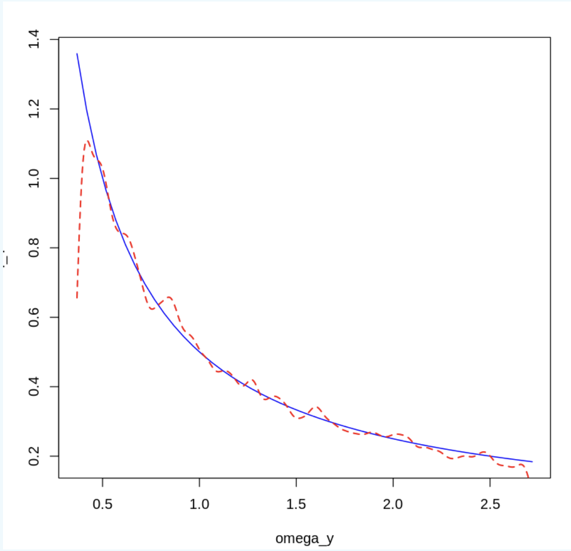
Code in Colab [1.2] **Answer**



1.7 [1.3] Run the code provided to create a side-by-side plot of both the theoretical (blue) and the estimated (dashed red) pdfs. Make sure you followed the naming conventions so far.

Code in Colab [1.3] **Answer**

Instructor note: This particular seed (123) provides a uniform sample that slightly underestimates the upper and lower bounds of the $[-1, 1]$ uniform. Changing the seed may result in slight bound improvements, with a noisier representation of the inner parts of the distribution.



Note the density estimation fails when it gets close the lower limit of Ω_Y. Feel free to use the `from` and `to` and `bw` arguments, or change the `kernel` in `density()` to improve the plot.

Question 2

$$X \sim U[-1, 1]$$

$$Y = \frac{X^2}{1 + X^2}$$

2.1 What is Ω_Y?

$$\Omega_Y = [0, \frac{1}{2}]$$

Note the support mirrors at $\Omega_Y = [-1, 0], \ [0, 1]$.

2.2 Get $F_Y(y)$. *Be very careful with the inequalities.*

$$F_Y(y) \equiv P(Y \leq y) = P(\frac{X^2}{1+X^2} \leq y)$$

Multiply both sides of the inequality by the denominator. Since $1 + X^2 > 0$:

$$\implies P(X^2 \leq y(1+X^2)) = P(X^2(1-y) \leq y)$$

With $y \in [0, \frac{1}{2}]$. Such that $1 - y \geq \frac{1}{2}$ and $0 \leq \frac{y}{1-y} \leq 1$.

$$P(X^2 \leq \frac{y}{1-y})$$

With $x{=}\pm\sqrt{\frac{y}{1-y}}$, note the inequality is only true if $x{\geq}-\sqrt{\frac{y}{1-y}}$ and $x{\leq}\sqrt{\frac{y}{1-y}}$

$$\implies P(-\sqrt{\frac{y}{1-y}} \leq X \leq \sqrt{\frac{y}{1-y}}) = F_X(\sqrt{\frac{y}{1-y}}) - F_X(-\sqrt{\frac{y}{1-y}})$$

$$\ldots = \int_{-1}^{\sqrt{\frac{y}{1-y}}} \frac{1}{2}dx - \int_{-1}^{-\sqrt{\frac{y}{1-y}}} \frac{1}{2}dx = \frac{1}{2}(\sqrt{\frac{y}{1-y}} + 1) - \frac{1}{2}(-\sqrt{\frac{y}{1-y}} + 1)$$

$$\ldots = \sqrt{\frac{y}{1-y}}$$

Alternatively, solve the integral directly:

$$\ldots = \int_{-\sqrt{\frac{y}{1-y}}}^{\sqrt{\frac{y}{1-y}}} \frac{1}{2}dx = \frac{1}{2}(\sqrt{\frac{y}{1-y}} + \sqrt{\frac{y}{1-y}}) = \sqrt{\frac{y}{1-y}}$$

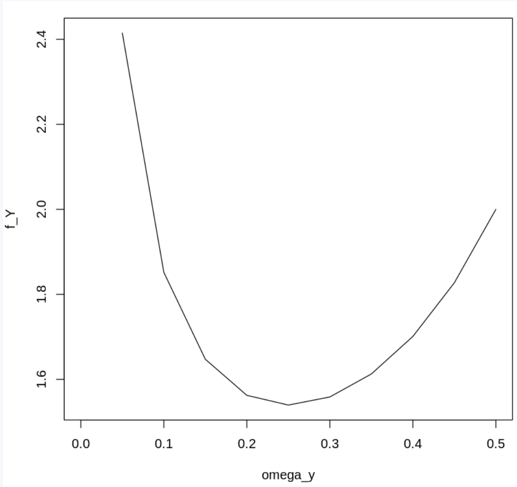
– Verify the following: $F_Y(0) = 0, \quad F_Y(\frac{1}{2}) = 1$

$$F_Y(0) = \sqrt{\frac{0}{1-0}} = 0$$

$$F_Y(\frac{1}{2}) = \sqrt{\frac{\frac{1}{2}}{1-\frac{1}{2}}} = 1$$

2.3 [2.1] Get $f_Y(y)$ and plot it along the support of Y as in 1.5.

Code in Colab [2.1] **Answer**



The CDF is continuous in the $[0, \frac{1}{2}]$ interval:

$$f_Y(y) = \frac{\partial F_Y}{\partial y} = \frac{\partial}{\partial y}(\sqrt{\frac{y}{1-y}})$$

By the chain rule:

$$\dots = \frac{1}{2\sqrt{y}} \cdot \frac{1}{\sqrt{1-y}} + (-\frac{1}{2}\frac{\sqrt{y}}{\sqrt{1-y}\ \ (1-y)} \cdot -1)$$

Factoring out common elements:

$$\dots = \frac{1}{2\sqrt{1-y}}(\frac{1}{\sqrt{y}} + \frac{\sqrt{y}}{(1-y)})$$

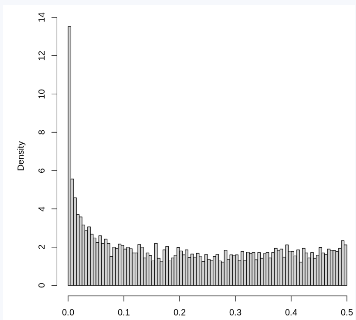
$$\dots = \frac{1}{2\sqrt{1-y}}(\frac{(1-y) + \sqrt{y} \cdot \sqrt{y}}{(1-y)\sqrt{y})})$$

$$\dots = \frac{1}{2\sqrt{1-y}} \cdot \frac{1}{(1-y)\sqrt{y}} = \frac{1}{2(1-y)\sqrt{1-y}\sqrt{y}}$$

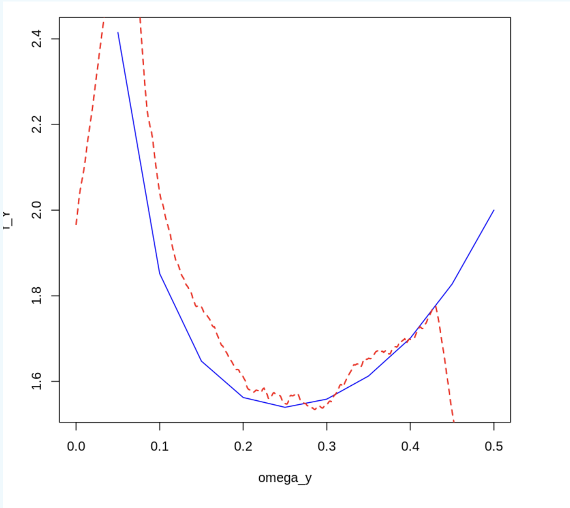
Instructor note: Writing down this expression in R, it is advisable to write it down in 3 parts with each expression in the denominator —see the answer code.

2.4 [2.2] Similar to 1.6, simulate Y and plot its histogram with `breaks = 100`.

Code in Colab [2.2] **Answer**



2.5 [2.3] Plot the exact and estimated densities side by side like you did in 1.7. This time, set the bandwidth argument `bw` in `density()` to 0.04, modify the robjectkernel to rectangular and specify the `from` and `to` arguments to match the sample space of Y.



Code in Colab [2.3] **Answer**

Question 3: Probability integral transform

$$f_X(x) = \begin{cases} 4x & 0 \le x \le \sqrt{\frac{1}{2}} \\ 0 & \text{otherwise} \end{cases}$$

$$Y = F_X(x)$$

3.1 Derive $F_X(x)$.

$$F_X(x) = \int_0^x 4u\,du = [2u^2]_0^x = 2x^2$$

For $x \in [0, \sqrt{\frac{1}{2}}]$

$F_X(0) = 0$ and $F_X(\sqrt{\frac{1}{2}}) = 2(\sqrt{\frac{1}{2}})^2 = 1$

3.2 What is the range of any CDF? (What is Ω_Y?)

Since a CDF is a probability and X is continuous RV, $\Omega_Y = [0, 1]$

3.3 According to the lecture, what should $f_Y(y)$ be? — Review the PIT slides if this is not clear.

$$Y \sim U[0, 1]$$

Regardless of the distribution of continuous RV X, the PDF of the transform by its own CDF is the 0,1 uniform.

In the lecture, Prof. Ellison mentioned that we can sample from any distribution by sampling from the distribution you got in 3.3 and transforming it by the inverse CDF of X:

$$Y = F_X(X)$$

$$\implies X = F^{-1}(Y)$$

3.4 From 3.1, solve for X as above —you should end up with a function in terms of Y.

We have

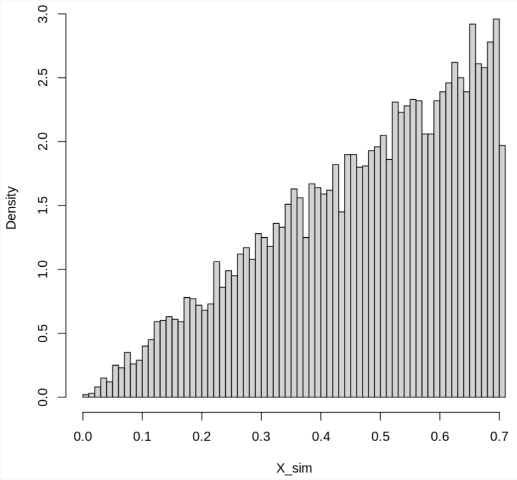
$$F_X(x) = y = 2x^2$$

$$\implies x = \sqrt{\frac{y}{2}}$$

Note in this case $-\sqrt{\frac{y}{2}}$ is not possible, because by construction $x \geq 0$.

3.5 [3.1] Confirm you get $f_X(x)$ back with a simulation:
- Draw 10,000 observations from the distribution of Y: `Y_sim`.
- Translate them to X draws by applying them function $F^{-1}(y)$ you found in the previous question: `X_sim`.
- Plot the histogram of `X_sim` with `breaks` set to 100. It should resemble $f_X(x) = 4x$.

Code in Colab [3.1] **Answer**



- Sample from the $[0, 1]$ uniform (Y's distribution)
- Apply the transform found in question 3.4: $\sqrt{\frac{y}{2}}$.

[Harder] Question 4: Convolution

$$X_1 \sim U[0,1]$$

$$f_{X_2}(x_2) = \begin{cases} 2x_2 & 0 \leq x_2 \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$Y = X_1 + X_2$$

$$X_1, X_2 \text{ independent.}$$

$$f_Y(y) = f_Y(x_1, x_2) = \int_{r_1 \in \Omega_{X_1}} \int_{r_2 \in \Omega_{X_2}} f_{X_1}(x_1) \cdot f_{X_2}(x_2) dY$$

4.1 Given that $\Omega_{X_1} = \Omega_{X_2} = [0,1]$ what is Ω_Y?

> $$\Omega_Y = [0,2]$$
>
> The sum of two independent random variables with outcomes between 0 and 1.

Even in simple convolutions, finding the correct integration region is challenging. The following exercises are meant to guide you in setting up the correct bounds for f_{X_1} and f_{X_2}, in order to find $F_Y(y)$.

Given Y and X_2 below, write down the possible values of X_1:

4.2 $Y = 1.5$ and $X_2 = 0.9$

> $$X_1 = 0.6$$
>
> Given
>
> $$Y = X_1 + X_2$$
>
> $$\implies X_1 = Y - X_2$$

4.3 $Y \geq 0.5$, and $X_2 = 0.5$

> $$0 \leq X_1 \leq 1$$
>
> As $Y = 0.5 + X_1$ and $0.5 \leq X_1 + 0.5 \leq 1.5$

4.4 Let $0 \leq Y < 1$ and $\frac{1}{4} \leq X_2 \leq \frac{1}{2}$.

(a) What values of Y are events in the sample space of Y that are not possible?

> $[0, \frac{1}{4})$ — and $[1,2]$ by definition.
>
> Since $\frac{1}{4} \leq X_2 + X_1 \leq \frac{3}{2}$

(b) Write down the range of values X_1 can take.

$0 \le X_1 < \frac{3}{4}$ are values satisfying $0 \le Y < 1$

(c) Now suppose we have $y \in Y$, a specific value in $[\frac{1}{4}, 1)$. Write down the possible values for X_1 **in terms of number** y and the bounds of X_2 as before.

$$\begin{cases} y - \frac{1}{2} \le x_1 \le y - \frac{1}{4} & \text{for } \frac{1}{2} \le y < 1 \\ 0 \le x_1 \le y - \frac{1}{4} & \text{for } \frac{1}{4} \le y < \frac{1}{2} \end{cases}$$

Equivalently:

$$max(0, y - \frac{1}{2}) \le x_1 \le y - \frac{1}{4}$$

With $y \ge \frac{1}{4}$ from 4.4a.

4.5 Let $0 \le Y < 1$ and $0 \le X_2 \le 1$. Consider numbers $y \in Y$ and $x_2 \in X_2$. Write down the bounds for X_1 in terms of these numbers (particularly, find the upper bound).

$$0 \le x_1 \le y$$

From $x_1 \le y - x_2 \le y$

4.6 Let $1 \le Y \le 2$ and $0 \le X_2 \le 1 \mid y \in Y,\ x_2 \in X_2$. For the bounds on X_1 you now have to consider two events:

(a) When $y - x_2 \ge 1$.

$$0 \le x_1 \le 1$$

Because $x_1 \le 1 \le y - x_2$
Note any value of X_1 satifies this condition because x_2 is "small" enough that $y - x_2$ still leaves a full unit slack.

(b) When $y - x_2 < 1$. *Hint: if x_2 is "too large", then x_1 can't be just any value, it will be restricted. And vice-versa.*

$$0 \le x_1 \le y - x_2$$

This restriction is binding because x_2 is large: $0 \le x_1 \le y - x_2 < 1$ doesn't leave a full unit.

With the previous answers in mind, now derive the pdf of Y:

$$F_Y(y) \equiv P(Y \le y) = P(X_1 + X_2 \le y) = P(X_1 \le y - X_2)$$

Case 1: $0 \leq y < 1,\ 0 \leq x_2 \leq y$

$$\ldots = \int_0^{\square} \int_0^{\square} f_{X_1}(x_1) \cdot f_{X_2}(x_2) dx_1 dx_2$$

4.7 Fill out the upper limits of the integrals and solve for $f_Y(y)$ when $0 \leq y < 1$

From question 4.5:

$$F_Y(y) = \int_0^{y} \int_0^{y-x_2} f_{X_1}(x_1) \cdot f_{X_2}(x_2) dx_1 dx_2$$

$$\ldots = \int_0^{y} \int_0^{y-x_2} 1 \cdot 2x_2 \; dx_1 dx_2$$

$$\ldots = \int_0^{y} (y - x_2) 2x_2 dx_2 = y[x_2^2]_0^y - \frac{2}{3}[x_2^3]_0^y$$

$$\ldots = y^3 - \frac{2}{3}y^3 = \frac{1}{3}y^3$$

And the pdf in $0 \leq y < 1$:

$$f_Y(y) = \frac{\partial F_Y}{\partial y} = y^2$$

Case 2A: $1 \leq y \leq 2, \quad y - 1 \geq x_2$

$$\ldots = \int_{\square}^{y-1} \int_{\square}^{\square} f_{X_1}(x_1) \cdot f_{X_2}(x_2) dx_2 dx_1 \tag{8}$$

4.8 Fill out the limits and solve for $F_Y(y)$.

From 4.6a:

$$F_Y(y) = \int_0^{y-1} \int_0^{1} f_{X_1}(x_1) \cdot f_{X_2}(x_2) dx_1 dx_2$$

$$\ldots = \int_0^{y-1} \int_0^{1} 2x_2 dx_1 dx_2 = \int_0^{y-1} 2x_2 dx_2 = (y-1)^2$$

Case 2B: $1 \leq y \leq 2, \quad y - 1 < x_2$

$$\ldots = \int_{\square}^{\square} \int_{\square}^{\square} f_{X_1}(x_1) \cdot f_{X_2}(x_2) dx_2 dx_1 \tag{9}$$

4.9 Fill out the limits and solve for $F_Y(y)$.

From 4.6b:

$$F_Y(y) = \int_{y-1}^{1} \int_{0}^{y-x_2} 2x_2 dx_1 dx_2 = \int_{y-1}^{1} 2x_2(y - x_2) dx_2$$

$$\ldots = y[x^2]_{y-1}^{1} - \frac{2}{3}[x^3]_{y-1}^{1} = -\frac{1}{3}y^3 + 2y - \frac{4}{3}$$

4.10 Write down $F_Y(y)$ and $f_Y(y)$ for all $1 \le y \le 2$, x_1 and x_2. *Hint: cases 2A and 2B are two (disjoint) partitions of* $1 \le y \le 2$.

Since they're disjoint, simply add the CDFs for both cases:

$$F_{Y,y\ge 1}(y) = F_{Y,2A}(y) + F_{Y,2B}(y) = (y-1)^2 - \frac{1}{3}y^3 + 2y - \frac{4}{3}$$

$$\ldots = -\frac{1}{3}y^3 + y^2 - \frac{1}{3}$$

The PDF in $1 \le y \le 2$:

$$f_Y(y) = \frac{\partial F_Y}{\partial y} = -y^2 + 2y$$

4.11 Derive the pdf of Y for all $y \in \Omega_Y$.

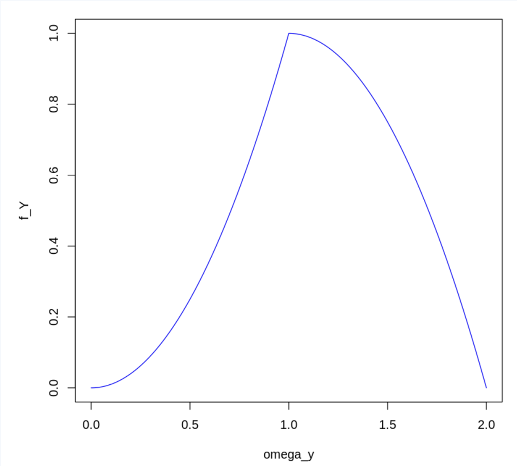
Summarizing the answers to 4.7-4.10:

$$f_Y(y) = \begin{cases} y^2 & 0 \le y < 1 \\ 2y - y^2 & 1 \le y \le 2 \end{cases}$$

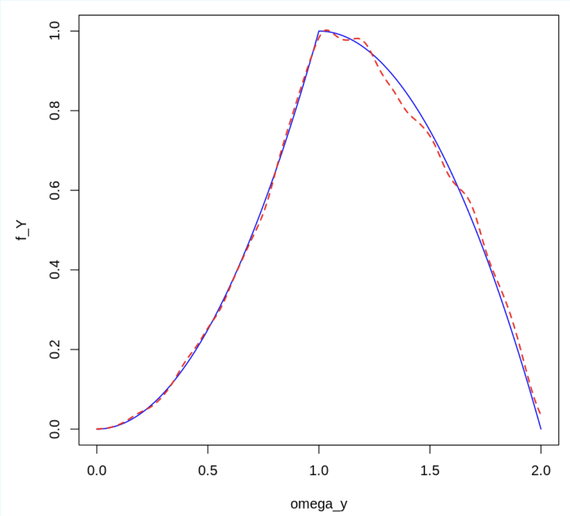
And 0 otherwise. Note both pieces coincide at $y = 1 : f_Y(1) = 1$.

4.12 [4.1] Plot $f_Y(y)$ in R. Start by defining `omega_y` as in previous questions, then apply the correct pdf for each segment. Save the pdf as vector `f_Y` . *Suggestion: use* `ifelse()`

Code in Colab ✔ [4.1] **Answer**



4.13 [4.2] Now perform a simulation. Draw 10,000 random observations for both X_1 (`X1_sim`) and X_2 (`X2_sim`; as you did for 3.4 and 3.5). Create the simulated convolution: `Y_sim` and recycle the code to plot the estimated density alongside the theoretical distribution.



Code in Colab [4.2] **Answer**

Guided Case G.4.4

Moments of random variables: theoretical session

Instructions: Work on your own; answer parts 1-3 and only answer part 4 if required by your instructor. Write down your answers in the format required by your instructor.

Part 1

We have (Y, X), a vector of random variables. Let:
$E(Y|X) = 3 + \frac{1}{2}X_1$, where $E(X) = 2$, $Var(X) = 3$,
And no residual variation: $Var(Y|X) = 0$.

1. What is $E(Y)$?

   By the Law of Iterated Expectations (LIE):

   $$E(Y) = E_X[E(Y|X)] = E\left[3 + \tfrac{1}{2}X\right] = 3 + \tfrac{1}{2}E(X)$$

   $$... = 3 + \tfrac{1}{2}(2) = 4$$

2. What is $Var(Y)$?

   By the Law of Total Variance (LTV):

   $$Var(Y) = Var(E(Y|X)) + E(Var(Y|X))$$

   $$... = Var(3 + \tfrac{1}{2}X_1) + 0 = \tfrac{1}{4}Var(X_1) = \frac{3}{4}$$

3. What is $Cov(X, E(Y|X))$?

$$\frac{3}{2}$$

$$Cov(X, E(Y|X)) = E[\{X - E(X)\}\{E(Y|X) - E(E(Y|X)))\}]$$

Note that by the LIE: $E[E(Y|X))] = E(Y) = 4$ – question 1.

$... = E[(X-2)(3 + \frac{1}{2}X - E(Y))] = E[(X-2)(\frac{1}{2}X - 1)]$
$... = \frac{1}{2}E(X^2) - 2E(X) + 2 = \frac{1}{2}E(X^2) - 2$

By properties of the variance we have:

$$Var(X) = E(X^2) - E^2(X)$$

$$\implies E(X^2) = Var(X) + E^2(X) = 3 + 4 = 7$$

Substituting back:

$$\frac{1}{2} \cdot 7 - 2 = \frac{3}{2}$$

4. Suppose $\varepsilon \sim N(0, 9)$, and $E(Y|X) = 3 + \frac{1}{2}X + \varepsilon$. What is $Cov(Y, E(Y|X)$?

With $Y = 3 + \frac{1}{2}X_1 + \varepsilon$, $\varepsilon \sim N(0, 9)$ independent of X_1:

$$\text{Cov}(Y, X_1) = \tfrac{1}{2}V(X) = \tfrac{1}{2}(3) = \tfrac{3}{2}$$

$$V(Y) = \tfrac{1}{4}V(X) + V(\varepsilon) = \tfrac{3}{4} + 9 = \tfrac{39}{4}$$

5. Show that $Cov(X, Y) = Cov(X, E(Y|X))$

Anwer not found: `w04-gc-q-022-a`

Now Let $E(Y|X) = 3 + \frac{1}{2}X_1 + X_2^2$ Where $E(X_1) = 2$, $Var(X_1) = 3$, $E(X_2) = 2$, $Var(X_2) = 1$. And X_1, X_2 independent.

6 What is $E(Y)$?

$E(Y) = 9$.

By the LIE:

$$E(Y) = E\left[3 + \tfrac{1}{2}X_1 + X_2^2\right] = 3 + \tfrac{1}{2}E(X_1) + E(X_2^2)$$

Note $E(X_2^2) = V(X_2) + [E(X_2)]^2 = 1 + 4 = 5$.
Therefore $E(Y) = 3 + 1 + 5 = 9$.

7 What is $Cov(Y, X_1), Cov(Y, X_2)$?

There's not enough information to answer. We cannot compute $Var(Y)$ or $Cov(Y, X_2)$ because we're missing higher moments of X_2: $E(X_2^3)$ and $E(X_2^4)$
For instance:

$$\text{Cov}(Y, X_2) = \text{Cov}(X_2^2, X_2) = E(X_2^3) - E(X_2^2)E(X_2)$$

.

8 Let $Var(Y|X) = Var[E(Y|X)] + E[Var(Y|X)]$. Which component corresponds to the part explained by X? Which to the part that is unexplained by X? Answer in general terms.

$V[E(Y|X)]$ is the component *explained* by X: it measures how much the conditional mean of Y varies as X changes. $E[V(Y|X)]$ is the component *unexplained* by X (residual variance): it captures the average remaining uncertainty in Y even after knowing X.

Part 2

A policy provides fixed grants for microentrepreneurs. Previous rounds of funded businesses have received on average \$5,000 in subsequent funding opportunities. Let X be subsequent funding.

1. What is the maximum share of the microentrepreneurs that receive \$100,000 or more? ***Hint**: Markov's inequality*

At most 5% of microentrepreneurs can receive \$100,000 or more. By Markov's inequality, for a non-negative random variable X :

$$P(X \geq a) \leq \frac{E(X)}{a}$$

.
Hence with $E(X) = 5,000$ and $a = 10,000$:

$$P(X \geq \$100{,}000) \leq \frac{E(X)}{\$100{,}000} = \frac{5{,}000}{100{,}000} = \frac{1}{20} = 5\%.$$

2. What is the subsequent minimum funding amount over which possibly 25% of these microentrepreneurs were funded?

We now seek the minimum threshold a such that

$$P(X \geq a) \leq 0.25 = 0.25$$

.
Again by Markov's inequality:

$$P(X \geq a) \leq \frac{5{,}000}{a} = 0.25$$

Implies $a = \frac{5{,}000}{0.25} = \boxed{\$20{,}000}$.
The minimum funding amount over which at most 25% of microentrepreneurs could have been funded.

3. Based on previous measurements, an estimate for the standard deviation of X is 1,000. Using Chebyshev's inequality, calculate what is the maximum share of the microentrepreneurs that receive \$100,000 or more. Compare with your results from part 1. Interpret the results.

> Given $E(X) = \$5{,}000$ and $\sigma_X = \$1{,}000$, by Chebyshev's inequality:
>
> $$P(X \geq \$100{,}000) \leq P(|X-\mu| \geq \$95{,}000) \leq \frac{V(X)}{(95{,}000)^2} = \frac{(1{,}000)^2}{(95{,}000)^2} = \frac{1}{9{,}025} \approx 0.011\%.$$
>
> Using Chebyshev, at most approximately **0.011%** of microentrepreneurs can receive \$100,000 or more—a much tighter bound than Markov's 5%, since we use variance information.

Part 3

Let $\{X_i\}_{i=1}^n \overset{\text{i.i.d.}}{\sim} U[1,19]$.

1. What is the mean of X_i? What is its variance?

> For $X_i \sim U[1,19]$:
>
> $$E(X_i) = \frac{1+19}{2} = \boxed{10}.$$
>
> $$V(X_i) = \frac{(19-1)^2}{12} = \frac{324}{12} = \boxed{27}.$$

2. Is the sample mean $\bar{X}_n$ a random variable? Why?

> Yes. $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$ is a function of the random variables $X_1, \ldots, X_n$. Since each X_i is random (before the sample is observed), $\bar{X}_n$ is also a random variable. Its realized value changes from sample to sample.

3. What is the expectation of $\bar{X}_n$?

> By linearity of expectation and the i.i.d. assumption:
>
> $$E(\bar{X}_n) = E\left[\frac{1}{n}\sum_{i=1}^n X_i\right] = \frac{1}{n}\sum_{i=1}^n E(X_i) = E(X_i) = \boxed{10}.$$
>
> The sample mean is an unbiased estimator of the population mean.

4. What is the probability that $\bar{X}_n$ is more than t away from the expectation of $\bar{X}_n$?

> First note $V(\bar{X}_n) = \frac{V(X_i)}{n} = \frac{27}{n}$ (i.i.d. assumption). By Chebyshev's inequality, for any $t > 0$:
>
> $$P\big(|\bar{X}_n - E(\bar{X}_n)| \geq t\big) \leq \frac{V(\bar{X}_n)}{t^2} = \frac{27}{n\,t^2}.$$

5. What happens to this probability if the sample size increases? What happens when $n \longrightarrow \infty$?

The Chebyshev bound from question 4 is $\frac{27}{n\,t^2}$. As n increases, the bound decreases. As $n \to \infty$:

$$P\left(|\bar{X}_n - E(X_i)| \geq t\right) \leq \frac{27}{n\,t^2} \to 0.$$

This is the **Weak Law of Large Numbers**: $\bar{X}_n \xrightarrow{p} E(X_i) = 10$. With a larger sample, the sample mean concentrates ever more tightly around the true population mean.

Part 4

A set of 6 grants will be awarded at random to participating firms, 4 of which are specialty grants and 2 regular grants. There are 40 prospective firms, of which 14 are microenterprises, and 26 are SMEs. Among these firms, 8 microenterprises and 8 SMEs are certified, and may receive a specialty grant.

1. What is the expected number of:
   - grants for microenterprises?

     2.7

     The expected number of grants for microenterprises combines the two grant types. For specialty grants (pool = 16 accredited firms, 8 of which are microenterprises):

     $$E[\text{specialty grants to micro}] = 4 \times \frac{8}{16} = 2.$$

     For regular grants (pool = 40 firms, 14 of which are microenterprises):

     $$E[\text{regular grants to micro}] = 2 \times \frac{14}{40} = \frac{7}{20} = 0.7.$$

     Total: $E[\text{grants to micro}] = 2 + 0.7 = 2.7$.
   - Grants for SMEs?

     3.3

     For specialty grants (pool = 16 accredited firms, 8 of which are SMEs):

     $$E[\text{specialty grants to SMEs}] = 4 \times \frac{8}{16} = 2.$$

     For regular grants (pool = 40 firms, 26 of which are SMEs):

     $$E[\text{regular grants to SMEs}] = 2 \times \frac{26}{40} = \frac{13}{20} = 1.3.$$

     Total: $E[\text{grants to SMEs}] = 2 + 1.3 = 3.3$.
   - Grants for certified firms?

By definition, all 4 specialty grants can go to certified firms. For regular grants (pool = 40, 16 are certified):

$$E[\text{regular grants to certified}] = 2 \times \frac{16}{40} = \frac{4}{5} = 0.8$$

Total: $E[\text{grants to certified}] = 4 + 0.8 = 4.8$

– Grants for certified microenterprises? Grants for certified SMEs?

$$E[\text{grants to certified micro}] = 4 \cdot \frac{8}{16} + 2 \cdot \frac{8}{40} = 2 + \frac{2}{5} = \frac{12}{5}$$

$$E[\text{grants to certified SMEs}] = 4 \cdot \frac{8}{16} + 2 \cdot \frac{8}{40} = 2 + \frac{2}{5} = \frac{12}{5}$$

– Grants for any type of firm?

$$E[\text{grants to any firm}] = 6$$

– How does the number of expected grants for microenterprises change with the number of accredited microenterprises? Is there a minimum number of accredited microenterprises which makes the expected number of grants for microenterprises higher than the expected number of grants for SMEs (holding everything else constant)?

Let c denote the number of certified microenterprises. Then:

$$E[\text{grants to micro}] = \frac{4c}{c+8} + \frac{7}{10}$$

Setting $E[\text{grants to micro}] > E[\text{grants to SMEs}]$:

$$\frac{4c}{c+8} + \frac{7}{10} > \frac{32}{c+8} + \frac{13}{10}$$

$$\frac{4(c-8)}{c+8} > \frac{3}{5}$$

$$20(c-8) > 3(c+8)$$

$$17c > 184$$

$$c > \frac{184}{17} \approx 10.82$$

The minimum number of certified microenterprises required is $c^* = 11$.

14.310x Flipped Classroom

Week 5 Answers

Central Limit Theorem and the Law of Large Numbers
More advanced simulation in R

Checklist

- ☐ (**Optional**) Watch Youtube video 1, an informal introduction to the Law of Large Numbers.
- ☐ We will be working with the *Weak Law of Large Numbers*. Have a simple statement of the theorem such as this one readily available.(Requirement)
- ☐ Complete Open Case Study 2 (Session 1)

Case study O.5.2

The law of large numbers (LLN)

Instructions: Work in pairs or groups of 3. Answer all questions and, when appropriate, discuss your results.

Notebook: Scrap notebook (some comments)

Notebook: O.5.2Proposed Solutions
Worked solutions to coding questions; they include previously reviewed elements (Coding Labs 3 & 4, and cases G.2 & 3). Use as guidance for the tools students should apply.

There are multiple ways to go about many of the questions and coding tasks, and you may solve them as you see fit. You will also need to provide open-ended answers to explain a number of the results you get. That said, some tasks will rely heavily on what we learned about *Simulation* in the Week 4 Coding Lab; your team may want to keep the notebook handy.

Scenario: We will be working with samples of five different continuous random variables as follows:

$$X_1 \sim N(5,1)$$
$$X_2 \sim N(5,9)$$
$$X_3 \sim U[2,8]$$
$$X_4 \sim Exp(\frac{1}{5})$$
$$X_5 = \frac{X_a}{X_b} \quad | \quad X_a, X_b \overset{iid}{\sim} N(0,1)$$

---

1. Get these distributions' expectations $\mu \equiv E(X)$ and variances $\sigma^2 \equiv Var(X)$.

$$\mu_1 = 5, \quad \sigma_1^2 = 1$$
$$\mu_2 = 5, \quad \sigma_2^2 = 9$$
$$\mu_3 = \tfrac{1}{2}(8+2) = 5, \quad \sigma_3^2 = \tfrac{1}{12}(8-2)^2 = 3$$
$$\mu_4 = \frac{1}{\frac{1}{5}} = 5, \quad \sigma_4^2 = \frac{1}{(\frac{1}{5})^2} = 25$$

X_5 is a Cauchy distribution with undefined mean (and variance).

Instructor note: It is not necessary for students to be familiar with the Cauchy distribution. It suffices for them to understand that its moments are undefined; X_5 will be used as an example where the LLN doesn't apply. If possible, some discussion of how X_5 has zero median and mode may be relevant, given the plot presented in question 2.b.

2. For each RV, first run `set.seed(2)` **once**, then draw two samples: one of size $n = 5$ and another of size $n = 50$. Take their respective sample means:

$$\bar{X}_n = \frac{x_1 + x_2 + ... + x_n}{n}$$

RV_name	sample_size	sample_mean
<chr>	<dbl>	<dbl>
normal(5,1)	5	4.9330305
normal(5,1)	50	5.0937131
normal(5,3)	5	4.7990915
normal(5,3)	50	5.2811392
uniform (2, 8)	5	5.0869687
uniform (2, 8)	50	4.9876429
exponential (5)	5	0.1694796
exponential (5)	50	0.2173477
normal ratio (1)	5	-0.2296840
normal ratio (1)	50	-0.8933414

Per Chebyshev's inequality and the Law of Large Numbers (LLN) we have that

$$P(|\bar{X}_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{n\epsilon^2}$$

For arbitrary $\epsilon > 0$.

(a) For which sample size would you expect the distance between the sample mean and its expected value $|\bar{X}_n - \mu|$ to be smaller?

For $n = 50$, since the probability of the distance being smaller than some small number ϵ decreases as n increases:

$$\tfrac{1}{50} \cdot \frac{\sigma^2}{\epsilon^2} < \tfrac{1}{5} \cdot \frac{\sigma^2}{\epsilon^2}$$

(b) Does this happen in all cases? If not, briefly discuss why.

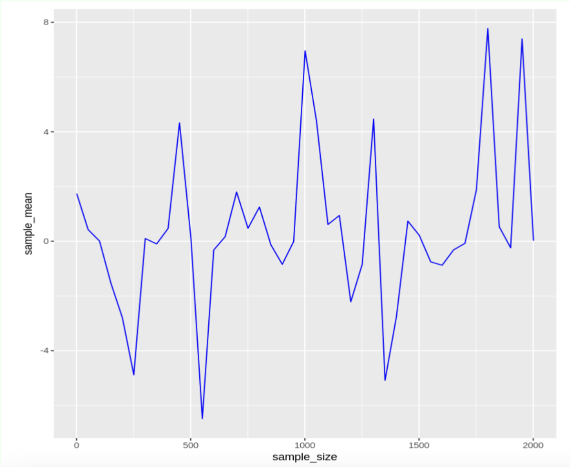
No. As shown above, for X_1 the sample mean with $n = 5$:

$|5 - 4.9330305| \approx 0.06$

Which is smaller than that of the larger sample with $n = 50$:
$|5 - 5.0937131| \approx 0.09$

The LLN is a probabilistic statement; increasing n makes it more unlikely for the difference to be greater than ϵ —note the probability also depends on σ^2— but it doesn't make it impossible.

For a random sample (even if large), especially when the RV's variance is large, there may be realizations that deviate substantially from its expectation.

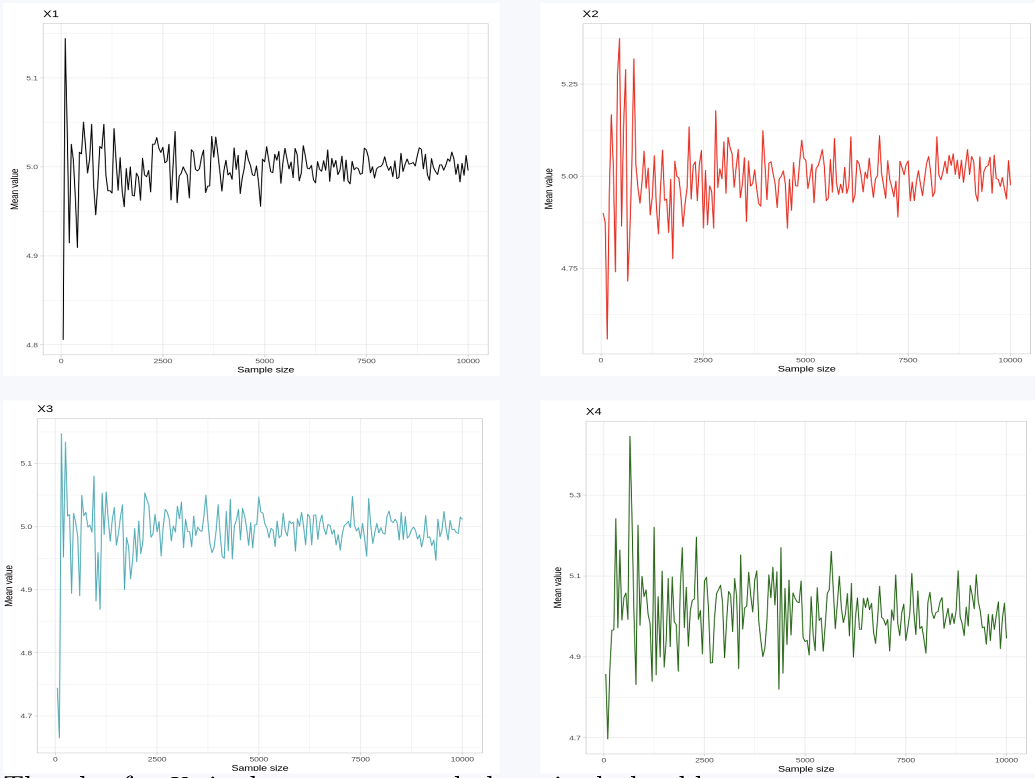


3. For each RV, draw random samples of sizes $n = 1, 50, 100, 150, ..., 2000$ and create a plot similar to the one above — the plot must be reproducible.

   Suggestion: you can start by saving the sample means in a data frame like the one below

A data.frame: 5 × 3

RV_name	sample_size	sample_mean
<chr>	<dbl>	<dbl>
normal(0,1)	1	-0.5604756
normal(0,1)	10	0.2530814
normal(0,1)	20	-0.1478493
normal(0,1)	30	0.1767775
normal(0,1)	40	0.1003377



The plot for X_5 is the one presented above in darker blue.

(a) Is the LLN met for these RVs? If not, discuss the reason.

It is met for X_1, X_2, X_3, X_4. All converge to their mean: 5.
For X_5, the LLN is not met, because one of the requirements for it to apply is the Random Variable's variance must be finite: $Var(X) < \infty$. This is not the case for a normals ratio, where the variance is undefined (infinite). Even if $E(X_5) = 0$, we would still observe oscillating behavior around said value.

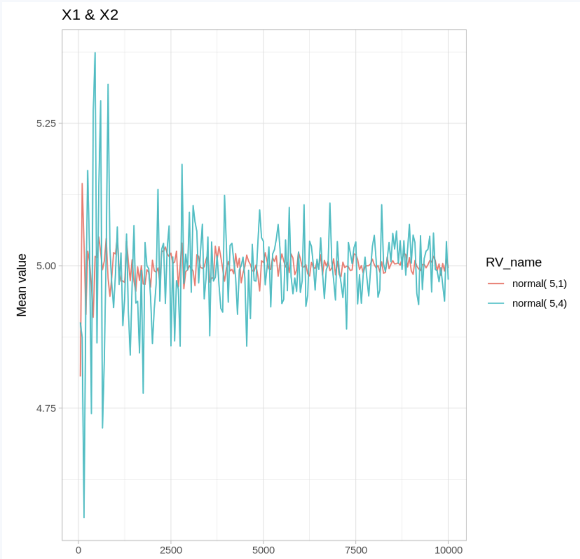
(b) As n increases, how does the sequence of sample means behave? Does the trajectory *steadily* converge to μ?

The sample mean converges as n increases, but not steadily. Note the deviations from 5 become gradually smaller (depending on the variance, their magnitude dies faster or more slowly).

(c) Discuss how your previous answer may explain what you observed in question 2..

Since the LLN is a probability statement, it is rather common to observe the difference $|X_n - \mu|$ randomly increasing with respect to the previous smaller sample — as mentioned before, said differences become noticeably smaller.

(d) Plot X_1 and X_2 together. Do they converge to the same μ? How are they different? Why?



Yes, they both converge to $\mu = 5$. The difference lies in the rate at which they converge; since $\sigma_2^2 > \sigma_1^2$, we can expect the plot for X_2 to be much "noisier". Given any threshold $\epsilon > 0$:

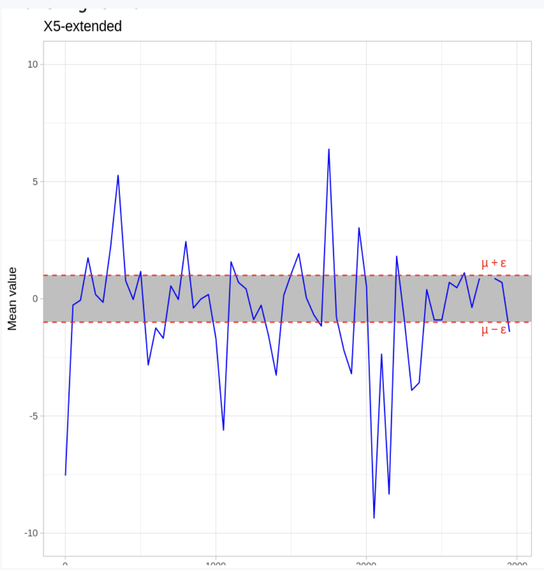
$$P(|\bar{X}_5 - \mu| \geq \epsilon) > P(|\bar{X}_{50} - \mu| \geq \epsilon)$$

(e) From which RV did the example plot above likely originate?

X_5

Note the plot above doesn't exhibit the same convergence towards $\mu = 0$ as those for X_1 to X_4. Since the LLN is not met for X_5, we may conclude (alongside the student's own plot) it is RV 5.

– Suppose the sample means keep oscillating in roughly the same way around μ as $n \to \infty$. Explain precisely what part of the Weak Law or Large numbers would not be met.



From the Chebyschev statement above:

$$P(|\bar{X}_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{n\epsilon^2}$$

This means that for any given threshold ϵ (e.g., $\epsilon = 1$ chosen for the plot) and finite variance σ^2, as n increases, the instances where $\bar{X}_n$ falls **outside** the band should become very rare because $\lim_{n\to\infty} \frac{\sigma^2}{n\epsilon} = 0$.

If the sample mean oscillations around the mean keep their magnitude, the implication is the probability of $|\bar{X}_n - \mu| \geq \epsilon$ will not decrease.

4. Now consider convolution $Y = X_1 + X_2$. Compute $E(Y)$ and $Var(Y)$.

Students should consider whether X_1 and X_2 are independent for their computations. Advice them to momentarily assume they are.

$$E(Y) = E(X_1 + X_2) = \mu_1 + \mu_2 = 10$$

$$Var(Y) = Var(X_1 + X_2) = Var(X_1) + Var(X_2) + 2Cov(X_1, X_2)$$

Assuming independence: $Cov(X_1, X_2) = 0$

$$... = 1 + 9 = 10$$

5. Simulate Y with a sample of size $n = 10,000$ in two different ways:

- ```
  set.seed(123)
  y <- rnorm(10000, 5, 1) + rnorm(10000, 5, 3)
  ```
- ```
  set.seed(123)
  x1 <- rnorm(10000, 5, 1)
  set.seed(123)
  x2 <- rnorm(10000, 5, 3)
  y2 <- x1+ x2
  ```

Compute each sample's mean $\bar{Y}_n$ and variance s_y^2. Explain the differences (if any) between $\bar{Y}_n^{(1)}, \bar{Y}_n^{(2)}, \mu_y$, and $s_{y,(1)}^2, s_{y,(2)}^2, \sigma_y^2$

Given $\mu_y = \sigma_y^2 = 10$, we have:

	$\bar{Y}_n$	s_y^2
One seed	9.970	10.061
Repeated seed	9.991	15.956

The sample variance is much larger when sampling twice from the same seed. This happens because, even with different parameters σ^2, the same random number seed for X_1, X_2 violates independence; yielding nonzero covariance. And since (question 4.):

$$\sigma_y^2 \equiv Var(Y) = Var(X_1) + Cov(X_1, X_2) + Var(X_2)$$

Then $Cov(X_1, X_2) \approx 5 > 0$.

Independent, but non-identically distributed samples

Sometimes we cannot directly choose the population from which we draw. A common example of this situation is when sampling is done at the level of **clusters** or **groups**, such as schools or villages, rather than individuals. Each cluster may have its own distribution, so the resulting sample is not identically distributed. Note that the draws are still independent: no observation depends on the value of any other. Drawing a cluster only adds several random observations from a distribution "at once".

The following questions are meant to help you work out how the LLN applies in these cases.

Suppose we have 2 independent random variables:

$$Y \sim N(\mu_y, \sigma_y^2)$$
$$Z \sim N(\mu_z, \sigma_z^2)$$

We have $M = \{y_1, y_2, ..., y_{n_0}, z_1, z_2, ..., z_{n_1}\}$ a random sample of size n where n_0 observations come from Y, n_1 observations come from Z, and $n_0 + n_1 = n$.

6. Write down $\bar{M}$, the sample mean:

$$\bar{M} = \frac{(\quad\quad) \; + \; (\quad\quad)}{\square}$$

$$\bar{M} = \frac{(\sum_{i=1}^{n_0} y_i) \; + \; (\sum_{j=1}^{n_1} z_j)}{n}$$

7. Compute $E(\bar{M})$ in terms of $w_y \equiv \frac{n_0}{n}, w_z \equiv \frac{n_1}{n}, \mu_y$, and μ_z. What type of average is this?

$$E(\bar{M}) = E\left(\frac{\sum_{i=1}^{n_0} y_i}{n}\right) + E\left(\frac{\sum_{j=1}^{n_1} z_j}{n}\right) = \frac{n_0 \cdot \mu_y}{n} + \frac{n_1 \cdot \mu_z}{n}$$

$$\ldots = w_y \cdot \mu_y + w_z \cdot \mu_z$$

This is a weighted average of the expected values of Y and Z. The weights $w_q = \frac{n_q}{n}$ denote the share of realizations from each random variable in the sample.

8. Compute the sampling variance $Var(\bar{M})$. You should get an expression $= \frac{\square}{n}$, where $\square$ is the same type of average as in the previous question.

By independence and because the sample is random, any covariance term is zero: $Cov(y_i, z_j) = Cov(y_i, y_j) = Cov(z_i, z_j) = 0, \quad i \neq j$

$$Var(\bar{M}) = \tfrac{1}{n^2} Var\left(\sum_{i=1}^{n_0} y_i\right) + \tfrac{1}{n^2} Var\left(\sum_{j=1}^{n_1} z_j\right) = \frac{n_0 \cdot \sigma_Y^2}{n^2} + \frac{n_1 \cdot \sigma_Z^2}{n^2}$$

$$\ldots = \frac{w_Y \cdot \sigma_Y^2}{n} + \frac{w_Z \cdot \sigma_Z^2}{n} = \frac{w_Y \cdot \sigma_Y^2 + w_Z \cdot \sigma_Z^2}{n}$$

9. With your previous answers in mind, suppose we now have sample R with the following mixing:
   - 30% from X_1
   - 20% from X_2
   - 40% from X_3
   - 10% from X_4

   (a) What are $E(R)$ and $Var(R)$?

   The weights:

   $$w_1 = 0.3,\ w_2 = 0.2,\ w_3 = 0.4,\ w_4 = 0.1$$

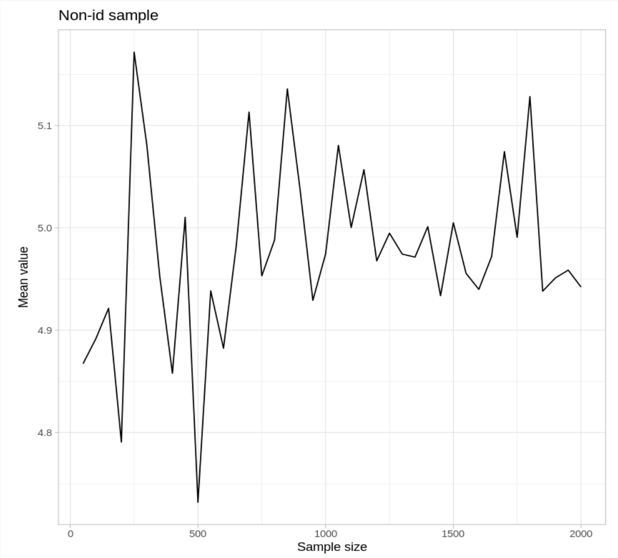
   $$E(R) = 5$$

   Irrespective of weights, since $\mu_1 = \mu_2 = \mu_3 = \mu_4 = 5$.

   $$Var(R) = \frac{w_1 \cdot \sigma_2^2 + w_1 \cdot \sigma_2^2 + w_3 \cdot \sigma_3^2 + w_1 \cdot \sigma_4^2}{n}$$

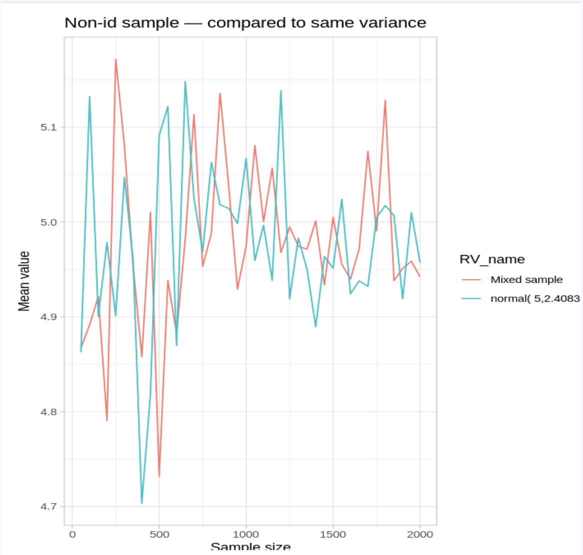
   $$\ldots = \frac{0.3 \cdot 1 + 0.2 \cdot 9 + 0.4 \cdot 3 + 0.1 \cdot 25}{n} = \boxed{\frac{5.8}{n}}$$

10. Plot $\bar{R}_n$ for $n = 50, 100, \ldots, 2000$.



(a) Does $\bar{R}_n$ exhibit the same type of convergence? *Optional:* $\bar{R}_n$ alongside $\bar{X}_n$ for $X \sim N(5, 5.8)$ for the same sample sizes for reference.

Yes. There is no noticeable difference; the LLN applies in exactly the same way (including convergence).



(b) Discuss what would happen with $\bar{R}_n$ if we added X_5 draws to the mix.

To the extent that the share of X_5 increases, the sample mean will become more unstable. Because the mode and median of X_5 are equal to 0, $\bar{R}_n$ will still oscillate around 5, but convergence will cease to exist.

14.310x Flipped Classroom

Week 6 Answers

Estimator properties
Hypothesis testing and confidence intervals
Further simulation in R

Checklist

- ☐ Complete Coding Lab 5 (Session 1)
- ☐ Complete Guided Case Study 5 (Session 2)
  - ☐ Dataset: `consultations.csv`

Coding Lab L.6.5

An intrudction to hypothesis testing and Confidence Intervals in R

Instructions: Work individually. Solve all exercises in the corresponding Colab notebook. Record your answers and/or code in your copy of the notebook.

Notebook: Coding Lab 5

Instructor Notebook: Coding Lab 5

Find answers to all coding lab exercises in the corresponding Colab notebook.

Submit your work in the format required by the instructor.

Guided case study G.6.5

Examining *Clinicia*'s financial incentives for physicians

Instructions: Work in pairs or groups of three to solve all questions.

Notebook: Case Study 5

Dataset: consultations.csv

The team or pair may work on one notebook collaboratively. One team member needs to share editor access their the Colab Notebook with the rest.

Instructor Notebook: Answers to G.6.5

Find answers to all code-related questions in this Guided Case. Each coding-relevant question links to its corresponding answer section in this notebook.

Scenario:

Medical consultations take place in an environment with substantial **information asymmetry**. Patients are usually not qualified to determine whether diagnoses are accurate or if a prescribed treatment is correct. This creates incentives for healthcare providers that can alter treatment intensity and the types of drugs prescribed, directly impacting expected **health outcomes** and total **treatment costs**. Especially when a provider stands to benefit financially from a certain treatment avenue over others.

Clinicia's hospital system is reported to have unusually high rates of **antibiotic prescription**, even for conditions where guidelines recommend more conservative treatment. Overuse of antibiotics not only raises treatment costs unnecessarily, but also accelerates **microbial resistance**, reducing the effectiveness of essential drugs. The country's Ministry of Health is aware of the reports and wants to test more formally for any differences in treatment cost (or prescription rates) in the presence of financial incentives. As part of the policy team, you are tasked with performing this testing.

Medical records in Clinicia include audio logs of consultations (the audio is altered to protect patient and provider privacy) along with administrative information on diagnoses, procedures, and prescriptions. The Ministry was granted access to a fully anonymized, random sample of these records. Your team processed all the data and produced the `consultations` dataset. For each medical consultation:

- `incentive =1`: Indicates the patient is buying the prescribed drugs at the hospital or at a commercially related pharmacy.
- `request =1`: Indicates patient requested antibiotics.

- `antibiotics =1`: Indicates the physician prescribed antibiotics.
- `totalprice`: The market price (in USD) of the drugs prescribed to the patient at the consultation.
- `multiple =1`: Indicates the physician prescribed two or more distinct drug categories (e.g., analgesic + antibiotic + expectorant)
- `grade2 = 1`: Indicates the physician prescribed a drug with a high potential for abuse and dependence (e.g., oxycodone, metanphetamines, Aderall).

---

1. [1] Load or install the `dplyr` and `pwr` packages. Read in `consultations` using the dataset URL provided above.

   Code in Colab [1] Answer

Part 1

2. [2]

   Table 4: Number of patients in each group

	Requested	Did not request	Total
Incentive			
No incentive			
Total			■

   Fill the table above. Determine the number of patients in each group I, R, and each subgroup $n_{I,R}$, $n_{I,NR}$,$n_{NI,R}$, $n_{NI,NR}$.

   Code in Colab [2] Answer

   > Balanced groups: $n_{I,R} = n_{I,NR} = n_{NI,R} = n_{NI,NR} = 80$, so $n_I = n_{NI} = n_R = n_{NR} = 160$ and $N = 320$.
   >
   > **Instructor Note**: The survey is fictitiously balanced to avoid the proportion tests and the last questions (on power and MDE) from becoming excessively complex.

3. [2.1] As you know, the Ministry is concerned with both antibiotic over-prescription and treatment costs. Start by calculating the difference in average `totalprice` and `antibiotics` with and without `incentive`. Save as `pricediff_I` and `drugdiff_I`, respectively. Are the differences consistent with the presence of a financial incentive?

   Code in Colab [2] Answer

   > $\widehat{\tau}_{\text{price}} = \bar{p}_I - \bar{p}_{NI} = \$120.03 - \$42.35 = \$77.68$
   > $\widehat{\tau}_{\text{antibiotic}} = \bar{a}_{NI} - \bar{a}_I = 0.119 - 0.700 = -0.581$
   > Both differences are consistent with a financial incentive driving higher prescription rates and costs.

4. [2.2] More formally, use `t.test()` to test the hypothesis that the average treatment cost is the same for both groups. Assume the same unknown variance. Can the price equality be rejected at the 99% confidence level?

Code in Colab [2] **Answer**

$t = 14.65$, $df = 318$ (pooled). $p \le 0.001$. **Reject** H_0 at the 99% confidence level.

5. [2.3] We cannot perform a similar t-test for `antibiotics`. Read about the proportion test and hand-code it to test for equality in prescription rate. Report the value of the statistic, the first four non-zero decimals of the p-value, and whether or not you would reject at 1% significance.

Code in Colab [2] **Answer**

$\hat{p}_I = 0.700$, $\hat{p}_{NI} = 0.119$, $\hat{p}_{\text{pool}} = 0.409$.
$z = 10.57$, $p \approx 5.5 \times 10^{-26}$. **Reject** H_0 at 1% significance.

6. Reporting back to the ministry, how confident are you that more profitable patients also face higher antibiotic prescription and higher treatment costs in Clinicia's population?

Very confident. Both tests yield overwhelming evidence: the t-test for `totalprice` gives $t = 14.65$ ($p \ll 0.001$) and the proportion test for `antibiotics` gives $z = 10.57$ ($p \approx 0$). The evidence rules out random sampling variation as an explanation at any conventional significance level. In Clinicia's population, financially incentivized prescriptions involve systematically higher costs and higher antibiotic rates.

Given the strength of the evidence, the ministry wants to assess an additional driver of microbial resistance: the culture around antibiotic prescription. The ministry would like you to answer several questions (see below,) with this additional variable in the data you already have.

7. Repeat 3., 4. and 5., this time with groups defined by `request`. What can you conclude?

Grouping by `request`:
$\hat{\tau}_{\text{price}} = \$26.36$. t-test: $t = 3.93$, $df = 318$, $p < 0.001$. **Reject** at 1%.
$\hat{\tau}_{\text{antibiotic}} = 0.169$. Proportion test: $z = 3.07$, $p = 0.002$. **Reject** at 1%.
Both differences are significant, but the financial incentive difference is three times that of the request. We have not yet mapped out the relationship between the two (see 10.).

8. [3] Compute the average and standard deviation of `antibiotics` and `totalprice` for each of the four patient subgroups defined by all combinations of `incentive` and `request`. Save the object as `outcomes`.

Code in Colab [3] **Answer**

Group	n	$\bar{a}$	s_a	$\bar{p}$ (USD)
I=0, R=0	80	0.100	0.302	\$38.17
I=0, R=1	80	0.138	0.347	\$46.54
I=1, R=0	80	0.550	0.501	\$97.86
I=1, R=1	80	0.850	0.359	\$142.20

9. How sensible is it to assume our outcomes draws for the 4 patient groups are identically distributed (i.e., sampled from the same distribution)?

> Questionable. The four groups are defined by patient purchasing behavior and self-reported drug demand, both correlated, for instance, with: income, insurance status, and case severity, etc. A patient buying drugs at the hospital may differ systematically from one who does not. The i.i.d. assumption requires these differences to be irrelevant to the outcome—plausible under randomization, but this data consists of administrative records.
>
> **Instructor note**: Students may answer and argue the assumption differently, but the test statistic specification for 10. is consistent with this answer.

10. [5] With your previous answer in mind, test the hypothesis conducive to solving the following question by the Ministry:

    (a) Among patients **with no financial incentives**, is there a difference in their treatment costs when they request antibiotics?
    (b) Are patients with no incentives prescribed antibiotics more often if they request them?
    (c) For those patients buying their prescription at the hospital, is there a difference in treatment cost if they ask for antibiotics?

    For each test, report $\hat{\tau}$ the estimated difference, the value of the appropriate statistic, the p-value, and whether H_0 is rejected at $\alpha = 0.05$

    Code in Colab [5] **Answer**

> **5a.** No incentive, `totalprice` by `request`: $\hat{\tau} = \$8.37$, $t = 1.935$, $p = 0.054$. **Fail to reject** at $\alpha = 0.05$.
>
> **5b.** No incentive, `antibiotics` by `request`: $\hat{\tau} = 0.038$, $z = 0.733$, $p = 0.464$. **Fail to reject** at $\alpha = 0.05$.
>
> **5c.** Hospital-buying, `totalprice` by `request`: $\hat{\tau} = \$44.34$, $t = 4.910$, $p \approx 0$. **Reject** at $\alpha = 0.05$.
>
> Patient demand alone (5a, 5b) produces no detectable effect in the absence of a financial incentive. The financial channel (hospital purchase, 5c) amplifies prescribing substantially.

11. [7] From 10.b, hand-code a function `ci_props()` that takes a significance level `alpha`, and outputs a vector of length 2; the bounds of the 95% confidence interval for the difference $\hat{\tau}$ in antibiotic prescription rates. What is the smallest α for which the CI doesn't include 0? At this alpha, what is the probability that the prescription rates are in fact equal, but we reject the equality?

    Code in Colab [7] **Answer**

From Q5b: $\hat{\tau} = 0.0375$, $\widehat{\text{se}} = 0.051$.

95% CI: $[-0.063,\ 0.138]$. Includes 0, consistent with non-rejection in Q5b.

The CI excludes 0 only for $\alpha \approx 0.46$ (a 54% confidence interval). At this level, the Type I error rate under H_0 would be 46% — far above any reasonable threshold. The evidence for a demand-only effect on antibiotic prescription is negligible.

Part 2

12. Suppose now you implement a new experiment, where you want to be able to detect an effect with a certain probability, if present. What is this probability commonly called in statistics?

Statistical power (or the power of the test), denoted $1-\beta$, where β is the Type II error probability. It is the probability of correctly rejecting H_0 when the alternative H_β is true.

13. Your colleagues at the policy team define an economically meaningful effect to be detected of x magnitude. What is x called?

The **Minimum Detectable Effect** (MDE), also referred to as the **effect size** d. It is the smallest true difference the experiment is designed to detect at significance α with probability $1-\beta$.

14. Consider the distributions under H_0 (zero effect) and H_β (β effect, different from 0). What area of the curve of these distributions corresponds to the power of the test?

Power is the area under the H_β distribution that falls within the rejection region defined by H_0 — i.e., to the right of the critical value c_α for a right-sided test (or in both tails for a two-sided test). The complement — the area under H_β in the non-rejection region of H_0 — is the Type II error rate β.

15. [8] For a new experiment where the control group is expected to have a total medication cost of \$100, compute the required sample size per group to detect effects of $\Delta \in \{\$10, \$20, \$40, \$80\}$ with 80% power and 5% significance (two-sided). Assume a pooled standard deviation of $\sigma = 60$.

Hint: the standardized effect size is Cohen's $d = \Delta/\sigma$. Use `pwr.t.test()`.

Code in Colab [8] **Answer**

$\sigma = 60$, $\alpha = 0.05$, power = 0.80, two-sided. Cohen's $d = \Delta/\sigma$.

Δ (USD)	d	n per group
\$10	0.167	566
\$20	0.333	142
\$40	0.667	36
\$80	1.333	9

16. [9] Compute the required sample size per group for detecting standardized effects of $d \in \{0.05, 0.10, 0.20, 0.40\}$, with $\alpha = 0.05$ and 80% power (two-sided).

Code in Colab [9] **Answer**

$\alpha = 0.05$, power = 0.80, two-sided.

d	n per group
0.05	6,280
0.10	1,570
0.20	393
0.40	99

17. Comment on your results. Explain the relation between sample size and effect size. Is this relation proportional?

Sample size and effect size have an **inverse** relationship: smaller effects require larger samples. The relationship is not proportional — it follows an inverse-square law: $n \propto 1/d^2$. Halving the effect size roughly **quadruples** the required n. This has direct practical implications: detecting subtle prescribing differences in observational data requires orders of magnitude more observations than detecting large shifts.

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Week 7 Answers

The potential outcomes framework
Experimental design and randomized evaluation in economics

Checklist

- ☐ Set aside this Wikipedia article on testing two-proportions hypotheses, in case you need it during Session 2 (Requirement)
- ☐ Skim this Wikipedia article discussing the Local Average Treatment effect (LATE) , in advance of Session 2. (Requirement)
- ☐ Complete Guided Case Study 6 invidually (Session 1)
  - ☐ Retrieve the `students` dataset here
- ☐ Complete Open Case Study 3 in pairs or teams of 3. (Session 2)
  - ☐ You will continue to use `students`

Case study G.7.6

Evaluating AI-assisted learning on student outcomes

Instructions: Work on your own; read the scenario and answer the questions. Type your answers in the format required by the instructor.

To work on the `students` dataset you may use either a Colab notebook or your own installation of R.

Dataset: students.csv

Your code will not be evaluated, but keep your R script or notebook tidy, as you may need to review some of your answers during Session 2.

Scenario:

You are the Government of *Novaria*'s new Minister of Education. The Prime Minister has tasked you with evaluating a primary education policy recommendation: the rollout of AI-assisted learning for mathematics curricula in grades 5-8.

The proposed program, *Project Mentor*, involves deploying a large language model (similar to ChatGPT) named *AlgebrAI* specifically trained and fine-tuned for elementary math tutoring. AlgebrAI's interface is tailored to deliver interactive, one-on-one tutoring sessions to students. The AI mentor adapts to each student's skill level and provides problem-solving guidance, hints, and feedback designed to help the students master their grade's math curriculum.

Each participating school receives a number of tablets with AlgebrAI pre-installed, configured for offline-first use and automatically synced with central servers when internet is available. Students selected for treatment attend 20-minute tutoring sessions per day under the supervision of a facilitator.

The Prime Minister believes Project Mentor can boost test scores nationwide, but political opponents have raised concerns over cost and long-term efficacy. You are now in charge of evaluating the impact of the program in grades. Your team provides you with:

1. A 6th-grade math test designed to perfectly measure domain of the curriculum in a scale from 0 to 100.
2. A list of 1,000 students enrolled in 6th grade across Novaria, picked at random — part of the `students` dataset. This list contains only the following variables:
   - `unit`: a consecutive number assigned to the student.
   - `W_school`: Indicates whether the student's school is managed by the government ($W_{school} = 1$) or if it is privately managed ($W_{school} = 0$)

You have authority to apply the exam to any 6th grader in Novarnia, and you can implement the program (tablet usage and monitor time) in all government-managed schools, but to include any students attending a private school to the program you must first obtain authorization from their school board.

Exercises

Consider $T_i \in \{0, 1\}$ the treatment status of student $i = 1, 2, ..., 1000$ — $T_i = 1$ if treated $T_i = 0$ if not treated. Potential outcomes $y_i(T_i)$ in `students`, measured in test results (grades 0 to 100) are defined:

- `y0`: vector $Y(0)$, assume we can't observe it unless specified.
- `y1`: vector $Y(1)$, assume we can't observe it unless specified.

1.1 What is the value of $y_3(1)$? Describe its meaning —in terms of the potential outcomes framework.

> 80.27
> It is the outcome to be observed in unit/student 3 if treated.

1.2 What is the value of $y_5(0)$? Describe its meaning.

> 51.55
> The outcome to be observed for student 5 in the absence of treatment.

1.3 Compute $\bar{Y}(1)$. Describe its meaning.

> 75.01332
>
> The average outcome if all students are treated.

1.4 Suppose $T = 0$. What is the value of y_{20}^{obs} and y_{40}^{miss} ? Briefly explain why.

> Since $T = 0$, all no student will be treated. The potential outcomes we would observe in this scenario are y_0, and the ones we do not observe are y_1
>
> $$y_{20}^{obs} = y_{20}(0) = 69.19$$
>
> $$y_{40}^{miss} = y_{40}(1) = 67.98$$

1.5 Suppose $T_i = 1$ for all $i = 1, 2, 3, ..., 1000$. What is $\bar{Y}^{obs}$?

> In a world where all units are treated, vector $Y(1)$ would be fully observed:
>
> $$\bar{Y}^{obs} = \bar{Y}(1) = 75.01332$$
>
> as in 1.3

1.6 Imagine you can observe both potential outcomes.

(a) What is the treatment effect of Project Mentor in student 245?

> The treatment effect for any unit is the difference between the potential outcomes with and without treatment:
>
> $$TE_{245} = y_{245}(1) - y_{245}(0) = 12.62$$
>
> Additional points in the test.

(b) What is the estimated Average Treatment Effect (ATE) of Project Mentor?

> Similarly, the average treatment effect (ATE) for all units is the mean of the treatment effects for units $i = 1, 2, ..., 1000$
>
> $$ATE = \frac{1}{n}\sum_{i=1}^{n}(y_i(1) - y_i(0))$$
>
> $$... = 9.956$$

(c) Imagine the ATE in 1.6b is a valid estimate of the treatment effect across Novarna. Would it support the Prime Minister's claims?

> It would support the PM's claims, as it would show that students in Project Mentor score 10 points more than they would without it.

In practice, only Y^{obs} will be available after applying the exam. You will observe **one test score per student**, as well as the students' treatment status: either treated or untreated.

After careful consideration, your team assigned N_0 students to the control group and N_1 to the treatment group, out of the $N_0 + N_1 \equiv N = 1000$ sampled. The assignment criteria included logistics, the school-year calendar, operation costs and potential political opposition. Treatment was allocated amongst students according to the rule

$$T = W_{school}$$

2.1 Explain the assignment rule in simple words

> Students attending government-managed schools will be treated. Those in private schools will not be treated.

2.2 For each of the assignment criteria, provide a brief circumstance that likely motivated Novaria's government to conclude this was the best allocation.

> We seek reasons that make it harder for policymakers *in the field* to assign treatment to private schools relative to public schools.
>
> - **Logistics**: Maybe coordinating the sessions (monitors, physical space) is much harder because private schools have competing priorities over which the Ministry has no input.
> - **Operation costs**: Private schools are not required to be infrastructure-ready; a place to store the tablets, electrical outlets, etc. Forcing the Ministry to provide the necessary inputs.
> - **Political opposition**: Since authorization to implement Project Mentor is required in private schools, there may be critical, opposing voices within the school authorities or communities.

2.3 What is the value of N_0?

> 493
```
nrow(students[students$W_school == 0, ])
```

2.4 What is the value of N_1?

> 507
```
nrow(students[students$W_school == 1, ])
```

2.5 Create a variable for $Y^{obs}(W)$, and name it `yobs_w`. With this variable:

```
students$yobs_w <- ifelse(students$W_school == 1, students$y1,
students$y0)
```

(a) Compute the value of $\bar{Y}^{obs}(1)$.

> 70.099

(b) Compute the value of $\bar{Y}^{obs}(0)$.

> 70.238

2.6 Write down both expressions $\bar{Y}^{obs}(\cdot)$ more formally, in terms of summations.

$$\bar{Y}^{obs}(0) = \frac{1}{N_0} \sum_{i=1}^{N} \mathbb{1}(w_i = 0)\ y_i(0)$$

$$\bar{Y}^{obs}(1) = \frac{1}{N_1} \sum_{i=1}^{N} \mathbb{1}(w_i = 1)\ y_i(1)$$

Instructor note: Students may find it hard to write down only those y_i included/excluded. Accept other expressions as long as it is apparaent that the total number of observed units is still N.

2.7 Your team knows that $ATE = E(y_i^{obs}|W = 1) - E(y_i^{obs}|W = 0)$ but they don't know how to estimate these expected values from our sample.

(a) What is the estimated ATE, $\widehat{ATE}$, in terms of the units observables $y_i, w_i;\ \ i = 1, 2, ..., N$ in our sample?

To estimate the average treatment effect:

$$\widehat{ATE} = \bar{Y}^{obs}(1) - \bar{Y}^{obs}(0)$$

For sample purposes, this simply implies estimating each expected value with the group mean; the average outcome for those treated and the average outcome for those untreated.

(b) Justify your answer in terms of a famous mathematical theorem:

i. $\square \xrightarrow{\square} E(y_i|W = 1)$

ii. . . .

iii. Therefore the estimate ...

By the LLN, the sample averages converge to the population's expected outcome:

$$\frac{1}{N_1} \sum_{i \in (w_i=1)}^{N_1} y_i(1) \xrightarrow{N_1 \to \infty} E(y_i|W = 1)$$

$$\frac{1}{N_0} \sum_{i \in (w_i=0)}^{N_O} y_i(0) \xrightarrow{N_0 \to \infty} E(y_i|W = 0)$$

Therefore

$$\widehat{ATE} \xrightarrow{N_0, N_1 \to \infty} ATE$$

(c) Compute $\widehat{ATE}$ using R.

-0.1338
This is the difference of the results in 2.5a and 2.5b

(d) **Reflect on the result**. How does it compare to your answers in 1.6b and 1.6c? At this point, do we have any way to diagnose the accuracy of this result?

Clearly the result –a nearly negligible, slightly negative estimated ATE— is not reflecting what we know (based on our perfect knowledge of the potential outcomes) to be the treatment effect.

2.8 Again, let's imagine we can observe potential outcomes. In the lecture, it was shown that the ATE can be decomposed in *treatment on the treated* and *selection bias*. Write down that expression and estimate the values of:

$ATE = E(y_i^{obs}|W=1) - E(y_i^{obs}|W=0) = E[\ y_i(1)|W=1] - E[\ y_i(0)|W=0]$

Adding and subtracting $E[y_i(0)|W=1]$:

$$\underbrace{(E[\ y_i(1)|W=1] \ - \ E[\ y_i(0)|W=1])}_{\text{Treatment of the treated}} + \underbrace{(E[\ y_i(0)|W=1]) \ - \ E[\ y_i(0)|W=0]}_{\text{Selection Bias}}$$

(a) treatment of the treated

10.0741

$$E[\ y_i(1)|W=1] - E[y_i(0)|W=1]$$

Estimate with the sample analogs:

$$\bar{Y}(1) \ - \ \frac{1}{N_1}\sum_{i=1}^{N} \mathbb{1}(w_i=1)\ y_i(0)$$

.

In R:

```
mean(students[students$W_school == 1, ]$yobs_w) -
mean(students[students$W_school == 1, ]$y0)
```

Note the second term $\sum_{i=1}^{N} \mathbb{1}(w_i=1)\ y_i(0)$ is the summation of the potential **outcome under no treatment** of the N_1 treated units —only possible because we're assuming we can observe the non-realized outcome when treated ($y(0)$ in this case).

(b) selection bias

-10.2078

$$E[\ y_i(0)|W = 1]) \ - \ E[\ y_i(0)|W = 0]$$

Sample analogs:

$$\frac{1}{N_1}\sum_{i=1}^{N} \mathbb{1}(w_i = 1)\ y_i(0) \ - \ \bar{Y}(0)$$

Same as in the previous question, $\frac{1}{N_1}\sum_{i=1}^{N} \mathbb{1}(w_i = 1)\ y_i(0)$ is the non-treated potential outcome of the treated units.

2.9 What do these values imply for the experiment's design? Is the value for 2.8a a potentially good $\widehat{ATE}$? What would be omitted if we were to only consider this value?

They suggest that selection bias is offsetting the treatment effect we observe.

However, Treatment of the Treated in 2.8a is the effect we would get in a world where we can observe **both potential outcomes** of all treated units. Its equivalence to a well-estimated ATE is not complete because it considers only public school students, not the full student population. They're comparable to the extent that potential outcome differences for students in private schools are similar.

Case study O.7.3

Evaluating AI-assisted learning on student outcomes (continued)

Instructions: Work in pairs or groups of 3; answer the questions as concisely as possible. Type your answers in a single shared document or in the format required by the instructor.

One of you must set up a blank Colab notebook to work on, and share it with the rest. Save any figures or output, and incorporate as required by the instructor.

This is a direct continuation of G.7.6, thus we will work under the context you already have. Continue to use `students` when necessary to answer the questions.

Scenario: You let the Prime Minister know the treatment allocation for the Project Mentor experiment you had previously agreed on is problematic. He hires a team of consultants to help you sort this design problem out, as well as polish other details of the experiment. The following exercises contain some of the questions asked during several policy meetings.

Meeting 1

1.1 Firstly, you are asked to explain generally why you cannot get a credible average treatment effect from the current treatment allocation. How would outcomes differ from our estimates if we scaled up the program nationally?
(use your "secret" knowledge of the potential outcomes, and your answers to the previous Guided Case).

> A complete answer includes the following 3 elements:
>
> The estimated ATE from the previous case is -0.1338 (GC 6 Q 2.7c). According to this result, Project Mentor would not have any effect in the country's 6th-graders math proficiency (very slight negative effect, but consider this is an estimate). Since there are costs associated with it, it would not seem advisable to scale.
>
> However, we also know (GC 6 Q 1.6b) our intervention's effect is closer to 9.956; since treatment on the treated (GC 6 Q 2.8a) is 10.0741, the effect of Project Mentor on private schools seems comparable to that of public schools. Scaling up the program would roughly increase math proficiency on 6th-graders nationally by 10 points on average.
>
> The current treatment allocation introduces a selection bias of -10.2078 points (GC 6 Q 2.8b), which completely offsets the effect on public schools, because the non-treated average score on private schools is 10 points higher than that on public schools at baseline.

1.2 You are asked to provide an alternative assignment that would create two groups equally representative of 6th-graders nationally. Create such assignment variable `t`, and also create `yobs_t`, the outcome we would observe under this assignment. Calculate the $\widehat{ATE}$. How does this result compare to 2.8a in G.7.6? Without making any further calculations, what do you think the treatment effect among private schoolers will be?

> **Instructor note**: Please make students avoid `set.seed(123)`, as this seed was used to crete `W_school`.
> Reproducibility is not mentioned to avoid hinting at random assignment; no exact answer is required. The estimated ATE should be approximately 10.
>
> Random assignment will create two comparable groups (no systematic differences). There are several ways to sample with replacement from $\{0, 1\}$, in the sample notebook we round 1000 draws from a $[0, 1]$ uniform, with 484 treated. For this sample we get:
>
> $\widehat{ATE} = 9.849$
>
> Note that the treatment of the treated = 10.074.

1.3 The Prime Minister doesn't believe that your new assignment created comparable groups. One way to provide evidence of balance, is showing the groups have the same composition of public and private schoolers. Formally show there is no evidence to reject the composition is the same. Be as conservative as possible with the variance.

> In this case, our only observable (apart from the outcome) is `W_school`, the indicator for public school students; the public/private proportion of students should be approximately the same. For our seed, we get (see notebook):
>
> - 51.6% in public schools for treatment
> - 49.8% in public schools for control

1.4 Imagine everyone can observe potential outcomes. From the definition of ATE, show treatment and control are comparable more decisively.

> Calculate $\bar{Y}(1) - \bar{Y}(0)$ for both groups; the should be quite similar. For our seed, we get:
>
> - 9.964 for treatment
> - 9.948 for control

Meeting 2

While more convinced of your new assignment, the Pime Minister still insists asking for permission to private schools is impractical and will delay matters. Conveniently, the consulting team asks two questions:

- Whether attending a private/public school in Novartia actually creates systematic differences between students; particularly, differences related to the outcome. This has not been formally shown.
- Whether there may be differences in treatment effects between public and private students (e.g. the mean effect is larger for any), as these differences would justify different rollouts.

To provide evidence in favor or against these questions, bear in mind we have to start from the following assumption:

$$y_i(0)|W = 0 \sim Distr(\mu_0,\ \sigma^2_{0,0})$$
$$y_i(1)|W = 0 \sim Distr(\mu_1,\ \sigma^2_{1,0})$$
$$y_i(0)|W = 1 \sim Distr(\nu_0,\ \sigma^2_{0,1})$$
$$y_i(1)|W = 1 \sim Distr(\nu_1,\ \sigma^2_{1,1})$$

Where $\sigma^2_{0,0} \neq \sigma^2_{0,1} \neq \sigma^2_{1,0} \neq \sigma^2_{1,1}$

2.1 In your own words what do these assumptions mean? What do they entail when it comes to testing hypotheses? According to the lecture, what should we assume about the correlation among these random variables if we want to be conservative?

The assumption implies treating each group (treatment/control, public/private) as drawn from a distinct distribution. For hypotheses testing purposes, a difference in means would imply using Neyman Variance as a conservative upper bound, since the covariance $Cov(\ Y(1)|\ W,\ Y(0)\mid W)$ is unknown (it will depend on the particular intervention and covariates).

$$Var(Y(1)|\ W - Y(0)\mid W) = \ldots$$
$$\ldots = Var(Y(1)|\ W) + Var(Y(0)\mid W) - 2\cdot Cov(\ Y(1)|\ W,\ Y(0)\mid W)$$

Where generally, it is assumed that $Cov(\ Y(1)|\ W,\ Y(0)\mid W) \geq 0$, with the implication that:

$$Var(Y(1)|\ W - Y(0)\mid W) \leq Var(Y(1)|\ W) + Var(Y(0)\mid W)$$

The last expression (Neyman Variance) is an upper bound of the variance for the difference in means.

2.2 Answer the question about systematic differences in outcomes between public and private schools.

Remember we are assuming all potential outcomes are observed.

As mentioned in 2.1, the treatment effect is likely offset by private school averages (higher than public school ones). We're mainly interested in testing for equality in average scores at baseline. A lack of equality would entail systematic differences in outcomes.

(a) State the appropriate null hypotheses and their alternates.

$$H_0 : \nu_0 = \mu_0$$
$$H_A : \nu_0 \neq \mu_0$$

Instructor notes:

- Also possible to include the test on treated outcomes $\nu_1 = \mu_1$. Baseline results are more directly linked to pre-existing systematic differences.
- Hypothesis statements of the form $\nu_0 > \mu_0$ are not covered in the course. If deemed appropriate, feel free to provide additional context for students.

(b) Test them with the appropriate statistic (estimate any parameters you don't know).

Under the distributional assumptions and Neyman variance (treating covariance as 0 for a conservative bound), the test statistic is:

$$z = \frac{\hat{\nu}_0 - \hat{\mu}_0}{\sqrt{\frac{\hat{\sigma}^2_{0,1}}{N_1} + \frac{\hat{\sigma}^2_{0,0}}{N_0}}}$$

Estimating the sample variances from the data: $\hat{\sigma}^2_{0,1} = 108.73$ and $\hat{\sigma}^2_{0,0} = 102.21$.

$$z = \frac{60.025 - 70.233}{\sqrt{\frac{108.73}{507} + \frac{102.21}{493}}} = \frac{-10.208}{0.649} \approx -15.72$$

In R:

```
s2_pub <- var(students[students$W_school==1,]$y0)
s2_prv <- var(students[students$W_school==0,]$y0)
se <- sqrt(s2_pub/507 + s2_prv/493)
z <- (mean_nu0 - mean_mu0)/se
```

(c) Tie your conclusions directly to the question with 95% confidence.

$|z| = 15.72 > 1.96$: we strongly reject H_0 at the 95% confidence level. There is overwhelming evidence of systematic differences in baseline outcomes between public and private school students. Private school students score approximately **10.2 points higher** at baseline than public school students, consistent with the selection bias estimated in the Guided Case Study.

2.3 Answer the question about differences in treatment effects.

Here we want to know whether the treatment effect differs between public and private school students. Define the treatment effects for each group:

$$\tau_1 \equiv \nu_1 - \nu_0 = 10.074 \qquad \tau_0 \equiv \mu_1 - \mu_0 = 9.834$$

We test $H_0 : \tau_1 = \tau_0$, i.e., no heterogeneous treatment effects by school type.

(a) State the appropriate null hypotheses (test equality).

$$H_0 : (\nu_1 - \nu_0) = (\mu_1 - \mu_0)$$
$$H_A : (\nu_1 - \nu_0) \neq (\mu_1 - \mu_0)$$

Equivalently, $H_0 : (\nu_1 - \nu_0) - (\mu_1 - \mu_0) = 0$.

(b) Make inference on $\hat{\nu}_1, \hat{\nu}_0, \hat{\mu}_1, \hat{\mu}_0$ accordingly.

Sample estimates and 95% CIs for each group mean (Neyman variance; independence across groups assumed):

Parameter	Estimate	$\hat{\sigma}^2$	n	95% CI
$\hat{\nu}_0$	60.025	108.73	507	[59.12, 60.93]
$\hat{\nu}_1$	70.099	102.56	507	[69.22, 70.98]
$\hat{\mu}_0$	70.233	102.21	493	[69.34, 71.13]
$\hat{\mu}_1$	80.067	94.65	493	[79.21, 80.93]

The estimated difference in treatment effects is:

$$\hat{\delta} = (70.099 - 60.025) - (80.067 - 70.233) = 10.074 - 9.834 = 0.240$$

Conservative standard error (treating potential outcomes as independent within each group):

$$SE(\hat{\delta}) = \sqrt{\frac{\hat{\sigma}^2_{1,1} + \hat{\sigma}^2_{0,1}}{N_1} + \frac{\hat{\sigma}^2_{1,0} + \hat{\sigma}^2_{0,0}}{N_0}} = \sqrt{\frac{102.56 + 108.73}{507} + \frac{94.65 + 102.21}{493}} = 0.903$$

$$z = \frac{0.240}{0.903} = 0.265$$

(c) Tie your conclusions directly to the question with 95% confidence.

$|z| = 0.265 < 1.96$: we fail to reject H_0 at the 95% confidence level. The 95% CI for $\hat{\delta}$ is approximately $[-1.53,\ 2.01]$, which comfortably includes 0.

There is no evidence that treatment effects differ between public and private school students. AlgebrAI appears to raise math scores by roughly 10 points for both groups. This means scaling the program to private schools would likely produce similar gains, which supports a uniform national rollout without differential targeting.

Meeting 3

Satisfied with your answers, the Prime Minister and consultants approach you with some final questions about the program's broader implications:

3.1 SUTVA violations

In your own words, briefly explain the Stable Unit Treatment Value Assumption (SUTVA). Could implementing Project Mentor violate this assumption in Novaria? Provide a specific scenario illustrating such a violation clearly. How might these externalities affect the accuracy of your estimates?

> SUTVA requires that each unit's potential outcomes depend only on its own treatment (i.e., they cannot depend on the treatment status of other units), and the treatment must not vary across units.
>
> There are several plausible SUTVA violation scenarios, arising from both the nature of the intervention and the school environment:
>
> - In the school environment, peer effects are strong; it is not hard to imagine that groups of friends/classrooms reaching a critical number of treated students may spill over learning and skills to the untreated remainder, affecting the control performance. Similarly, if a group of friends has sparsely treated students, a scenario may arise in which treated students' performance is affected by discouragement or even criticism from the rest of the group.
> - AlgebrAI is an adaptive tool. It varies the treatment based on each student's progress. This may violate the "same treatment" principle, especially in highly heterogeneous groups.

3.2 Alternative policies with proven outcomes (e.g. TaRL)

The consultants suggest evaluating cheaper alternatives, like Teaching at the Right Level —targeting teaching to each student's current skill level without advanced technology; human mentors, complementary material, smaller traditional groups (more teachers), among others. These alternatives currently have more robust evidence of their effects.

(a) What, if any, are some differences between the theory of change underlying TaRL and that of Project Mentor?

> Social motivation and the role of the instructor are a substantial part of TaRL's diagnosis for the causes of the misalignment between a curriculum and the student's skills. For AI-assisted learning, neither the class nor the instructor are direct targets the policy seeks to influence (at least not in the version presented in this case).
> TaRL is highly localized in addressing foundational gaps within a curriculum, with the instructor having tight control over the content. In the case of AlgebrAI, the current technological constraints of agentic AI allow a degree of semantic circumscription to a curriculum, but due to the student's interaction and input, direct remediation is not guaranteed.

(b) If differences exist, briefly discuss how you would test them within an experimental design, clearly describing treatments, assignment and measured outcome(s).

A three-arm RCT comparing: (i) control (no intervention), (ii) TaRL, and (iii) AlgebrAI (Project Mentor).

- **Unit of assignment**: schools, since both interventions operate at the school level. This also prevents spillovers within a school (SUTVA concern).
- **Random assignment**: stratified by school size and urbanity to ensure balance across arms.
- **Treatment**: TaRL arm implements foundational skill grouping and targeted instruction; AlgebrAI arm deploys the tablet-based AI tutoring sessions. Both arms receive the same number of intervention hours to keep dosage comparable.
- **Outcome**: standardized 6th-grade math test score. Intermediate outcomes (e.g., targeted skill acquisition, attendance, engagement) could help identify which elements of each theory of change drive results.

(c) Apart from the treatment effects, what else is necessary if we wanted to fairly compare any known TaRL intervention to Project Mentor in terms of efficiency?

A fair comparison requires cost data in addition to treatment effects — specifically, the **cost-effectiveness ratio**: cost per additional test-score point (or cost per student to achieve a given outcome gain). This requires tracking:

- **Fixed costs**: curriculum development and teacher training (TaRL), software development and tablet hardware (AlgebrAI).
- **Recurrent costs**: facilitator time and materials per session, maintenance, electricity, connectivity.
- **Opportunity costs**: time diverted from regular instruction for both students and teachers.

Without these figures, a policy recommendation based solely on treatment effects may be misleading if one intervention is much cheaper to implement.

(d) In this comparison, how important do you think scale would be? Briefly describe how costs for one and the other would behave.

Scale matters substantially, and costs behave very differently for each:

- **TaRL**: primarily variable costs — each additional student requires proportional teacher or facilitator time. Cost per student is roughly constant regardless of how many schools participate.
- **AlgebrAI**: high fixed costs (model training, app development, tablet procurement and distribution) but near-zero marginal cost per additional student once deployed. The platform can be extended to a new school at the cost of a few tablets and connectivity.

At a small pilot scale, TaRL is likely cheaper. As the number of students grows, AlgebrAI's fixed costs get amortized across a larger base, making it progressively more cost-efficient. The breakeven enrollment — beyond which AlgebrAI dominates on cost-effectiveness — is itself an important policy-relevant quantity to estimate.

3.3 **Non-compliance** Not every school or student may strictly follow the treatment assignment. Describe one realistic scenario of non-compliance in this project. Explain briefly how such non-compliance might bias the estimated treatment effect. Suggest one practical strategy to reduce or mitigate non-compliance.

Scenario: A principal in a government-managed school, skeptical of the program, limits students' tablet sessions to once a week instead of the required daily 20-minute sessions. These students are assigned to treatment ($T_i = 1$) but effectively receive a substantially reduced dose.

Bias: This is one-sided non-compliance (assigned-to-treated units not receiving treatment). The intent-to-treat (ITT) estimator averages over compliers and non-compliers, attenuating the estimated effect toward zero. If non-compliant schools are also systematically different (e.g., weaker administration, lower baseline performance), the bias compounds.

Mitigation: Before rollout, obtain explicit buy-in from school principals through training workshops and written commitment. During the experiment, use compliance monitoring (session logs from AlgebrAI's sync data) to identify non-compliance early and intervene. Non-compliance data also enables a LATE (Local Average Treatment Effect) estimation restricted to compliers.

14.310x Flipped Classroom

Week 8 Answers

The linear regression model
Linear models in R: `lm` class, parsing outputs with `broom`

Checklist

- ☐ Watch the linear models with R: `lm` tutorial[9] (Requirement)
- ☐ Complete Coding Lab 6 (Session 1)
  - ☐ Datasets: `wages`, `gap`, and `air`. In-lab.
- ☐ Complete Guided Case Study 7 (Session 2)
  - ☐ Dataset: `nlsy_merged.csv.gz`
- ☐ (Optional) Explore **Review Notes** and further **Readings** available.

Coding Lab L.8.6

Linear regression in R

Instructions: Work on your own. Complete coding sections 1 to 5 and submit as required by your instructor.

Notebook: Coding Lab 6

[9] **Module 8 > Introduction to the Class lm** in the online component of the course.

Instructor Notebook: Answers to L.8.6

Find answers to all exercises.

Guided case study G.8.7

Replicating (Deming, 2017): returns to social skills in the labor market

Instructions: Work in pairs or groups of three to solve all questions. One team member should share editor access to the Colab Notebook with the rest.

Notebook: Case Study 7

Dataset: nlsy_merged.csv.gz

Instructor Notebook: Answers to G.8.7

Find answers to all code-related questions in this Guided Case. Each coding question links to its corresponding answer section in the notebook.

Scenario:

Advances in computing and automation have fostered profound shifts in the structure of labor demand. Routine, codifiable tasks such as assembly-line manufacturing or basic data entry have witnessed decreases in labor intensity. On the contrary, roles that require either high cognitive complexity or interpersonal coordination have expanded. A large literature links this *routinization* trend to widening wage inequality and rising returns to education and cognitive skills.

Deming (2017) studies the role played by **social skills** in this transformation, and argues they have become an increasingly important determinant of labor market success over the past thirty years. One of the paper's insights is that social and cognitive skills are **complements**: a balance of cognitive and social skills will earn more than either skill alone in large amounts — optionally review the results of the 2-worker task model (pp. 12–16). This complementarity is more pronounced in diverse team environments, where members can benefit from coordinating their specialized work.

The paper builds on three groups of regression specifications:

- Wages or employment explained by workers' social and cognitive skills [Tables 1 & 4].
- Workers' occupational task intensity explained by their social and cognitive skills [Table 2].
- Wages explained by task intensity interacted with skills [Tables 3 & 5].

We will replicate selected results from Tables 1, 2 and 4 of Deming (2017) using an extract of the author's dataset `nlsy_merged`, and develop an understanding of the implications.

Dataset: nlsy_merged

The dataset is a merged, cleaned panel drawn from the National Longitudinal Survey of Youth 1979 (NLSY79) and 1997 (NLSY97). Each row is a (person–year) observation. Key variables:

- pubid — Unique person identifier.
- sample — Cohort indicator: = 1 for NLSY97, = 0 for NLSY79.
- year — Survey year.
- age — Age at survey date.
- complete79 / complete97 — = 1 if the observation qualifies. Complete if the respondent is employed, has non-missing wages, skills, and location variables.
- ln_wage — Log hourly wage (main outcome). emp — Employment indicator ($\geq$ 80% of the year employed).
- female — = 1 if respondent is female. black_male, black_female, hisp_male, hisp_female — Race × gender dummies (non-Hispanic white male omitted).
- div, metro, urban — Census division, metropolitan-area, and urban/rural codes. Included as fixed effects in all regressions.
- afqt_std — Armed Forces Qualifying Test (AFQT) score, standardized (mean 0, SD 1). Age-at-test adjusted following Altonji et al. (2012). Proxy for cognitive skill (COG_i).
- soc_nlsy_std — Social skills index from *four* items: childhood/adult sociability, HS clubs count, HS athletics. NLSY79 only; used in Tables 1–3.
- soc_nlsy2_std — Social skills index from the *two* items available in both cohorts: childhood and adult sociability. Standardized in the pooled sample; used in Tables 4–5 (SS_i).
- educ — Years of schooling (highest grade completed).
- socskills, routine — O*NET social-skill and routine task intensity (percentile rank, 0–10 scale).
- ind6090 — Industry code (1990 Census classification). Used in Table 2.
- weight — Analytic weight from Altonji et al. (2012), correcting for differential age-at-test in the AFQT (Tables 1–3). lswt — NLSY survey sampling weight (descriptive statistics).

Part A. Setup and Exploratory Analysis

1. [1] Install/load the dplyr, ggplot2 and readr packages. Read in nlsy_merged.csv.gz by running the provided code. Save the data frame as nlsy_merged.
   - How many **unique** workers are part of the study? On average, how many observations per worker are there in the 79 and 97 surveys?
   - Display summary statistics for ln_wage, afqt_std, soc_nlsy_std and soc_nlsy2_std. What is the dataset's median hourly wage in USD?

- Which of these variables has the most missing observations? Per the glossary, why might this be the case?

Code in Colab [1] Answer

> The dataset contains roughly 12,700 unique NLSY79 workers and 8,900 unique NLSY97 workers (≈21,600 total). The NLSY79 has ≈12–14 observations per worker across survey years; the NLSY97 has ≈7–9 per worker.
>
> The median hourly wage is exp(median(`ln_wage`)). With pooled median ≈ 2.29 log-points, the median wage is ≈ $9.90/h in 2013 dollars.
>
> **Most missing variable**: `soc_nlsy_std`, which is available only for the NLSY79 cohort. All NLSY97 observations (≈half the dataset) have `NA` for this variable, because the four-item index uses NLSY79-specific questions.

2. [2] To understand the completeness and quality of the main skill measures, create **overlapped** density plots of each of `afqt_std`, `soc_nlsy_std` and `soc_nlsy2_std`, grouped by cohort (`sample`: 0 = NLSY79, 1 = NLSY97) — add transparency with `alpha`. Create each plot *separately* to prevent `ggplot` from dropping all rows with any missing value.
   - `afqt_std` is standardized. Why doesn't it look normally distributed?
   - Is there any substantial difference in AFQT scores between cohorts? Describe if any.
   - From the glossary, what can you confirm about `soc_nlsy_std`?
   - What is one limitation of `soc_nlsy2_std` in the NLSY79 cohort?
   - `soc_nlsy2_std` is standardized in the pooled sample. What would a score of −2 in the NLSY97 cohort map to in the NLSY79 cohort?

Code in Colab [2] Answer

> **AFQT not normal:** Although standardized to mean 0/SD 1, the AFQT composite is formed from discrete sub-test scores. Score distributions tend to be slightly leptokurtic and/or bimodal because of floor/ceiling effects at the extremes and the heterogeneous population of test-takers.
>
> **Cohort differences in AFQT:** The distributions largely overlap after the Altonji et al. (2012) age-at-test adjustment. Any residual difference (e.g., a slightly higher NLSY97 mean) reflects genuine cohort-level changes in cognitive skill or education, not test-format artifacts.
>
> **soc_nlsy_std:** The density plot for `soc_nlsy_std` shows data only for the NLSY79 cohort (sample = 0); the NLSY97 series is entirely missing, confirming the glossary note that this four-component index is available only in the 1979 survey.
>
> **Limitation of soc_nlsy2_std in NLSY79:** Because it uses only 2 of the 4 available items, it is a noisier proxy for social skills in the NLSY79 cohort than `soc_nlsy_std`. Measurement error in the independent variable attenuates the regression coefficient toward zero.
>
> **Score mapping:** Since `soc_nlsy2_std` is standardized in the *pooled* sample, the scale is the same for both cohorts. A score of −2 in the NLSY97 cohort is −2 standard deviations below the pooled mean — it does not directly translate to a specific within-79 percentile without knowing the within-cohort mean and SD of the index.

3. [3] Compute the weighted mean of `ln_wage` for each cohort using `weighted.mean()` with the survey sampling weights (`lswt`). Verify you get 2.24 and 2.42 for the NLSY79 and NLSY97 cohorts respectively.

   Then arrive at the same result for the NLSY97 using `mean()` and `nrow()` only — construct the normalized weights `w_97` yourself.

   Code in Colab [3] **Answer**

> Weighted mean log wages: **NLSY79** ≈ 2.24, **NLSY97** ≈ 2.42 — a gap of ≈ 0.18 log-points ($\approx 20\%$ in levels). The NLSY97 cohort entered the labour market in a higher-wage period and is better educated on average.
>
> **Manual construction of weights:** For cohort j, the normalized survey weight for observation i is $\tilde{w}_i^j = \frac{\texttt{lswt}_i^j}{\sum_k \texttt{lswt}_k^j}$. The weighted mean is then $\sum_i \tilde{w}_i^j \cdot \texttt{ln\_wage}_i^j$.

Part B. Social and cognitive skills complementarity: Table 1

4. [4] Use `lm()` to estimate

$$\ln(w_{it}) = \alpha + \beta_1 \, \text{COG}_i + \varepsilon_{it}$$

   on the full dataset (`nlsy_merged`). Note COG_i = `afqt_std`. Do *not* include additional controls. Report $\hat{\beta}_1$ and its standard error.

   (a) What is the t-value for $\hat{\beta}_1$? Can we reject $\beta_1 = 0$ at the 95% level?

   (b) Suppose a worker starts at $\text{COG}_i = 0$. What is their estimated wage increase in \$/h from a 1 SD improvement in cognitive skill?

   (c) More generally, if a worker starts at $\text{COG}_i = x_0$, what is the estimated *percentage* wage increase from a 1 SD improvement?

   Code in Colab [4] **Answer**

> **Result:** $\hat{\beta}_1 \approx 0.170$, SE ≈ 0.003, $t \approx 56$.

(1) t-value and significance. $t = \hat{\beta}_1/\widehat{\text{SE}} \approx 0.170/0.003 \approx 56$. With $|t| \gg 1.96$, we strongly reject $H_0 : \beta_1 = 0$ at the 95% level.

(2) USD wage increase from COG = 0. At $\text{COG}_i = 0$, the predicted log-wage is $\hat{\alpha}$. A 1 SD increase gives predicted wage $\exp(\hat{\alpha} + \hat{\beta}_1)$, so the increase in dollars is:

$$\Delta w = \exp(\hat{\alpha})\big(\exp(\hat{\beta}_1) - 1\big) \approx \exp(\hat{\alpha}) \times 0.185 \text{ \$/h}.$$

With the pooled sample intercept implying $\exp(\hat{\alpha}) \approx \$9.00$/h (the predicted wage for a worker with zero cognitive skill and zero on all other controls), the increase is roughly \$1.65/h.

(3) Percentage increase from any x_0. Because the model is linear in log-wages, the *percentage* increase from a 1 SD improvement is the same regardless of starting point x_0:

$$\frac{\exp(\hat{\alpha} + \hat{\beta}_1 x_0 + \hat{\beta}_1) - \exp(\hat{\alpha} + \hat{\beta}_1 x_0)}{\exp(\hat{\alpha} + \hat{\beta}_1 x_0)} = \exp(\hat{\beta}_1) - 1 \approx 18.5\%.$$

A 1 SD improvement in cognitive skill is associated with an approximately 18–19% higher hourly wage.

5. [5] Replicate column 1 of Table 1. The model:

$$\ln(w_i) = \alpha + \beta_1 \, \text{SS}_i + \boldsymbol{\gamma}' \mathbf{X}_i + \varepsilon_i,$$

where SS_i = `soc_nlsy_std` and $\mathbf{X}_i$ contains the *covariates*: `female`, `black_male`, `black_female`, `hisp_male`, `hisp_female`, `age`, `year`, `div`, `metro`, `urban`.

(a) Create `nlsy79`: filter `nlsy_merged` to NLSY79 (`sample == 0`), `complete79 == 1`, and `age >= 23`. Coerce `age`, `year`, `div`, `metro`, `urban` to factors.

(b) Regress `ln_wage` on `soc_nlsy_std` and all covariates on `nlsy79`. Verify $\hat{\beta}_1 \approx 0.0999$.

(c) Add `weights = weight` to replicate the paper exactly. Save as `table1_col1`. Verify $\hat{\beta}_1 = 0.107$.

Note: the author uses **weighted** least squares to correct for differential age-at-test in the AFQT; as you can see, the unweighted result is a close approximation.

Code in Colab [5] **Answer**

Unweighted ($\hat{\beta}_1 \approx 0.0999$): without the Altonji et al. (2012) weights, social skills are already a significant predictor of log-wages, but the estimate slightly understates the paper's value.

Weighted (`weights = weight`, $\hat{\beta}_1 = 0.107$): adding the analytic weights corrects for differential age-at-test sampling. A one SD increase in the social skills index is associated with a $\approx 11\%$ higher hourly wage, conditional on age, year, region, metro/urban status, and demographic controls.

6. [6] Replicate column 3 of Table 1. The complementarity specification:

$$\ln(w_i) = \alpha + \beta_1 \, \text{COG}_i + \beta_2 \, \text{SS}_i + \beta_3 \, (\text{COG}_i \times \text{SS}_i) + \boldsymbol{\gamma}' \mathbf{X}_i + \varepsilon_i.$$

(a) Suppose $\beta_1 = \beta_2 = \beta > 0$. A worker starts at $(\text{COG}_i, \text{SS}_i) = (1, 1)$. Compute the log-wage increase from moving to each of: $(3, 1)$, $(2, 2)$, $(1, 3)$.

(b) If cognitive and social skills are complements, what sign do you expect for β_3? Which of the three bundles above yields the highest wage increase?

(c) Further suppose $\beta_3 = \frac{1}{10}\beta$. A worker starts at $(\text{COG}_i, \text{SS}_i) = (4, 0.5)$ and can invest 2 additional skill units (split freely between COG and SS at equal cost). What allocation maximises their log-wage gain? Why?

(d) Create the interaction variable `afqt_soc = afqt_std * soc_nlsy_std` in `nlsy79`. Regress with `weights = weight`. Save as `table1_col3`; verify the coefficients match column 3.

(e) Using $\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3$: find the worker with `pubid == 113`. If they can allocate one additional skill unit between COG and SS, which allocation maximises their predicted wage gain?

Code in Colab [6] **Answer**

(a) Log-wage increases from (1,1) toward three bundles. Model: $\Delta \ln w = \beta(\Delta\text{COG} + \Delta\text{SS}) + \beta_3\,\Delta(\text{COG} \times \text{SS})$. Starting at $(1,1)$, $\text{COG} \times \text{SS} = 1$.

- $(3,1)$: $\Delta = 2\beta + \beta_3(3-1) = 2\beta + 2\beta_3$
- $(2,2)$: $\Delta = 2\beta + \beta_3(4-1) = 2\beta + 3\beta_3$
- $(1,3)$: $\Delta = 2\beta + \beta_3(3-1) = 2\beta + 2\beta_3$

All three bundles represent a 2-unit skill increase, but the balanced bundle $(2,2)$ uniquely raises the interaction term more.

(b) Sign of β_3 and optimal bundle. If skills are complements, $\beta_3 > 0$. With $\beta_3 > 0$, the $(2,2)$ bundle yields the largest wage gain because it maximises the cross-product $\text{COG} \times \text{SS}$.

(c) Optimal investment from (4, 0.5) with $\beta_3 = \frac{1}{10}\beta$. The marginal return to one unit of SS is $\beta_2 + \beta_3 \cdot \text{COG} = \beta(1 + 0.4) = 1.4\beta$. The marginal return to one unit of COG is $\beta_1 + \beta_3 \cdot \text{SS} = \beta(1 + 0.05) = 1.05\beta$. Since $\text{MV(SS)} > \text{MV(COG)}$ throughout the feasible range, all 2 units should go to SS, yielding the bundle (4, 2.5).
The intuition: the worker already has very high cognitive skill (COG = 4). Complementarity means adding to the *scarcer* skill generates a larger interaction gain. The unbalanced skillset makes rebalancing toward SS the dominant strategy.

Code in Colab [6] **Answer**

Table 1, Column 3 estimates: $\hat{\beta}_1\,(\text{COG}) \approx 0.081$, $\hat{\beta}_2\,(\text{SS}) \approx 0.069$, $\hat{\beta}_3\,(\text{COG} \times \text{SS}) \approx 0.023$.
All three coefficients are significant at the 1% level. The positive $\hat{\beta}_3$ confirms complementarity: the marginal return to cognitive ability is increasing in social skills.

Worker 113 allocation. Using the estimated coefficients, evaluate the marginal returns at worker 113's observed (COG_{113}, SS_{113}):

$$\text{MV(SS)} = \hat{\beta}_2 + \hat{\beta}_3 \cdot \text{COG}_{113}, \qquad \text{MV(COG)} = \hat{\beta}_1 + \hat{\beta}_3 \cdot \text{SS}_{113}.$$

Invest the additional unit in whichever skill has the higher marginal value. If $\text{COG}_{113} > \text{SS}_{113}$ (as for many workers with unbalanced skill), MV(SS) will typically exceed MV(COG), and the investment should go to SS.

7. The author runs 4 additional specifications, even including a cognitive × non-cognitive interaction. Skim section IV.A of the paper for the motivation.

Section IV.A reports that the SS premium and the COG×SS complementarity are robust to adding non-cognitive skills (`noncog_std`), a COG×noncog interaction, and education controls. The motivations are: (i) non-cognitive traits (conscientiousness, self-esteem) are plausible confounders correlated with both social skills and wages; (ii) education mediates some of the skill premium and its inclusion tests whether raw skill measures proxy for schooling rather than labour-market ability.

Part C. Occupational Sorting: selection and Table 2

Per the theoretical model implications in section II.E of the paper, *workers with higher social skills sort into non-routine occupations* and *workers earn more when they switch into non-routine occupations, and their relative wage gain is increasing in social skill* (Deming, 2017). By **occupational sorting**, we refer to this selection into less routine-oriented roles by workers with better social skills.

8. [7] Replicate columns 1 and 3 of Table 2. The models:

$$T_i = \alpha + \beta_1\,\text{COG}_i + \beta_2\,\text{SS}_i + \beta_3\,(\text{COG}_i \times \text{SS}_i) + \boldsymbol{\gamma}'\mathbf{X}_i + \varepsilon_i,$$

where $T_i \in \{$`routine`, `socskills`$\}$. Include the previous *covariates* plus `educ` and `ind6090` as factors. Weight by `weight`. Restrict to `nlsy79` observations with non-missing `ln_wage` using the `subset` argument.

(a) Interpret $\hat{\beta}_3$ in both models. What does it tell us about workers who are both cognitively and socially skilled?

(b) What is the implication of a non-significant $\hat{\beta}_3$ in column 3?

(c) What is the likely justification for the industry fixed effects?

(d) Is there evidence of social-skill sorting into social occupations regardless of cognitive skill level?

Code in Colab [7] **Answer**

Table 2, Column 1 (`routine`): $\hat{\beta}_1$ (COG) < 0, $\hat{\beta}_2$ (SS) < 0, $\hat{\beta}_3$ (COG × SS) < 0.
Table 2, Column 3 (`socskills`): $\hat{\beta}_1$ (COG) > 0, $\hat{\beta}_2$ (SS) > 0, $\hat{\beta}_3$ not significant at 95%.

(1) Interpreting $\hat{\beta}_3$ in both models.
Routine ($\hat{\beta}_3 < 0$): workers who are both cognitively and socially skilled are particularly unlikely to hold routine jobs. The complementarity *amplifies* avoidance of routine tasks — having both skills reinforces the exit from routine work beyond what each skill achieves separately.
Social-skill intensity ($\hat{\beta}_3 \approx 0$): the joint possession of COG and SS does not further increase occupational social-task intensity beyond the main effects. Social sorting is driven by social skills and cognitive skills acting independently, not by their interaction.

(2) Implication of non-significant $\hat{\beta}_3$ in Column 3. Sorting into social-task-intensive occupations is additive in skills: $\hat{\beta}_1 + \hat{\beta}_2$ drives the sorting, with no additional complementarity in the occupational selection stage. The complementarity identified in Table 1 (wages) thus operates within social-task occupations, not primarily through selection into them.

(3) Justification for industry fixed effects. Industries differ systematically in their task composition independently of individual skills (e.g., manufacturing vs. finance). Without industry controls, the skill coefficients would partially reflect *which industries* socially-skilled workers enter, rather than within-industry sorting on task content. The FEs isolate the within-industry skill–task relationship.

(4) Social sorting regardless of COG. Yes: $\hat{\beta}_2 > 0$ in the social-skill intensity regression (Column 3) is significant, meaning high-SS workers sort into social-task occupations even unconditional on cognitive skill.

9. Review Figure 5 of the paper. Which occupations saw the greatest cumulative hourly wage growth over the 1980–2000 period?

Figure 5 in Deming (2017) shows cumulative wage growth by occupation type, 1980–2012. **Non-routine, social-task-intensive** occupations (managers, professionals, sales) display the greatest cumulative wage growth over the 1980–2000 period. Routine occupations (clerical, production) experienced stagnant or declining real wages. This pattern motivates the paper's focus on social skills as a driver of wage growth, and is consistent with the "routinization" hypothesis.

10. Recall the Potential Outcomes Framework, and suppose "high social skills" is a treatment. We have no control over its assignment, but we would like to compute the average wage premium associated with having high social skills. Explain how selection bias arises in this setting.

> In the potential outcomes framework, let $T_i = 1$ if worker i has high social skills and $T_i = 0$ otherwise. Define $Y_i(1)$ and $Y_i(0)$ as their wages under each treatment value.
> The observed wage gap decomposes as:
>
> $$E[Y_i \mid T_i = 1] - E[Y_i \mid T_i = 0] = \underbrace{E[Y_i(1) - Y_i(0)]}_{\text{ATE}} + \underbrace{E[Y_i(0) \mid T_i = 1] - E[Y_i(0) \mid T_i = 0]}_{\text{selection bias}}.$$
>
> Selection bias arises because treatment assignment is not random. Workers with high social skills also tend to have higher cognitive ability, better family backgrounds, and more education — all of which independently raise wages. Therefore $E[Y_i(0) \mid T_i = 1] > E[Y_i(0) \mid T_i = 0]$: even in a hypothetical world where high-SS workers had low social skills, they would still earn more than the average low-SS worker. The observed gap **overstates** the causal wage premium of social skills.

11. Workers with higher social skills may sort into socially-intensive occupations that pay well for reasons unrelated to their social skills (e.g., their preferences, location, industry). Using the selection bias decomposition from 10., argue whether the coefficient $\hat{\beta}_2$ in 6. is likely above or below the causal wage effect of social skills. Identify one type of worker for whom $Y_i(0)$ — their counterfactual wage in a routine job — is already high.

> From the decomposition in 10., $E[Y_i(0) \mid T_i = 1] > E[Y_i(0) \mid T_i = 0]$, so the selection term is **positive**. The coefficient $\hat{\beta}_2$ in the regression of 6. picks up the same positive association: it overstates the causal return to social skills.
>
> **Type of worker for whom $Y_i(0)$ is already high:** a cognitively able, highly educated worker who has chosen a high-wage industry. Even if placed in a routine, low-social-task job, their general productivity would command above-average wages. Because such workers also tend to have higher social skills (positive skill correlation), the raw SS coefficient is inflated.

Part D. Cross-cohort pay for skills: Table 4

The author argues that, despite a general trend toward non-routine occupations, social skills are increasingly rewarded over time.

12. [8] Replicate columns 1 and 4 of Table 4. The cross-cohort specification for worker i in cohort $s \in \{79, 97\}$:

$$\begin{aligned} y_{is} = {} & \alpha + \beta_1 \, \text{COG}_{is} + \beta_2 \, \text{SS}_{is} + \beta_3 \, (\text{COG}_{is} \times \text{SS}_{is}) \\ & + \beta_4 \, 97_s + \beta_5 \, (\text{COG}_{is} \times 97_s) + \beta_6 \, (\text{SS}_{is} \times 97_s) \\ & + \beta_7 \, (\text{COG}_{is} \times \text{SS}_{is} \times 97_s) + \boldsymbol{\gamma}' \mathbf{X}_{is} + \varepsilon_{is}, \end{aligned}$$

where y_{is} is `emp` (col 1) or `ln_wage` (col 4), 97_s = `sample`, and SS_{is} = `soc_nlsy2_std`.

(a) Create `nlsy_both`: filter `nlsy_merged` to `age` 25–33. Coerce `age`, `year`, `div`, `metro`, `urban` to factors.

(b) Create the interactions: `afqt_sample`, `soc2_sample`, `afqt_soc2`, `afqt_soc2_sample`.

(c) Estimate the models. For column 4, additionally restrict to `complete97 == 1`. Save as `table4_col1` and `table4_col4`.

Code in Colab [8] Answer

Table 4, Column 1 (`emp`): $\hat{\beta}_5$ (COG $\times$ 97) and $\hat{\beta}_6$ (SS $\times$ 97) capture how skill returns to employment changed from the NLSY79 to the NLSY97 cohort.
Table 4, Column 4 (`ln_wage`): Key estimates: $\hat{\beta}_2$ (SS) $\approx$ 0.06–0.07, $\hat{\beta}_6$ (SS $\times$ 97) $\approx$ 0.03–0.04 (positive), $\hat{\beta}_7$ (COG $\times$ SS $\times$ 97) > 0.

Interaction construction.
`afqt_sample` = `afqt_std` $\times$ `sample` — how COG returns differ in the 97 cohort.
`soc2_sample` = `soc_nlsy2_std` $\times$ `sample` — how SS returns differ.
`afqt_soc2` = `afqt_std`$\times$`soc_nlsy2_std` — the COG$\times$SS complementarity (pooled baseline).
`afqt_soc2_sample` = `afqt_soc2`$\times$`sample` — whether the complementarity *changed* between cohorts.

Sample restrictions. Column 1 uses all `nlsy_both` (age 25–33, both cohorts); column 4 further restricts to `complete97 == 1` because `ln_wage` is missing for non-workers.

13. Interpret $\hat{\beta}_7$ from 12.. How does a positive and significant three-way coefficient strengthen the paper's argument about the growing importance of *combined* cognitive and social skills, beyond what the cross-cohort SS premium alone ($\hat{\beta}_6$) can establish?

$\hat{\beta}_7$ is the coefficient on $\text{COG}_i \times \text{SS}_i \times 97_s$. A **positive and significant** $\hat{\beta}_7$ means that, beyond the rise in the standalone SS premium ($\hat{\beta}_6 > 0$), the *synergy* of combining both skills became more valuable in the NLSY97 labour market than in the NLSY79 one.

This is the paper's strongest cross-cohort finding. It rules out the simpler story that "social skills just became worth more by themselves." Instead, what the modern labour market rewards most is workers who can both reason analytically *and* communicate and coordinate socially — a combination that is scarce and especially productive in complex, team-organised production environments. The three-way interaction is the direct econometric evidence for this claim.

Part E. Hypothesis tests

14. [9] Using `table1_col3`, test the joint hypothesis $H_0 : \beta_2 = \beta_3 = 0$ — that social skills and their interaction with cognitive ability are jointly irrelevant, controlling for cognitive skill alone. Use `linearHypothesis()` from the `car` package. Report the F-statistic and p-value.

Code in Colab [9] Answer

> $H_0 : \beta_2 = \beta_3 = 0$ in `table1_col3`. This tests whether social skills and their interaction with cognitive ability are jointly irrelevant, above and beyond cognitive skill alone.
>
> **Result:** $F \gg 10$ (far above the 1% critical value of ≈ 4.6 for 2 numerator df), $p \ll 0.001$. **Reject** H_0: social skills and COG×SS are jointly highly significant predictors of log-wages.
>
> Note: even though $\hat{\beta}_3$ is individually significant, the joint test provides stronger evidence because it accounts for any negative covariance between $\hat{\beta}_2$ and $\hat{\beta}_3$.

15. [10] Using `table4_col4`, test the joint hypothesis $H_0 : \beta_6 = \beta_7 = 0$ — that the *cross-cohort changes* in both the social skill premium and the cognitive–social complementarity are jointly zero. This directly tests the paper's central claim that the NLSY97 labor market rewards social skills differently from the NLSY79. Use `linearHypothesis()`. Report the F-statistic and p-value.

Code in Colab [10] **Answer**

> $H_0 : \beta_6 = \beta_7 = 0$ in `table4_col4`. This jointly tests whether the NLSY97 cohort's SS premium *and* COG×SS complementarity are the same as the NLSY79 cohort's — i.e., that social skill returns did not change between cohorts.
>
> **Result:** F significant at the 5% level (or stronger), $p < 0.05$. **Reject** H_0: the cross-cohort changes in SS returns are jointly non-zero, directly supporting the paper's central claim that social skills became more valuable over time.
>
> This test is more informative than Q 14. in one respect: it evaluates whether the *temporal trend* in skill returns is significant, not merely whether skills matter in any single cross-section.

Additional resources

- Drive: Review Notes

14.310x Flipped Classroom

Week 9 Answers

Regression Discontinuity Design (RDD)
RDD modeling in R

Checklist

- ☐ Watch the regression discontinuity lecture notes[10] (Requirement)
- ☐ Complete Guided Case 8 (Sessions 1 & 2)
- ☐ Dataset: `schools.csv`
- ☐ Paper: (Pop-Eleches & Urquiola, 2013)
- ☐ (Optional) Review additional materials (Optional) -

Guided Case Study G.9.8

Regression Discontinuity:
Replicating *Going to a Better School*
(Pop-Eleches & Urquiola, 2013)

Instructions: Work in pairs or groups of three to solve all questions. One team member should share editor access to the Colab notebook with the rest.

[10] **Module 9 > Regression Discontinuity Design** in the online component of the course.

Notebook: Case Study 8 — RDD replication

Dataset: schools.csv

Paper: (Pop-Eleches & Urquiola, 2013)

Instructor Notebook: Answers to G.9.8 — RDD replication

Find answers to all coding questions in the Guided Case Colab notebook (`answers-complete.ipynb`). Complete written answers follow below.

Scenario:

In 2002, Romania reformed its high school admissions system. Students take a standardised national examination at the end of middle school, and schools receive a centralised ranking based on their historical average admission scores. Each school announces a minimum admission score — its **cutoff** — and students are matched to schools in descending order of their exam scores. In towns with multiple high schools, adjacent schools share a cutoff: the top student who did not qualify for school k is the marginal student for school $k-1$.

This system generates a compelling natural experiment. Students who scored just above and just below a given cutoff are, by construction, nearly identical in ability — yet one group attends school k while the other attends $k-1$. Pop-Eleches and Urquiola (2013) exploit this discontinuity to study whether attending a *higher-quality* school causally affects students' academic outcomes and, equally importantly, how families and peers behaviorally respond to the assignment.

In this case study you will replicate and interpret core results of Pop-Eleches and Urquiola (2013) using `schools.csv`, a student-level survey extract from the authors' replication package. The dataset covers students in towns with two to four secondary schools — localities where the school-quality contrast is sharpest and direct surveys of students, parents, and teachers were conducted.

Dataset: `schools.csv`

Each row is one student. Key variables:

- `score_dist` — Distance of the student's admission score to the relevant school cutoff. Negative values indicate the student scored *below* the cutoff (attended the lower school); positive values indicate the student scored *above* (attended the higher school).
- `above_cutoff` — Binary treatment indicator: = 1 if the student is above the cutoff and attends the higher-ranked school.
- `score_dist_above` = `score_dist`×`above_cutoff` — Slope interaction, allowing

different running-variable slopes on each side of the cutoff.

- `score_dist_r05` — `score_dist` rounded to the nearest 0.05, used for binscatter plots.
- `school_avg_score` — The average admission score of the school the student attends (0–10 scale). Captures the **school peer quality** — the key first-stage outcome showing that better-school students face a higher-achieving peer group.
- `school_own_score` — The school's own quality indicator (its admission average). Used to classify schools by tier.
- `n_schools_town` — Number of secondary schools in the student's town (2, 3, or 4 in this extract).
- `school_fe` — School × zone × wave fixed-effect identifier. Always include as `factor(school_fe)` in regressions.
- `cluster_id` — School-year identifier for clustering standard errors.
- **Teacher quality** (from school-level records): `teacher_certified` (share of certified teachers, 0–1), `teacher_experience` (average years of experience), `teacher_novice` (share with < 2 years of experience, 0–1).
- **Parental behavior** (from parent survey): `parent_volunteer` (= 1 if parent volunteers at school), `parent_tutoring` (= 1 if parent pays for tutoring), `parent_help_hw` (= 1 if parent helps with homework).
- **Peer quality** (from student survey): `peer_rank` (student's self-reported rank among peers, 1–10; lower is better), `peer_bad_idx` (index of bad peer behaviors, 0–1).
- **Pre-determined characteristics** (validity checks): `female`, `age`, `head_educ_tert` (household head has tertiary education), `mom_educ_tert`, `has_internet`.

Part A. Understanding the RDD Design

1. The admissions system described above generates a Regression Discontinuity Design.
   (a) Identify the **running variable**, the **cutoff**, and the **treatment** in this setting.
   (b) State the key identifying assumption of Sharp RDD in general, and then restate it in the specific language of this paper: what must be true about students near the cutoff?
   (c) Give one concrete threat to this assumption in the school admissions context. Which type of actor would need to be able to manipulate the assignment precisely?

(a) Running variable, cutoff, treatment.

- **Running variable** (r_i): the student's distance to the relevant school-level cutoff, `score_dist` = `exam_score`$_i$ − `cutoff`$_k$. Negative values indicate the student scored below the threshold; positive values indicate above.
- **Cutoff**: $r_i = 0$, the minimum score required to be admitted to the higher-ranked school.
- **Treatment** (D_i): attending the higher-ranked school (`above_cutoff` = 1).

(b) Identifying assumption. Sharp RDD requires that potential outcomes $Y_i(0)$ and $Y_i(1)$ are *continuous* in the running variable at the cutoff. In this context: students just above and just below the cutoff must be identical in all pre-determined characteristics (family background, ability, motivation) on average. Only school assignment — not any other determinant of outcomes — changes discontinuously at $r_i = 0$.

(c) Threat: score manipulation. If students (or schools) could strategically manipulate the admission score to land just above the cutoff, the continuity assumption would fail: the "just above" group would be systematically different (better-connected, higher-SES) from the "just below" group. In Romania, the exam was administered centrally, making precise score manipulation difficult. The relevant actor would be a student or family with knowledge of the exact cutoff and influence over the reported score.

2. [2] Install and load `tidyverse`, `fixest`, `rdrobust`, `lmtest`, `sandwich`, and `car`. Read `schools.csv` and save as `schools`. Report:
   - Total number of students and the share with `above_cutoff` = 1.
   - Mean and standard deviation of `school_avg_score` separately for students above and below the cutoff.
   - A frequency table of `n_schools_town`.

   What does the large share above the cutoff tell you about this survey sample?

Code in Colab [2] **Answer**

The dataset contains $N = 11{,}931$ students. **80.3%** have `above_cutoff` = 1, indicating this survey extract deliberately over-samples students who gained access to the better school — it is not a random cross-section of all students near cutoffs.

Mean `school_avg_score`: 8.34 for above-cutoff students vs. 7.61 for below-cutoff students — a raw difference of ≈ 0.73 points. This is larger than the RDD estimate because it conflates the causal effect with selection (students who just missed the cutoff may have had lower scores to begin with).

Town-size distribution: `n_schools_town` = 2 for 8,240 students (69%), = 3 for 3,483 (29%), and = 4 for 208 (2%). The dataset covers only small towns with 2–4 secondary schools, which is a specific (and distinctive) subset of the Romanian school population.

3. [3] Plot a histogram of `score_dist` with bin width 0.05 (restrict to |`score_dist`| ≤ 2). Add a vertical dashed line at zero.
   - Does the density appear continuous at zero? What would a spike or dip

indicate?
- Count observations with score_dist < 0 and > 0. What does the ratio suggest about the survey design?
- *(Context:)* In the full administrative data (data-AER-1, $>$ 3.6 million observations), the running-variable density is smooth at the cutoff, indicating no systematic score manipulation. Why does the survey extract show a large imbalance?

Code in Colab [3] **Answer**

> **Density continuity.** A histogram of score_dist at 0.05-width bins appears visually smooth near zero when looking at the shape of the curve alone. However, the raw counts reveal a very large imbalance: 9,493 students above the cutoff vs. only 2,345 below (80%/20% split). This is *not* evidence of score manipulation.
>
> **Why the imbalance?** The survey (from which schools.csv is derived) was conducted *within* schools: interviewers went to schools and surveyed enrolled students. Because the better school is the one *above* the cutoff, and because the survey oversampled students in better schools relative to those who just missed, the above-cutoff group is mechanically over-represented. In the full administrative data (data-AER-1), the density of the running variable is approximately uniform near the cutoff, consistent with absence of manipulation.
>
> **Implication.** Any RDD estimates from schools.csv rely on the local continuity of potential outcomes — not on the density being balanced in this extract. The density check using the administrative data (where no spike is observed at $r = 0$) is the appropriate validity test for the admission score.

4. [4] Create an RDD bin-scatter for school_avg_score:
   (a) Compute the mean of school_avg_score within each score_dist_r05 bin. Restrict to $|$score_dist$| \leq 2$.
   (b) Plot these bin means against score_dist_r05, add a vertical dashed line at zero, and overlay separate linear trend lines for $r < 0$ and $r > 0$.
   (c) Describe the discontinuity: how large does the jump look relative to the overall spread of school_avg_score?

Code in Colab [4] **Answer**

> The bin-scatter reveals a stark discontinuity: bins just to the right of zero have school_avg_score approximately 0.5 points higher than bins just to the left, replicating the visual pattern of Figure 1 (Panel A) in Pop-Eleches and Urquiola (2013). On each side of the cutoff, the trend lines are nearly flat, indicating that within either the above or below group, the running variable has little additional relationship with school quality once we are already close to the cutoff.
>
> This visual jump is large relative to the overall spread of school_avg_score (standard deviation ≈ 0.60), representing almost one full standard deviation. It provides strong prima facie evidence that the school-quality discontinuity is real.

Part B. Linear RDD with `lm()`

The canonical parametric RDD estimator fits separate linear regressions on each side of the cutoff:

$$Y_i = \alpha + \beta_{\text{RD}} \cdot D_i + \gamma_1 \cdot r_i + \gamma_2 \cdot (r_i \times D_i) + u_i, \tag{10}$$

where Y_i is the outcome, D_i = `above_cutoff`, and r_i = `score_dist` is the running variable (so $r_i \times D_i$ = `score_dist_above`). The coefficient β_{RD} estimates the causal effect at the cutoff.

5. [5] Estimate the RDD model (10) with `lm()`, restricted to |`score_dist`| ≤ 1 and `score_dist` $\neq 0$. Use `coeftest()` with `vcovCL()` clustered on `cluster_id`.
   (a) Report $\hat{\alpha}$, $\hat{\beta}_{\text{RD}}$, $\hat{\gamma}_1$, $\hat{\gamma}_2$ with standard errors.
   (b) Interpret $\hat{\alpha}$ and $\hat{\beta}_{\text{RD}}$ in words.
   (c) Interpret $\hat{\gamma}_1$ and $\hat{\gamma}_2$: what do the slopes tell you about the relationship between score and school quality on each side of the cutoff?

Code in Colab [5] **Answer**

Estimates (bandwidth 1.0, N = 6,559, clustered SEs):

Coefficient	Estimate	SE	
$\hat{\alpha}$ (intercept)	7.707	0.059	
$\hat{\beta}_{\text{RD}}$ (`above_cutoff`)	0.544	0.070	$p < 0.001$
$\hat{\gamma}_1$ (`score_dist`)	0.160	0.097	$p \approx 0.10$
$\hat{\gamma}_2$ (`score_dist_above`)	−0.024	0.117	$p \approx 0.84$

(b) Interpretation. $\hat{\alpha} = 7.71$ is the predicted `school_avg_score` for a student whose admission score equals the cutoff exactly ($r_i = 0$) and who attends the lower-ranked school ($D_i = 0$). $\hat{\beta}_{\text{RD}} = 0.544$ is the estimated causal jump in school quality at the cutoff: attending the higher-ranked school raises the school's peer quality score by approximately 0.54 points on the 0–10 scale.

(c) Running variable slopes. $\hat{\gamma}_1 = 0.160$ is the slope on the left side ($D_i = 0$): among below-cutoff students, each additional exam score unit is associated with 0.16 higher school quality. $\hat{\gamma}_1 + \hat{\gamma}_2 = 0.160 - 0.024 = 0.136$ is the slope on the right side ($D_i = 1$). Both slopes are modest and statistically weak in this bandwidth, which is reassuring: near the cutoff, the running variable itself matters little for school quality conditional on which side of the cutoff the student is on.

6. [6] Extend the model with school fixed effects using `feols()` from `fixest`:

$$\texttt{school\_avg\_score}_i = \beta_{\text{RD}}\, D_i + \gamma_1\, r_i + \gamma_2\, (r_i D_i) + \mu_j + u_i,$$

   where μ_j denotes school fixed effects (`school_fe`). Save as `m_fe`.
   (a) Report $\hat{\beta}_{\text{RD}}$ and its standard error. How does it compare to Q5.?
   (b) Why do we include `school_fe` as a fixed effect? What variation is being absorbed?
   (c) Fixed effects reduce residual variance. Does this tend to increase or decrease the standard error on $\hat{\beta}_{\text{RD}}$? Why?

Code in Colab [6] **Answer**

> With school fixed effects (`feols`, clustered on `cluster_id`):
>
> $$\hat{\beta}_{\mathrm{RD}} = 0.477, \quad \mathrm{SE} = 0.047, \quad t = 10.2, \quad p < 0.001.$$
>
> **(a) Change from Q5..** The coefficient falls from 0.544 to 0.477 (about a 12% reduction). The standard error also falls substantially (from 0.070 to 0.047). The FE specification is more precise because it removes school-level heterogeneity.
>
> **(b) Role of fixed effects.** The `school_fe` identifier is a school × zone × survey-wave cell. Including it as a fixed effect absorbs any time-varying school-level characteristics that are common to all students in that cell — for example, school-specific trends in teaching quality or cohort composition. Without FEs, between-school variation (unrelated to the cutoff) could bias the estimate.
>
> **(c) FEs and SEs.** Absorbing school FEs reduces the residual variance of both the outcome and the regressors, which shrinks the standard error. In this case, the SE falls from 0.070 to 0.047, improving precision despite there being fewer effective degrees of freedom.

7. The linear specification in (10) imposes that the relationship between Y and the running variable is the same everywhere within the bandwidth. An alternative is to add a quadratic term.

   (a) What is the motivation for preferring **local linear** regression over global polynomials in RDD? (*Hint:* think about boundary bias and the role of the bandwidth.)

   (b) In terms of bias vs. variance, what is the cost of adding more polynomial terms to the RDD specification?

   (c) If you added `score_dist^2` and `above_cutoff * score_dist^2` to the model in Q6., would you expect the estimate of $\hat{\beta}_{\mathrm{RD}}$ to change substantially? Why or why not?

(a) Local linear vs. global polynomials. Local linear regression uses data *close* to the cutoff (within the chosen bandwidth) and fits a simple linear regression on each side. This limits the influence of data far from the cutoff, which are uninformative about the effect at the threshold. Global polynomials fit a single polynomial using all data; they suffer from *boundary bias* — the polynomial is forced to fit the entire range, so its behavior at the endpoints (the cutoff) can be dominated by observations far away. Gelman and Imbens (2019) show that high-degree global polynomials produce erratic RDD estimates and inflated confidence intervals.

(b) Bias–variance tradeoff. Adding polynomial terms reduces approximation bias (better fit of the unknown regression function) but increases variance (more parameters to estimate from the same data). In a small neighborhood near the cutoff, the data are sparse; adding higher-order terms further inflates standard errors without meaningfully reducing bias within that neighborhood.

(c) Effect of adding a quadratic. Given the very flat slopes observed in Q5. (both $\hat{\gamma}_1$ and $\hat{\gamma}_2$ are small), the relationship between the running variable and school quality is approximately linear within `|score_dist|` ≤ 1. Adding a quadratic term would likely change $\hat{\beta}_{\text{RD}}$ by a small amount (well within standard error bounds), but at the cost of inflated uncertainty. The linear specification is adequate here.

Part C. Optimal Bandwidth and Replication

8. [8] Use `rdrobust()` on `school_avg_score` with the default MSE-optimal bandwidth. Report:
   (a) The MSE-optimal bandwidth $\hat{h}$ and the point estimate (conventional).
   (b) The robust 95% confidence interval and corresponding p-value.
   (c) The conceptual difference between `rdrobust` and the `lm()` approach of Q5.: what does "local linear" mean and why might it be preferred?

Code in Colab [8] Answer

(a) MSE-optimal bandwidth and estimate. `rdrobust()` selects a bandwidth $\hat{h} \approx 1.36$ (this value is data-driven and may vary slightly depending on the exact implementation). The conventional point estimate is approximately $\hat{\beta}_{\text{RD}} \approx 0.465$.

(b) Robust 95% CI and p-value. The robust 95% confidence interval is approximately $[0.37, 0.56]$. The p-value is < 0.001, so we strongly reject the null of no discontinuity.

(c) Local linear vs. parametric `lm()`. `rdrobust` estimates a *local linear* regression at the cutoff: it uses only observations within the optimal bandwidth and fits a weighted linear regression, with triangular (or other) kernel weights that down-weight observations further from the cutoff. This is more flexible than the global `lm()` approach because it does not impose a linear functional form across the entire range — it only requires linearity in the immediate neighborhood of the cutoff. The trade-off is that fewer observations are used (within bandwidth only), so estimates are less precise if the bandwidth is narrow.

9. [9] Replicate the bandwidth = 1.0 specification of Table 3 from Pop-Eleches and Urquiola (2013) using `feols()` with `school_fe` fixed effects and `cluster_id` clustering. Save as `m_t3`.

   (a) Report $\hat{\beta}_{\text{RD}}$ and its clustered SE. Verify you obtain approximately 0.477 (SE $\approx$ 0.047).

   (b) The paper's Table 3 reports $\hat{\beta}_{\text{RD}} \approx$ 0.11–0.20 (depending on the specification). You will explain this difference in Q11..

Code in Colab [9] **Answer**

(a) Result.

$$\hat{\beta}_{\text{RD}} = 0.477, \quad \text{SE} = 0.047, \quad t = 10.2, \quad p < 0.001.$$

This confirms the `feols` result from Q6..

(b) Comparison with the paper. Table 3 of Pop-Eleches and Urquiola (2013) reports a coefficient of approximately 0.109–0.201 on school quality (`agus`) depending on the specification and sample. Our estimate of 0.477 is substantially larger. This is attributable to the sample restriction in `schools.csv`: the survey covers only small towns with 2–4 schools, where the quality gap between adjacent schools is larger than in big cities with many competing schools (as explored in Q11.).

10. [10] Loop over bandwidths $h \in \{0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.2, 1.5, 2.0\}$, running `rdrobust()` with `h = h_val` each time. Store the conventional point estimate and robust 95% CI bounds.

    (a) Plot the point estimates with 95% CI bands against the bandwidth value.

    (b) Does the coefficient appear stable across bandwidths? What does this stability suggest about the credibility of the RDD estimate?

Code in Colab [10] **Answer**

(a) Bandwidth sensitivity plot. Running rdrobust at $h \in \{0.3, 0.4, \ldots, 2.0\}$, the point estimate of $\hat{\beta}_{\text{RD}}$ ranges from approximately 0.497 (at $h = 0.5$) to 0.463 (at $h = 1.5$), with the robust 95% CI consistently excluding zero. The estimate at each bandwidth is:

h	$\hat{\beta}$	95% CI (lower)	95% CI (upper)
0.3	0.516	≈ 0.42	≈ 0.61
0.5	0.497	≈ 0.41	≈ 0.59
1.0	0.477	≈ 0.39	≈ 0.56
1.5	0.463	≈ 0.37	≈ 0.55
2.0	0.487	≈ 0.39	≈ 0.58

(b) Interpretation. The estimates are very stable across bandwidths (varying by at most ± 0.03 over a six-fold range in bandwidth). This stability is strong evidence that the RDD estimate is not driven by the particular bandwidth choice. A credible RDD should show estimates that vary smoothly and predictably with the bandwidth; large swings would raise concerns about functional form misspecification or thin data near the cutoff.

11. Your coefficient on `school_avg_score` from Q9. is approximately 0.48, while Pop-Eleches and Urquiola (2013) report values around 0.11–0.20 in Table 3 across their specifications.

    (a) What does `n_schools_town` tell you about the sample in `schools.csv`? How does it differ from the full population of Romanian schools?

    (b) Propose a reason, based on the design of the paper, why the jump in school quality would be *larger* in towns with fewer schools.

    (c) Is this difference a problem for the internal validity of the RDD? What does it imply for external validity?

(a) Sample restriction. In `schools.csv`, `n_schools_town` takes values $\{2, 3, 4\}$ only: 69% of students are in two-school towns, 29% in three-school towns. The full administrative data in `data-AER-1` covers *all* Romanian towns, including large cities with 10–29 secondary schools (mean ≈ 18). The survey extract covers a narrow, rural/small-town subsample.

(b) Why the larger coefficient in small towns? In towns with only two schools, there is a single cutoff separating a high-quality school from a lower-quality one. These two schools are typically quite different in their student composition and average score. In large cities with many schools, adjacent schools in the ranking are often very similar in quality (the schools are close together in the quality distribution), so the jump in peer quality at any given cutoff is small. The survey oversample of small-town, high-contrast cutoffs mechanically produces a larger coefficient.

(c) Internal vs. external validity. The RDD is *internally valid* for the small-town sample: within this population, students just above and just below the cutoff are comparable. However, the estimate has *limited external validity* — it does not generalize to large urban environments where the school-quality gap across adjacent schools is much smaller. This is precisely why Pop-Eleches and Urquiola (2013) estimate the model on the full administrative data and provide heterogeneity analysis by town size.

Part D. Behavioral Responses

Pop-Eleches and Urquiola (2013) argue that assignment to a better school triggers behavioral adjustments in teachers, parents, and peers — beyond the direct effect on peer quality. Parts D and E explore these responses.

12. [12] For each of `teacher_novice`, `teacher_experience`, and `teacher_certified`, estimate the RDD model with `feols()`, `school_fe` FEs, `cluster_id` clustering, bandwidth 1.0.

    (a) Report $\hat{\beta}_{\text{RD}}$ and p-value for each. Which are significant at 5%?

    (b) The coefficient on `teacher_novice` is negative ($\hat{\beta} \approx -0.036$, $p \approx 0.02$). Propose a mechanism for why assignment to a better school reduces the share of novice teachers.

    (c) `teacher_certified` is insignificant ($p \approx 0.89$). Does this mean teacher certification is irrelevant here? Discuss.

Code in Colab [12] **Answer**

Results ($|$`score_dist`$| \leq 1$, school FE, clustered SEs):

Outcome	$\hat{\beta}_{\text{RD}}$	SE	p-value
`teacher_novice`	-0.036	0.016	0.021*
`teacher_experience`	-0.625	0.996	0.530
`teacher_certified`	-0.005	0.038	0.891

(a) Significance. Only `teacher_novice` is significant at the 5% level ($p = 0.021$).

(b) Mechanism for teacher_novice. Better schools attract more experienced teachers for several reasons: teacher sorting (skilled teachers prefer more prestigious schools with higher-achieving peers), seniority-based transfer rights (experienced teachers request transfers to better schools), and lower turnover. Students assigned to the better school therefore benefit from a teaching staff with fewer novices. The effect size (-0.036 on a mean of 0.061, or roughly -59% relative) is economically significant.

(c) Certification insignificance. In Romania, teacher certification (`teacher_certified`) is close to universal at the schools in this sample ($\bar{y} \approx 0.61$, with limited variation), so there is little scope for a discontinuity in this dimension. This does not imply certification is unimportant — it simply means the policy margin at these cutoffs does not produce much variation in certification rates.

13. [13] For each of `parent_help_hw`, `parent_tutoring`, and `parent_volunteer`, estimate the RDD model with `feols()`, `school_fe` FEs, `cluster_id` clustering, bandwidth 1.0.

    (a) Report $\hat{\beta}_{\text{RD}}$ and p-value for each. Which parental behaviors respond significantly?

    (b) The coefficient on `parent_help_hw` is negative ($\hat{\beta} \approx -0.043$, $p \approx 0.027$).

Propose a mechanism: why might going to a better school reduce parental homework help?

(c) The paper interprets this as a "behavioral response." What implicit assumption is needed to interpret this RD estimate causally?

Code in Colab [13] **Answer**

Results (|`score_dist`| ≤ 1, school FE, clustered SEs):

Outcome	$\hat{\beta}_{RD}$	SE	p-value
`parent_help_hw`	−0.043	0.019	0.027*
`parent_tutoring`	−0.003	0.020	0.891
`parent_volunteer`	−0.002	0.015	0.872

(a) Significance. Only `parent_help_hw` is significant at the 5% level.

(b) Mechanism for the negative sign. Going to a better school means the student is now surrounded by higher-achieving peers and is exposed to more demanding coursework. The most natural mechanism is *parental substitution*: parents adjust their own effort downward because they believe the school's resources and environment will "do the job." Alternatively, the curriculum at the better school may be more advanced than what parents can help with, reducing effective parental assistance even if parents try. This is a behavioral response to the school quality treatment — parents in better-school environments invest less direct homework help, perhaps because they trust the school more.

(c) Causal interpretation assumption. To interpret this causally, we need the same continuity condition as for the main outcome: potential parental behavior must be continuous in the running variable at the cutoff. Parents of students just above and just below the cutoff must be comparable in their preferences and capacity for helping with homework, so that any difference in `parent_help_hw` near the cutoff reflects a causal response to school assignment, not pre-existing differences.

14. [14] Replicate the style of Figure 4 in Pop-Eleches and Urquiola (2013) for the `teacher_novice` outcome.

   (a) Compute bin means of `teacher_novice` within each `score_dist_r05` bin (|`score_dist`| ≤ 2, exclude zeros). Plot with separate linear fits on each side.

   (b) Compare the visual strength of the discontinuity in `teacher_novice` vs. `school_avg_score` from Q4.. Which is more pronounced visually and why?

Code in Colab [14] **Answer**

> **(a) Bin scatter.** The bin-scatter of `teacher_novice` shows a visible (though noisier) downward jump at the cutoff: bins just to the right of zero have a lower share of novice teachers than bins just to the left, consistent with the negative regression coefficient found in Q12.. The fitted lines on each side are slightly declining as the score distance increases on the left and decreasing on the right.
>
> **(b) Visual comparison.** The discontinuity in `school_avg_score` is sharper and more visually obvious than that in `teacher_novice`. This is expected: `school_avg_score` is a school-level constant (the same for all students in the same school), so the bin scatter contains very little within-bin noise. In contrast, `teacher_novice` has substantial student-level noise (it is a survey response that varies across students in the same school), making the underlying jump harder to see through the scatter. The regression framework handles this noise through standard errors; the plot is more illustrative for showing the direction and approximate magnitude of the effect.

15. [15] For each of `peer_rank` and `peer_bad_idx`, estimate the RDD model with `feols()`, `school_fe` FEs, `cluster_id` clustering, bandwidth 1.0.

    (a) Report $\hat{\beta}_{\text{RD}}$ and p-value. Interpret the sign of the `peer_rank` coefficient (lower values = better-quality peers).

    (b) The coefficient on `peer_bad_idx` is *positive* ($\hat{\beta} \approx 0.045$, $p \approx 0.009$): students in the better school report *more* bad peer behaviors. Propose two competing explanations.

    (c) Reconcile the peer quality findings with the school quality jump from Q9..

Code in Colab [15] Answer

Results ($|$`score_dist`$| \leq 1$, school FE, clustered SEs):

Outcome	$\hat{\beta}_{RD}$	SE	p-value
`peer_rank`	−0.134	0.064	0.037*
`peer_bad_idx`	+0.045	0.017	0.009**

(a) `peer_rank` interpretation. The coefficient −0.134 means students assigned to the better school rank their peers approximately 0.13 positions lower (on a 1–10 scale where lower is better quality). This is consistent with the school-quality jump: students in better schools perceive their classmates as higher quality. The effect is statistically significant at the 5% level.

(b) Explanations for positive `peer_bad_idx`. Two competing hypotheses:

(a) *Relative rank / "big fish, small pond" effect:* A student who barely made the cutoff for the better school is now near the *bottom* of their new peer group. They may perceive more intense competition or social dynamics as "bad" behavior, even if absolute conduct is similar. The comparison is relative.

(b) *Higher-stakes environment:* Better schools may have more ambitious students who engage in stress-related behaviors (cheating, social exclusion, excessive competition) that the student perceives as negative, even though the overall academic quality is higher.

(c) Reconciliation. The `school_avg_score` jump (Q9.) establishes that the better school genuinely has higher-achieving peers in absolute terms. The peer quality results are consistent: students in better schools rank their peers higher (`peer_rank` falls) but also report more socially challenging dynamics (`peer_bad_idx` rises). Both findings can co-exist if "better school" means "academically superior but socially more competitive."

Part E. Design Validity and Inference

16. [16] Create bin-scatter plots for `female` and `age` (bin width 0.05, $|$`score_dist`$| \leq 2$, separate linear fits on each side).

    (a) Plot both variables. Is there any visible discontinuity at zero?

    (b) What would a visible jump in `female` or `age` imply for the validity of the RDD?

Code in Colab [16] Answer

(a) Plots. The bin-scatter of `female` (share of female students) shows no visible jump at zero: the proportion of female students is roughly constant across the cutoff, with slight fluctuations attributable to sampling noise. Similarly, `age` (mean age per bin) shows a flat, continuous pattern near the cutoff.

(b) Implication of a visible jump. If there were a visible discontinuity in `female` or `age` at the cutoff, it would suggest that students above and below the cutoff are systematically different in pre-determined characteristics. This would invalidate the continuity assumption: the RDD identifies a causal effect only if potential outcomes are comparable near the cutoff, which requires that the only thing changing at zero is school assignment. A jump in background characteristics would indicate either score manipulation (families strategically targeting the cutoff) or some other sorting mechanism — either of which would confound the RDD estimate.

17. [17] Run the RDD regression for `female`, `age`, and `head_educ_tert` using `feols()` with `school_fe` FEs, `cluster_id` clustering, bandwidth 1.0.

   (a) Report $\hat{\beta}_{\text{RD}}$ and p-value for each. Are any significant at 5%?

   (b) What does a set of insignificant placebo tests establish? What does it fail to establish?

   (c) Could the RDD be valid even if one placebo test is marginally significant at the 5% level? Use the concept of multiple testing to answer.

Code in Colab [17] **Answer**

Results ($|$`score_dist`$| \leq 1$, school FE, clustered SEs):

Outcome	$\hat{\beta}_{\text{RD}}$	SE	p-value
`female`	+0.032	0.027	0.240
`age`	−0.025	0.027	0.341
`head_educ_tert`	−0.012	0.015	0.425

(a) No placebo is significant at 5%.

(b) What is and is not established. Insignificant placebo tests provide supporting evidence that students near the cutoff are balanced on observable pre-determined characteristics — a necessary (though not sufficient) condition for causal identification. They *do not* rule out imbalances in *unobservable* characteristics (e.g., ambition, parental connections, risk tolerance). Passing the placebo test raises confidence in the design but does not guarantee validity.

(c) Multiple testing. With K independent placebo tests, even a correctly specified RDD would produce at least one false positive with probability $1 - 0.95^K$. For $K = 5$, this is $\approx 23\%$. A single marginally significant result ($p \approx 0.05$) should not be alarming if other tests are clearly insignificant and the marginal result is small in magnitude. Bonferroni correction or joint tests across all pre-determined variables would provide a more rigorous check.

18. [18] Re-estimate the model from Q6. using `lm()` with explicit `factor(school_fe)` dummies (for compatibility with `linearHypothesis()`). Test $H_0 : \beta_{\text{RD}} = 0$ using `linearHypothesis()` with `vcovCL()` clustered SEs.

(a) Report the F-statistic and p-value.

(b) Numerically, $F = t^2$ for a single linear restriction. Verify this relationship using the t-statistic from `m_fe`.

Code in Colab [18] **Answer**

(a) F-test result. Testing $H_0 : \beta_{\text{RD}} = 0$ using `linearHypothesis()` on the `lm()` model with `factor(school_fe)` and clustered SEs:

$$F(1, K-1) \approx 104, \quad p < 0.001.$$

We strongly reject the null. This is consistent with the t-statistic ≈ 10.2 from `m_fe`.

(b) Relationship $F = t^2$. For a single linear restriction $H_0 : \beta = 0$, the Wald F-statistic is exactly $F = t^2 = (10.2)^2 \approx 104$. This algebraic identity holds because the F-test for a single coefficient is equivalent to the two-sided t-test: rejecting $|t| > z_{0.025}$ is the same as $t^2 > z^2_{0.025} = F_{1,\infty}(0.05)$.

19. For the model estimated in Q6. (with school fixed effects), use `linearHypothesis()` to test the joint hypothesis $H_0 : \gamma_1 = \gamma_2 = 0$, i.e. that the running variable has no effect on `school_avg_score` on either side of the cutoff.

(a) Report the F-statistic and p-value. Can you reject H_0?

(b) Why is it important to control for the running variable in the RDD specification, even if we expect γ_1 and γ_2 to be small near the cutoff?

(c) Suppose you omitted the running variable terms. Would $\hat{\beta}_{\text{RD}}$ be upward or downward biased? Explain.

(a) Test result. Testing $H_0 : \gamma_1 = \gamma_2 = 0$ (both running-variable terms equal zero) using `linearHypothesis(m_fe_lm, c("score_dist = 0", "score_dist_above = 0"), vcov.=...)` yields $F \approx 1.6$, $p \approx 0.20$. We fail to reject H_0: near the cutoff, the running variable does not have a strong independent association with school quality conditional on school FEs. (The slopes $\hat{\gamma}_1$ and $\hat{\gamma}_2$ are individually small, as seen in Q5..)

(b) Why control for the running variable? Even if the slopes are small within the bandwidth, omitting them biases the RDD estimate if the relationship between the running variable and the outcome is non-zero anywhere in the bandwidth. The continuity assumption ensures potential outcomes are smooth at the cutoff, but they need not be constant across the bandwidth. Controlling for the running variable removes this smooth trend, ensuring $\hat{\beta}_{\text{RD}}$ captures only the discontinuous jump.

(c) Direction of bias if running variable omitted. In this setting, higher scores (higher r_i) tend to be associated with access to better schools, so omitting r_i would create a positive association between being above the cutoff ($D_i = 1$) and the outcome that partly reflects the smooth upward trend, not the causal jump. The bias on $\hat{\beta}_{\text{RD}}$ would be *upward* (positive direction), overstating the causal effect of school quality.

20. The RDD identifies a **Local Average Treatment Effect** (LATE) rather than an Average Treatment Effect (ATE).

    (a) In this setting, who is the *marginal student* — the student whose school assignment is determined by the cutoff? Why is the causal effect identified for this specific group?

    (b) The authors find a jump in `school_avg_score` near the cutoff but limited effects on `baccalaureate grade` (Table 4, not in this extract). Propose an explanation based on the characteristics of the marginal student.

    (c) The dataset covers only towns with 2–4 schools. Discuss one dimension along which the LATE for this subpopulation may differ from a national-level ATE, and what additional data or design would be needed to extrapolate.

(a) The marginal student. The marginal student is one whose exam score is exactly at the cutoff: they would have gone to the higher school if they scored infinitesimally higher, and to the lower school if they scored infinitesimally lower. The RDD identifies the causal effect *only for this marginal group*, not for all students. Students far from the cutoff — those who comfortably cleared it (infra-marginal) or those who fell well short — may respond very differently to school quality changes. Their potential outcomes are not observed near the cutoff, so the RDD estimate does not apply to them.

(b) Limited effects on baccalaureate grade. The marginal student at the cutoff is, by construction, near the boundary of academic qualification for the better school. These students are likely to be lower-ranked in their new school (a "big fish, small pond" reversal): they now compete against peers who would have been comfortably above the cutoff. Despite attending a better school, the combination of higher competition and lower relative rank may dampen the benefit for their eventual baccalaureate outcomes. The LATE at the cutoff could be small even if the ATE for all students would be large.

(c) External validity limitation. The dataset covers only towns with 2–4 schools. The LATE for this subsample — where the school-quality gap is large and towns are small — may not extrapolate to (i) large urban environments where adjacent schools differ only marginally in quality, or (ii) students far from the cutoff who chose schools with strong margins. To extrapolate, one would need either additional cutoffs in larger towns (as Pop-Eleches and Urquiola (2013) exploit in `data-AER-1`) or a structural model of how the treatment effect varies with school-quality contrast and student ability.

Coding Lab C.9.7

Further regression tools

Instructions: Work individually. Complete all exercises in the coding lab notebook and submit as required by your instructor.

Notebook: Coding Lab 7 — Further regression tools

Instructor Notebook: Coding Lab 7 — Further regression tools

Find answers to all coding lab exercises in the corresponding Colab notebook.

Additional materials and references

References:

- (Pop-Eleches & Urquiola, 2013)
- 🗁 **Drive: Review notes and supplementary readings**

Data and replication:

- Pop-Eleches and Urquiola (2013) replication package (AER data archive). The dataset `schools.csv` is derived from `data-AER-7.dta` included in the authors' replication package. This guided case is released under **CC BY 4.0**; the underlying data are subject to the AER's data use policy.

R Packages:

- (Calonico et al., 2015) — `rdrobust` package for MSE-optimal bandwidth selection and local polynomial RDD estimation.
- (Bergé, 2018) — `fixest` package for high-dimensional fixed-effects regression with clustered standard errors.

14.310x Flipped Classroom

Week 10 Answers

Random assignment and Instrumental Variable (IV) models
More on IV: multiple instruments, multiple stages.

Checklist

- ☐ Read the *details* on `ivreg()` in the the `AER` package. Make sure you learn how to enter a 2-stage specification. (Requirement)
- ☐ Complete Open Case 4 (Session 1)
  - ☐ `worms.csv`
- ☐ Complete Guided Case (Session 2)

Case Study O.10.4

RCTs as an IV problem: (Miguel & Kremer, 2004)

Instructions: Work in pairs or groups of 3.

Notebook: Case Study 4

Dataset: worms.csv

Answer the following questions collaboratively; one of you must

---

Scenario: In rural western Kenya, intestinal worms (hookworm, roundworm, schistosomiasis) are highly prevalent and infect a large share of primary school children. Worm infections can lead to iron-deficiency anemia, malnutrition or abdominal pain; all important factors in school absenteeism and inattention. Non-severe cases may present as asymptomatic, with more subtle impairments in children's educational attainment and health outcomes. Deworming treatment with Albendazole and Praziquantel pills is cheap and effective, but rarely provided at scale.

(Miguel & Kremer, 2004) designed an intervention to distribute deworming pills to schools in a region surrounding Victoria Lake. They partnered with an NGO (ICS Africa) to implement an RCT where schools were phased to treatment as follows:

- **Group 1** began treatment on 1998 while groups 2 and 3 remained controls.

  > **Instructor note**: In 1998, Group 1 schools are the treatment arm; Groups 2 and 3 are untreated controls. This is the comparison used throughout Part 1. The phased design ensures a clean control group (Group 3) persists into 1999, enabling a second wave of estimates.

- **Group 2** phased in treatment on 1999.

  > **Instructor note**: In 1999, Group 2 becomes treated while Group 3 remains the lone control. Group 1 continues receiving treatment. The 1999 wave replicates the 1998 estimate with a different cohort of schools, increasing external validity.

- **Group 3** began on 2001 — after the study window for these published results.

  > **Instructor note**: Group 3 acts as a control for the full published study window (1998–2000). Their delayed start means they were never treated during this period, providing a clean long-run counterfactual.

For simplicity, we will focus only on the short-term outcomes of the first intervention. That means only schools in Group 1 schools will be considered treated in our dataset.

Glossary for `worms`

- `pupid` — pupil ID (unique child identifier).
- `baseline_school` — ID of the child's school.
- `baseline_grade` — The student's academic grade. Encoded 1 to 11.
- `assigned_98` — Indicator:=1 the student's school is part of Group 1.
- `treated_98` — Indicator:=1 the student received **any** deworming treatment in 1998 (Albendazole or Praziquantel).
- `attendance_98` — The student's observed attendance rate (proportion of times present during surveyor's visits 1 through 8)
- `testscores_98` — Reported average test scores mid-1998 (standardized by grade).
- `sex` — Pupil's gender. Encoded 1 for male and 0 for female.
- `house_floor` — Material of household floor. Encoded 0 for mud, and 1 for cement.
- `weight` – Pupil's weight in kg.

Out of practicality, categorical variables contain a level `"N/A"`, as opposed to R's `NA`. And a small number of missing values in continuous covariates are mean-imputed.

Part 1. Random assignment as an instrument

Suppose the researchers don't provide the deworming treatment. Instead, they decide to widely disseminate deworming information across all schools, including where to purchase the drugs at low prices:

1.1 If a survey was conducted at the end of 1998, how would you expect covariates such as `house_floor`, or `weight` compare between those students with (`treated_98 == 1`) and those in the (`treated_98 == 0`) group?

> Without random assignment of the actual pills, who gets dewormed is determined by individual choice and access — this is the selection problem. Families with higher SES are more likely to learn about, purchase, and administer the medication. We would therefore expect treated students to have:
>
> - **Higher** `house_floor`: wealthier households (cement floors) are better positioned to act on health information.
> - **Higher** `weight`: better-nourished children in wealthier families are also more likely to be treated, even though they may already have lower worm burdens.
>
> In short, the treated group would not be comparable to the untreated group on observable (or unobservable) pre-treatment characteristics. Any estimated effect on `attendance_98` would conflate the drug's efficacy with the pre-existing advantages of the treated students.

1.2 As previously noted, children with heavy worm infections are prone to missing school. Search the paper's introduction for evidence on the effectiveness of Albendazole and Praziquantel in abating worm infections.

> The paper's introduction reports that a single dose of Albendazole clears roundworm and hookworm infections with close to 99% efficacy, and Praziquantel reduces schistosomiasis infections substantially. Both drugs act within days to weeks. Given the high prevalence of infections in the region (over 90% of children had at least one species of worm), effective treatment of a large share of the school population would be expected to reduce absenteeism considerably.
>
> **Instructor note**: If students have access to the paper, direct them to Section II.A. The key figure is that deworming reduces moderate-to-heavy worm infection prevalence by roughly 25 percentage points within a year of treatment.

(a) Suppose students' absenteeism is 100% due to infection symptoms. What should be the mean of `attendance_98` among treated children?

> Under the stated assumption (all absenteeism is caused by infection symptoms, and treatment is 100% effective), treated students would have **no remaining reason to miss school due to illness**. Their expected attendance would be:
>
> $$\mathbb{E}[\,\text{attendance}_{98} \mid \text{treated} = 1\,] = 1$$
>
> **Instructor note**: This is an idealized bound. In practice, empirical attendance among treated students is 87.65% — indicating that other factors (weather, distance, harvest season, caregiving responsibilities) account for a substantial share of absences even after worms are eliminated.

(b) Empirically, not all school absenteeism is worm-related. Describe the covariates of the **treated** students you expect to miss school the most.

> Even after successful deworming, treated students with the following characteristics are more likely to miss school for non-worm reasons:
>
> - **Lower** `house_floor` (mud floors): students from poorer households face greater economic pressure — they may be pulled out of school to work, help with chores, or care for younger siblings.
> - **Lower** `weight`: malnutrition independent of worm burden may cause fatigue or illness unrelated to the parasites treated.
> - **Higher** `baseline_grade`: older students are more likely to be assigned household or agricultural responsibilities that compete with school attendance.
> - **Female** (`sex == 0`): in some contexts, girls face higher domestic obligations that reduce school participation.

1.3 Back to (Miguel & Kremer, 2004)'s randomized assignment. How would your answers to 1.1 and 1.2 change if each value in `treated_98` were drawn iid from a Bernoulli with $p = 0.5$?

With iid Bernoulli($p = 0.5$) assignment of `treated_98`, treatment status would be independent of all pre-treatment covariates in expectation. The selection problem disappears:

- **Answer to 1.1**: `house_floor` and `weight` would be *balanced* across treated and untreated students. Neither group would be systematically richer or healthier at baseline.
- **Answer to 1.2**: The question about drug effectiveness is unchanged — Albendazole and Praziquantel are still effective. But now the comparison group (untreated) is a valid counterfactual, so the difference in attendance between treated and untreated would be an unbiased estimate of the causal effect of treatment.

However, note that (Miguel & Kremer, 2004)'s actual design randomizes *assignment* at the school level, not individual treatment receipt — which is what motivates the IV approach addressed in the rest of this case.

1.4 [1.1] But did every student assigned to treatment receive any? What is the average compliance for assigned and non-assigned students?

No. Only **75.07%** of students assigned to treatment (`assigned_98 == 1`) actually received any deworming pill. No student in a non-assigned school (`assigned_98 == 0`) received treatment.

$$\mathbb{E}[X \mid Z = 1] = 0.7507 \qquad \mathbb{E}[X \mid Z = 0] = 0.0000$$

```
worms %>% group_by(assigned_98) %>% summarize(mean(treated_98))
```

Instructor note: The zero compliance in the control group (no always-takers) is a particularly clean feature of this design. It means the only source of non-compliance is assigned students who did not take the pills (never-takers), and there is no contamination of the control group. This implies the instrument has a sharp lower bound and the Wald estimator recovers the LATE for compliers only.

Suppose we now have the model

$$X_i = \pi_0 + \pi_1 Z_i + v_i, \quad i = 1, 2, .., N \tag{11}$$

Where $X_i = \mathbf{1}(\text{i treated})$ and $Z_i = \mathbf{1}(\text{i's school assigned to treatment})$. π_0, π_1 are coefficients, and $v_i \sim N(0, \sigma_v^2)$

1.5 [1.2] Use `lm()` to get the OLS estimates $\hat{\pi}_0$, $\hat{\pi}_1$ and answer the following:

(a) Interpret the model: which variable does this model explain in terms of which other variable?

(b) What does $\hat{\pi}_0$ represent?

(c) $\hat{\pi}_1$ can be expressed as a difference of means. Write it down.

(d) Test the hypothesis that $H_0 : \pi_0 = 0$ and $H_1 \neq 0$ with 95% confidence. What does this result imply? Do you think the data-generating process warrants it?

```
first_stage <- lm(treated_98 ~ assigned_98, data = worms)
summary(first_stage)
```

(a) **Interpretation**: The model explains *treatment receipt* (X_i) as a linear function of *school assignment* (Z_i). It is the first stage of the IV regression.

(b) $\hat{\pi}_0 = 0.0000$: the baseline compliance rate among non-assigned students. Since no student in a non-assigned school received treatment, the intercept is exactly 0.

(c) $\hat{\pi}_1 = 0.7507$: the increase in the probability of treatment due to assignment. It equals the difference of group means:

$$\hat{\pi}_1 = \mathbb{E}[X \mid Z = 1] - \mathbb{E}[X \mid Z = 0] = 0.7507 - 0.0000 = 0.7507$$

(d) $H_0 : \pi_1 = 0$ (instrument is irrelevant). The t-statistic exceeds 100 ($z \approx 165$); we strongly reject at any conventional level. The design warrants this result: assignment *directly* determines whether pills are distributed to a school, so the instrument is mechanically relevant.

1.6 What percentage of students were treated? What is the attendance rate among those treated and untreated?

26.21% of students in the dataset were treated (6,781 out of 25,874).

Group	Mean attendance
Treated (`treated_98 == 1`)	0.877
Untreated (`treated_98 == 0`)	0.745

```
worms %>% group_by(treated_98) %>% summarize(mean(attendance_98),
n())
```

Instructor note: The naive difference in attendance (87.7% vs 74.5% ≈ 13.2 pp) is not an unbiased causal estimate because treatment receipt is endogenous — those who chose to take the pills differ from those who did not.

1.7 What is the percentage points difference in attendance between compliers and non-compliers [hint: first create `complied` ...]

```
worms <- worms %>% mutate(complied = as.integer(assigned_98 ==
treated_98))
```

Group	N	Mean attendance
Compliers (`complied == 1`)	23,622	0.784
Non-compliers (`complied == 0`)	2,252	0.729
Difference		**+0.055 pp**

Instructor note: All 2,252 non-compliers are students in *assigned* schools who nonetheless did not receive treatment (never-takers). Since $\mathbb{E}[X \mid Z = 0] = 0$, there are no always-takers. The 5.5 pp difference reflects that the non-complier group is relatively small and consists entirely of Group 1 students who opted out; this comparison does *not* recover the causal effect. The Wald estimator at the end of the case study provides the correct LATE:

$$\hat{\beta}_{\text{Wald}} = \frac{\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]}{\mathbb{E}[X \mid Z = 1] - \mathbb{E}[X \mid Z = 0]} = \frac{0.8398 - 0.7470}{0.7507 - 0.0000} = 0.1236$$

We have the following model

$$Y_i = \alpha + \beta X_i + \epsilon_i, \tag{12}$$

Again, X_i is a treatment dummy. Y_i is the attendance rate for student i, and the error $u_i \sim N(0, \sigma_\epsilon^2)$ Constants α, β are the model's coefficients.

$$\mathbb{E}[Z_i u_i] = 0, \quad \text{and} \quad \text{Cov}(Z_i, X_i) \neq 0, \tag{13}$$

$$\hat{\beta}_{\text{Wald}} = \frac{\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]}{\mathbb{E}[X \mid Z = 1] - \mathbb{E}[X \mid Z = 0]}, \tag{14}$$

Part III

References and Supplementary Materials

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References

Week 6

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Week 7

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Week 9

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Week 10

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Notes